

Report prepared for the WHO Consultation on the Composition of Influenza Virus Vaccines for the Northern Hemisphere 2024/2025

Worldwide Influenza Centre, WHO CC for Reference and Research on Influenza, The
Francis Crick Institute, 1 Midland Road, NW1 1AT

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Prof. Nicola Lewis (Director - WIC)

Dr. Ruth Harvey (Deputy Director)

Dr. Monica Galiano (Head of Molecular Testing and Genomics)

Dr. Alex Byrne,

Dr. Zheng Xiang,

Ms. Christine Carr, Ms. Burcu Ermetal

Dr. Karen Cross, Ms. Aine Rattigan

Ms. Alice Lilley, Mr. Michael Bennett

Ms. Chandrika Halai, Ms. Becky Clark, Mr. Lorin Adams

Dr. Tanya Mikael, Ms. Abi Lofts, Dr. Alize Proust

Tel (1): 020 3796 7254 (NL) Tel (2): 020 3796 0563 (WIC)

Email: Nicola.Lewis@crick.ac.uk; Ruth.Harvey@crick.ac.uk; whocc@crick.ac.uk

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Additional Personnel from the Francis Crick Institute

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Influenza Epidemiology in the WHO European Region

The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and contributed by Margaux Mesle, Piers Mook and Marc-Alain Widdowson.

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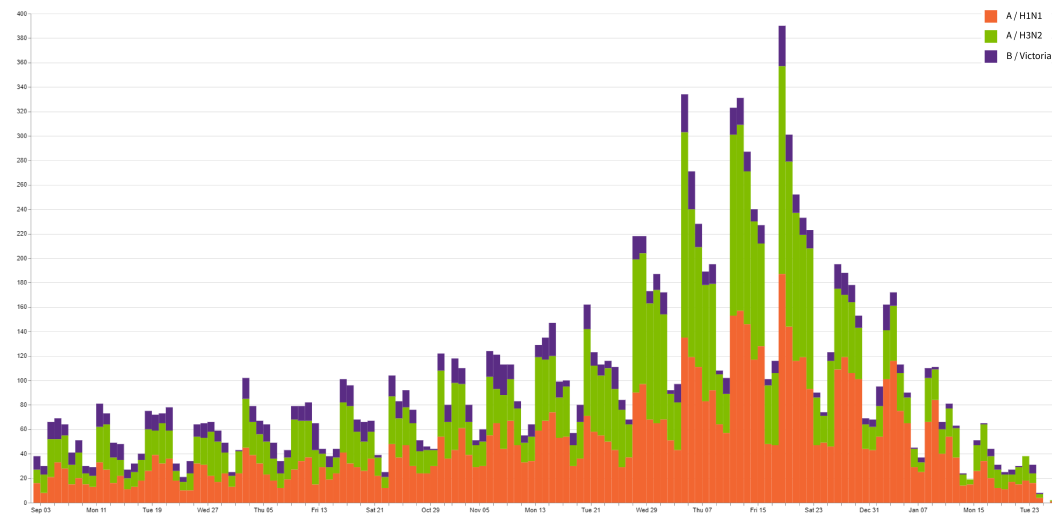
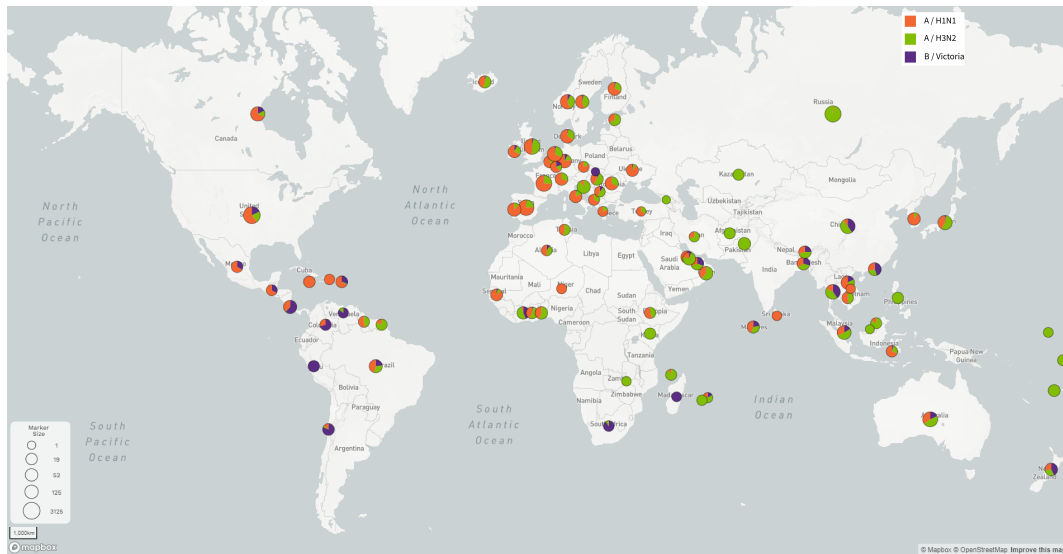
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Global influenza activity

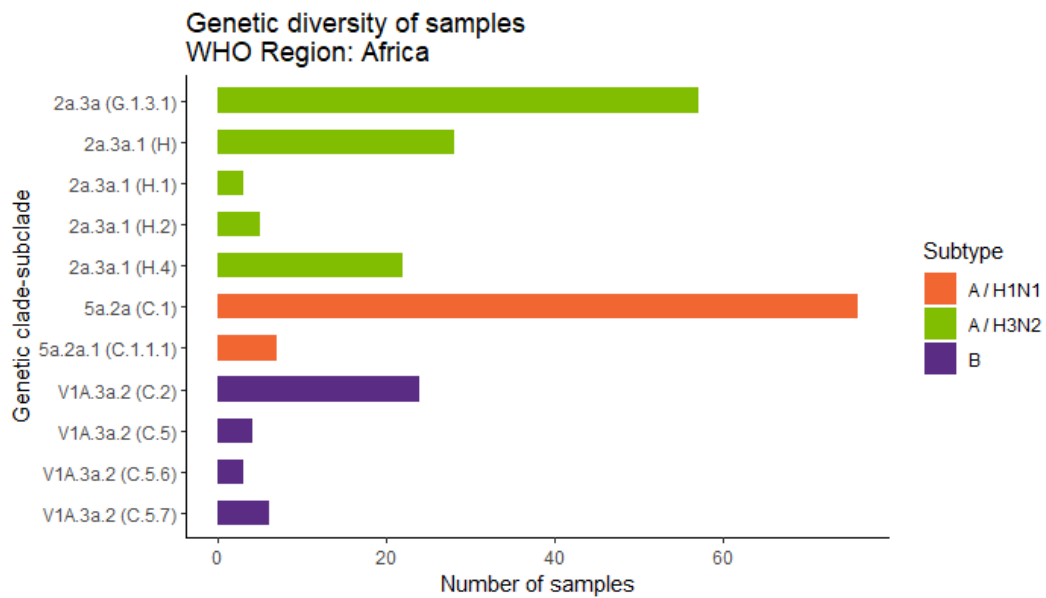
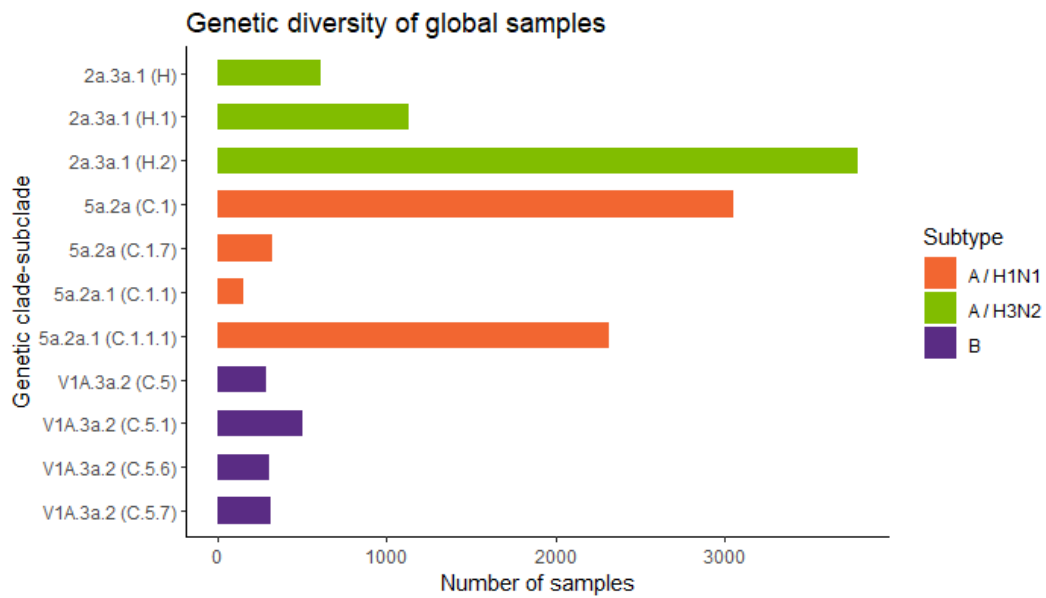
Influenza by type/subtype

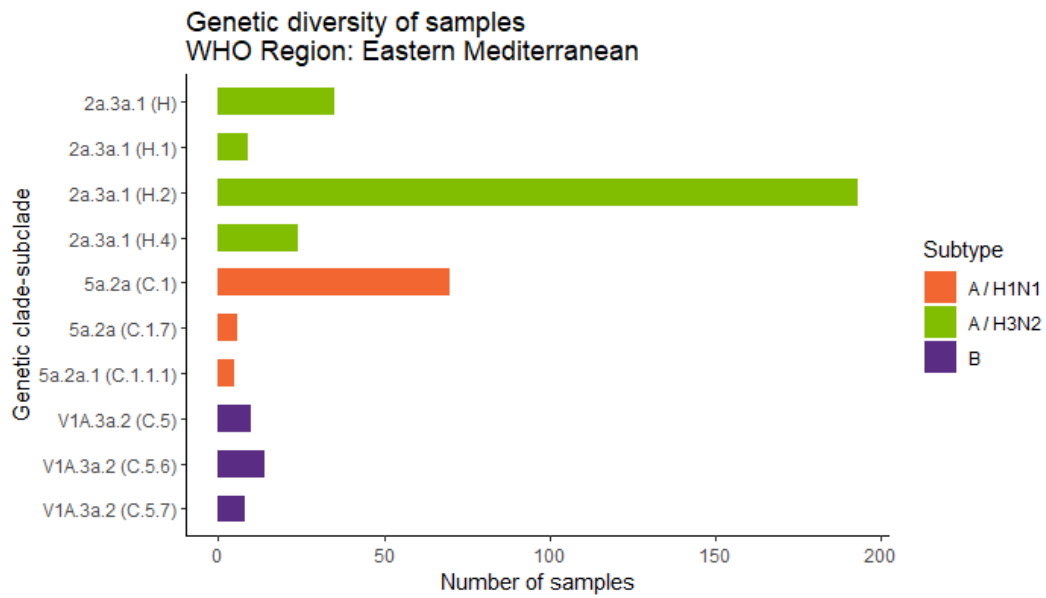
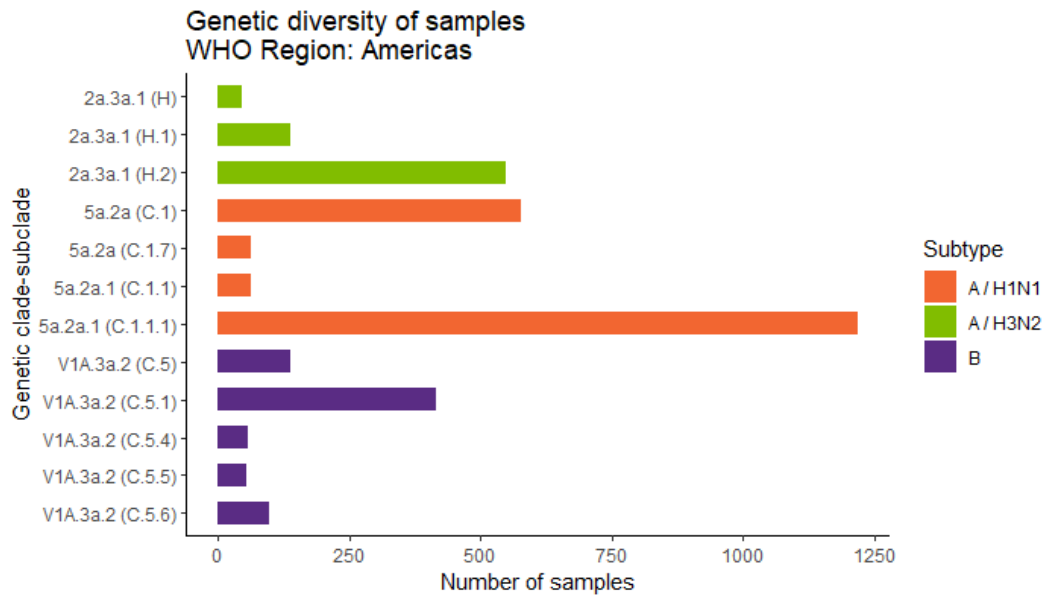
MicroReact map for influenza viruses with collection dates from 1st September 2023 through to 8th February 2024 as deposited in GISAID, coloured by Type/subtype and downloaded on 8th February 2024. Geographic markers scaled to detection proportions. Full length HA.

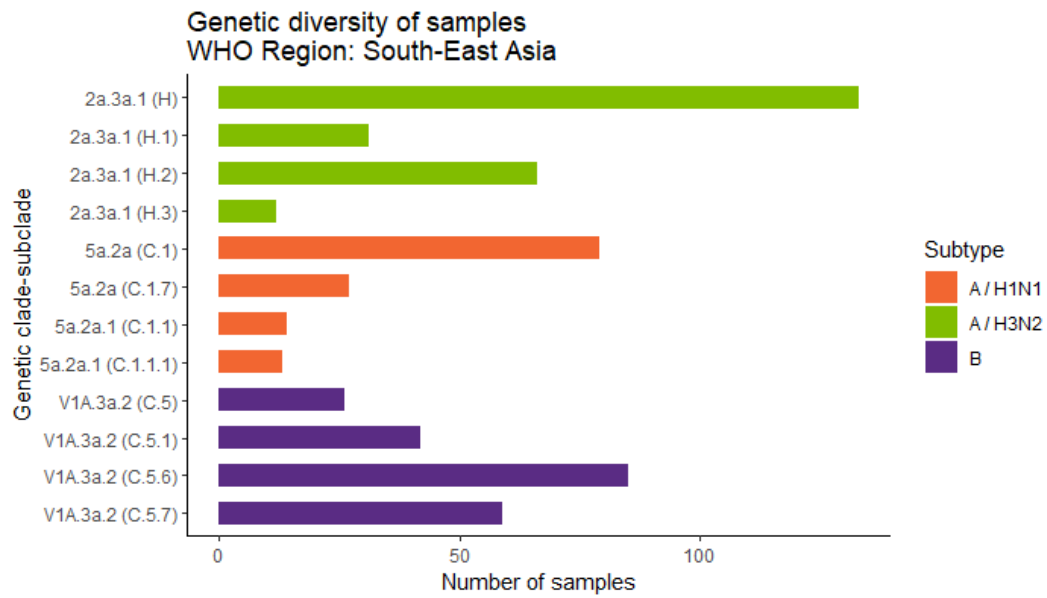
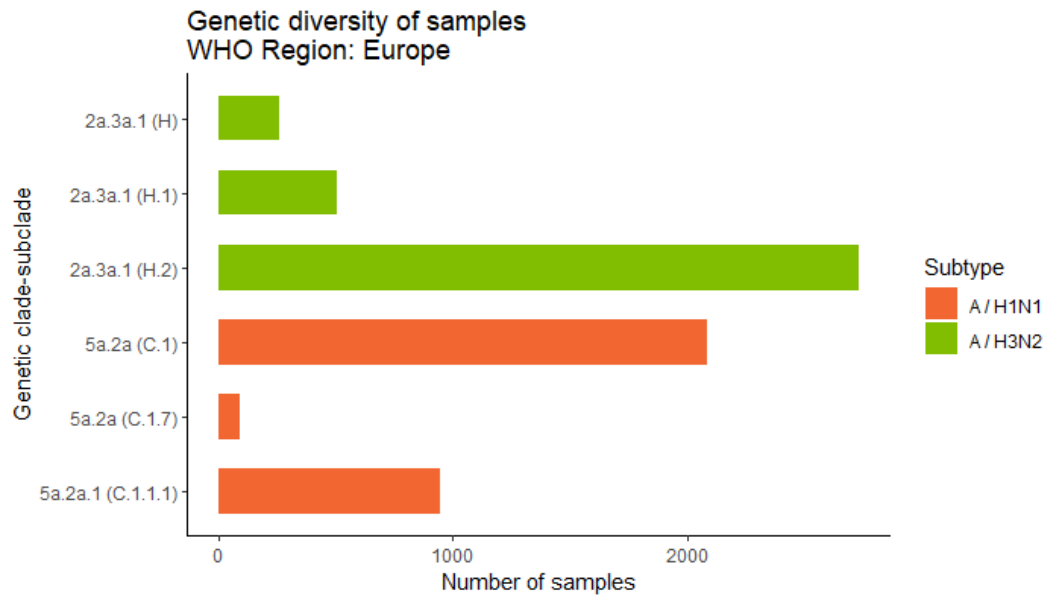


Regional influenza activity

Genetic diversity by Type/Lineage and group







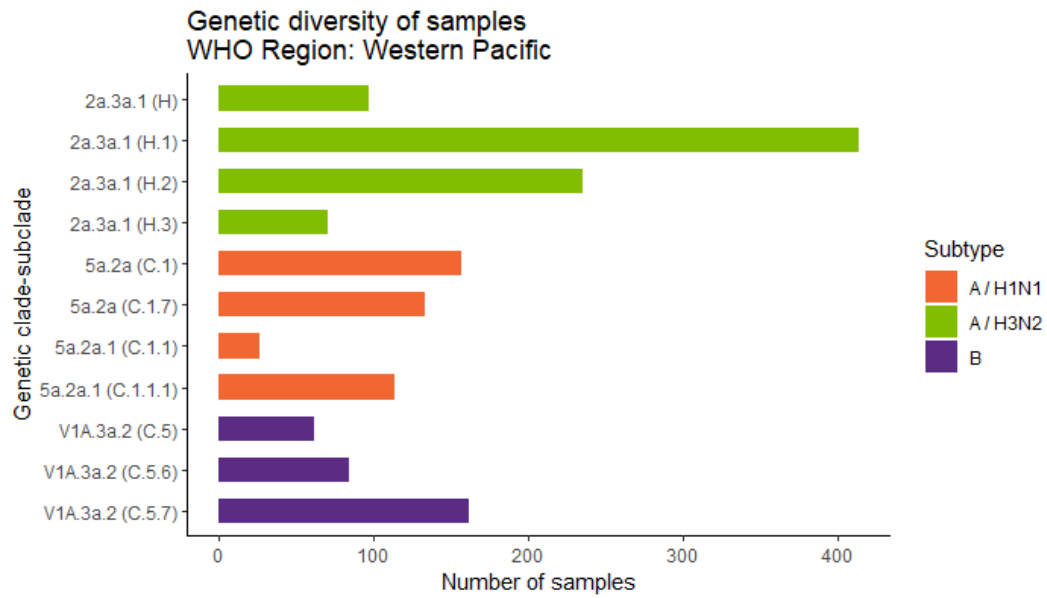


Table of shipments to the WIC

country	# samples received	A	B	BV	H1pdm	H3	# shipments
Algeria	10			1	5	4	1
Armenia	20				1	19	1
Austria	33	1			27	5	1
Belgium	18				10	8	1
Bosnia and Herzegovina	8			1		7	1
Bulgaria	33			3	26	4	1
Cameroon	52	36			7	9	1
Croatia	21				4	17	1
Cyprus	30				29	1	1
Czech Republic	8				7	1	1
Denmark	55			3	25	27	3
Egypt	10	4			6		1
Estonia	20				10	10	1
France	63			2	36	25	2
Georgia	20	2			12	6	1
Germany	46			8	24	14	2
Ghana	20			1	9	10	1
Greece	35	1	2		28	4	2
Hong Kong	30			10	10	10	1
Hungary	20			3	5	12	1
Iceland	45			1	20	24	1
Iran	12	2			5	5	1
Iraq	12				6	6	1
Ireland	59	1	1		22	35	2
Italy	15			2	12	1	1
Kazakhstan	40					40	1
Kyrgyzstan	54			4		50	2
Latvia	20				8	12	1
Luxembourg	17	1	1		12	3	1
Madagascar	79	4		13	11	51	1
Mauritius	48			17	4	27	1
Montenegro	15				10	5	1
Netherlands	36			5	13	18	2
Norway	69	1		12	35	21	2
Oman	100			7	45	48	1
Poland	24	4	1		18	1	1
Portugal	25			2	21	2	1
Qatar	93	5			17	71	1
Romania	35			1	25	9	1
Senegal	30	1			26	3	1
Serbia	10			1	4	5	1
Slovenia	32			3	15	14	1
Spain	336	19	4	15	242	56	7
Sweden	23			1	13	9	1
Switzerland	32			2	16	14	2
Togo	93				48	45	1
Tunisia	32				13	19	1
Ukraine	20	18	2				1
United Kingdom	14	3			3	8	2
TOTALS	1972	52	59	121	945	795	66

Epidemiology

Summary of influenza activity in the WHO European Region, 2nd October 2023 - 14th January 2024

Distribution of viruses

Between weeks 40/2023 and 03/2024, a total of 135,705 influenza detections from sentinel and non-sentinel primary care sources were reported to the European Surveillance System (TESSy) from countries and areas in the Region (Table E1). This is compared to 162,669, for the same period in 2022-2023, representing a 17% decrease (Table E1). However, this decrease occurred in the context of a 2% decrease in the number of specimens tested in 2023-2024 compared to 2022-2023 (1,184,874 and 1,208,962 specimens, respectively) (Table E1).

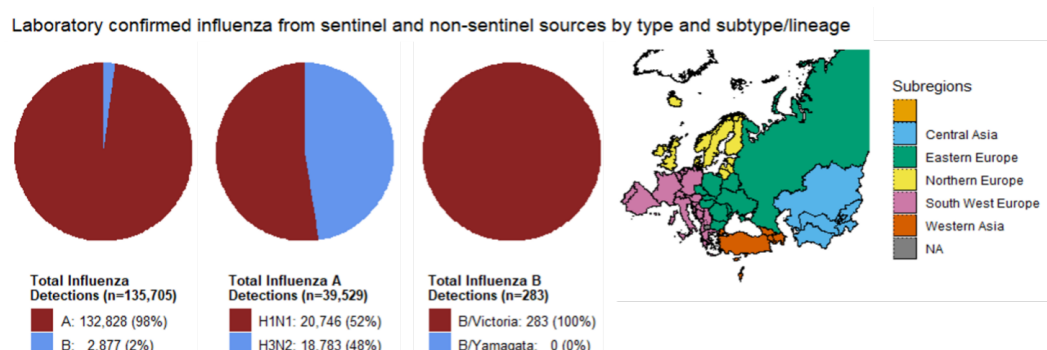
Table E1. 2023-2024 Influenza Season in the WHO European Region: weeks 40/2023-03/2024

Virus type/subtype/lineage	Cumulative number of detections for weeks 40/2023-03/2024				Cumulative number of detections for weeks 40/2022-03/2023			
	Sentinel sources	Non-sentinel sources	Totals	%	Sentinel sources	Non-sentinel sources	Totals	%
Influenza A	9,591	123,237	132,828	98	13,584	135,434	149,018	92
A(H1N1)pdm09	5,107	15,639	20,746	52	2,851	23,241	26,092	48
A(H3N2)	2,503	16,280	18,783	48	8,395	19,481	27,876	52
A not subtyped	1,981	91,318	93,299		2,338	92,712	95,050	
Influenza B	297	2,580	2,877	2	1,181	12,470	13,651	8
Victoria lineage	66	217	283	100	411	1,045	1,456	100
Yamagata lineage	0	0	0	0	0	0	0	0
Lineage not ascribed	231	2,363	2,594		770	11,425	12,195	
Total detections	9,888	125,817	135,705		14,765	147,904	162,669	
Total tested	80,882	1,103,992	1,184,874		63,137	1,145,825	1,208,962	

*Percentages are shown for total detections (types A & B [in bold type], and for viruses ascribed to influenza A subtype and influenza B lineage). Ratios are given for type A:B [in bold type], A(H3N2):A(H1N1) pdm09 and Victoria:Yamagata lineages.

Combined sentinel and non-sentinel data for influenza virus detections in the Region showed that most viruses detected were type A (98% compared to 2% type B) (Table 1). This was consistent across the Region with all subregions detecting at least 96% type A. Combined reports of sentinel and non-sentinel data for influenza type A viruses showed similar proportions of A(H1N1)pdm09 subtype (52%) and A(H3N2) viruses (48%) (Table E1). This was consistent with the observed distribution in northern Europe. However, in south west Europe, influenza A(H1N1)pdm09 viruses dominated (82%) and in central Asia, west Asia and eastern Europe influenza A(H3N2) viruses dominated to varying degrees (>99% for central Asia, 91% for eastern Europe and 65% in west Asia) (Table E2-A). Of 283 type B viruses ascribed to a lineage, all were B/Victoria and no B/Yamagata were detected. When the analysis was restricted to sentinel data only for weeks 40/2023 to 03/2024, the regional distribution of influenza by type was similar: of 9,888 specimens testing positive for influenza, the majority were type A (97%). Of type A viruses, the majority were A(H1N1)pdm09 subtype (67%) and of the type B viruses ascribed to a lineage, all were B/Victoria (Table E2-B). In south west Europe, western Asia and Eastern Europe A(H1N1)pdm09 viruses dominated (83%, 66% and 62%, respectively). However, similar proportions of influenza subtypes were observed in Northern Europe (52% of A(H1N1)pdm09) whereas A(H3N2) dominated in central Asia (99%). Of the total sentinel detections, 5,211 (53%) were from south west Europe and 2,823 (29%) were from northern Europe (Table E2-B)

Table E2. Distribution of Influenza Detections in the WHO European Region, weeks 40/2023-03/2024



A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South West Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	57,807 (99%)	18,613 (98%)	54,964 (97%)	487 (96%)	957 (>99%)	132,828 (98%)
Influenza A subtyped	16,441	11,085	10,861	358	784	39,529
A(H1N1)pdm09	13,421 (82%)	985 (9%)	6,104 (56%)	234 (65%)	2 (<1%)	20,746 (52%)
A(H3N2)	3,020 (18%)	10,100 (91%)	4,757 (44%)	124 (35%)	782 (>99%)	18,783 (48%)
Influenza B	821 (1%)	287 (2%)	1,749 (3%)	18 (4%)	2 (<1%)	2,877 (2%)
Influenza B lineage determined	84	14	185	0	0	283
B/Yamagata-lineage	0 (0%)	0	0 (0%)	0	0	0 (0%)
B/Victoria-lineage	84 (100%)	14 (100%)	185 (100%)	0	0	283 (100%)
Total	58,628	18,900	56,713	505	959	135,705

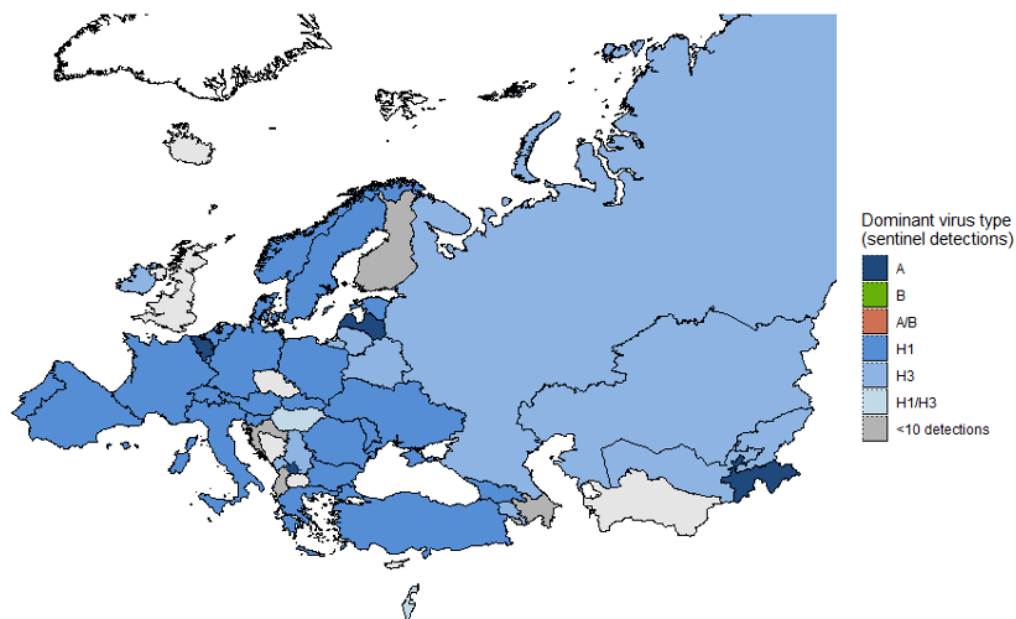
WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: (<http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>)

B. Laboratory confirmed influenza from sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	<u>South West</u> Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	5,117 (98%)	865 (95%)	2,677 (95%)	413 (97%)	519 (100%)	9,591 (97%)
Influenza A subtyped	4,034	581	2,307	311	377	7,610
A(H1N1)pdm09	3,336 (83%)	362 (62%)	1,204 (52%)	204 (66%)	1 (<1%)	5,107 (67%)
A(H3N2)	698 (17%)	219 (38%)	1,103 (48%)	107 (34%)	376 (>99%)	2,503 (33%)
Influenza B	94 (2%)	45 (5%)	146 (5%)	12 (3%)	0 (0%)	297 (3%)
Influenza B lineage determined	47	10	9	0	0	66
B/Yamagata-lineage	0 (0%)	0	0	0	0	0 (0%)
B/Victoria-lineage	47 (100%)	10	9	0	0	66 (100%)
Total	5,211	910	2,823	425	519	9,888

A dominant virus (sub)type (comprising at least 60% of influenza virus detections from sentinel sources) could be assigned by 43 countries and areas within the Region, with six (Albania, Azerbaijan, Croatia, Finland, Malta and United Kingdom (Northern Ireland)) having insufficient numbers of detections (fewer than ten) for a robust dominance calculation. All countries and areas reported predominance of influenza type A viruses, none reported predominance of influenza type B viruses and none reported co-dominance of type A and type B viruses (Figure E1). Four (Belgium, Latvia, Tajikistan and Kosovo (in accordance with United Nations Security Council Resolution 1244 (1999)) reported dominance of type A without specifying a subtype, and 39 were able to ascribe dominance (60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes for the season so far: 24 reported A(H1N1)pdm09 dominance, 11 (reported A(H3N2) dominance, and four (Hungary, Israel, United Kingdom (England) and United Kingdom (Scotland) reported co-dominance of A(H1N1)pdm09 and A(H3N2) viruses.

Figure E1. Influenza virus dominance in the WHO European Region by (sub)type and Country, territory or area, 2023-2024 season



Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 41% and 59% of detections. Calculations are only considered valid if at least 10 viruses were detected.

Timing and spread

Historically, a threshold of 10% of influenza-like illness (ILI) or acute respiratory infection (ARI) specimens from sentinel primary care sites testing positive for influenza viruses for two consecutive weeks has been used to indicate the start of the influenza season in the Region. The 2023-2024 seasonal influenza epidemic in the WHO European Region started in week 49/2023, within range compared with the five previous seasons: ranging from week 45 (2022-2023 season) to 49 (2021-2022 season) (Figures E2-A/B). Between weeks 40/2023 and 03/2024, the proportion of specimens testing positive for influenza on a weekly basis reached was highest at 31% in week 52/2023, which was lower than seasonal peaks in recent seasons (38-59%; 2016-2017 to 2022-2023 seasons) (Figures E2-A/B). The proportion of subtyped influenza viruses that were A(H1N1)pdm09 has increased over the course of season so far and there has also been a small increase in proportion of influenza type B virus detections. In non-sentinel settings, a change in proportion of subtyped viruses was observed, shifting from A(H1N1)pdm09 to A(H3N2) from week 02/2024 (Figure E2-C). SARS-CoV-2 and Respiratory Syncytial Virus (RSV) co-circulated with influenza virus between weeks 40/2023 and 03/2024. The SARS-CoV-2 test positivity from sentinel specimens peaked at 18% in week 50/2023 and has decreased to 7% up to week 03/2024. The RSV test positivity peaked at 11% in week 49/2022 before starting to slowly decrease (Figure E3). Based on weekly reporting of the intensity of influenza activity by each of the countries and areas, there was no apparent geographical pattern to initial sustained increases in activity across the Region (Figure E4). Over the season, a total of 49 countries and areas reported activity above their calculated baselines, with 31 having reported medium or higher activity in at least 1 week. A total of two countries reported very high intensity in some weeks (Finland and Russian Federation). (Note: this indicator is typically based on ILI and/or ARI activity but at least one influenza virus detection per week is required to meet the definition for an intensity level above baseline. Not all countries, territories or areas reporting intensity above baseline have reported detections) (Figure E4).

Figure E2. Seasonal influenza activity in the WHO European Region: weeks 40/2023-03/2024

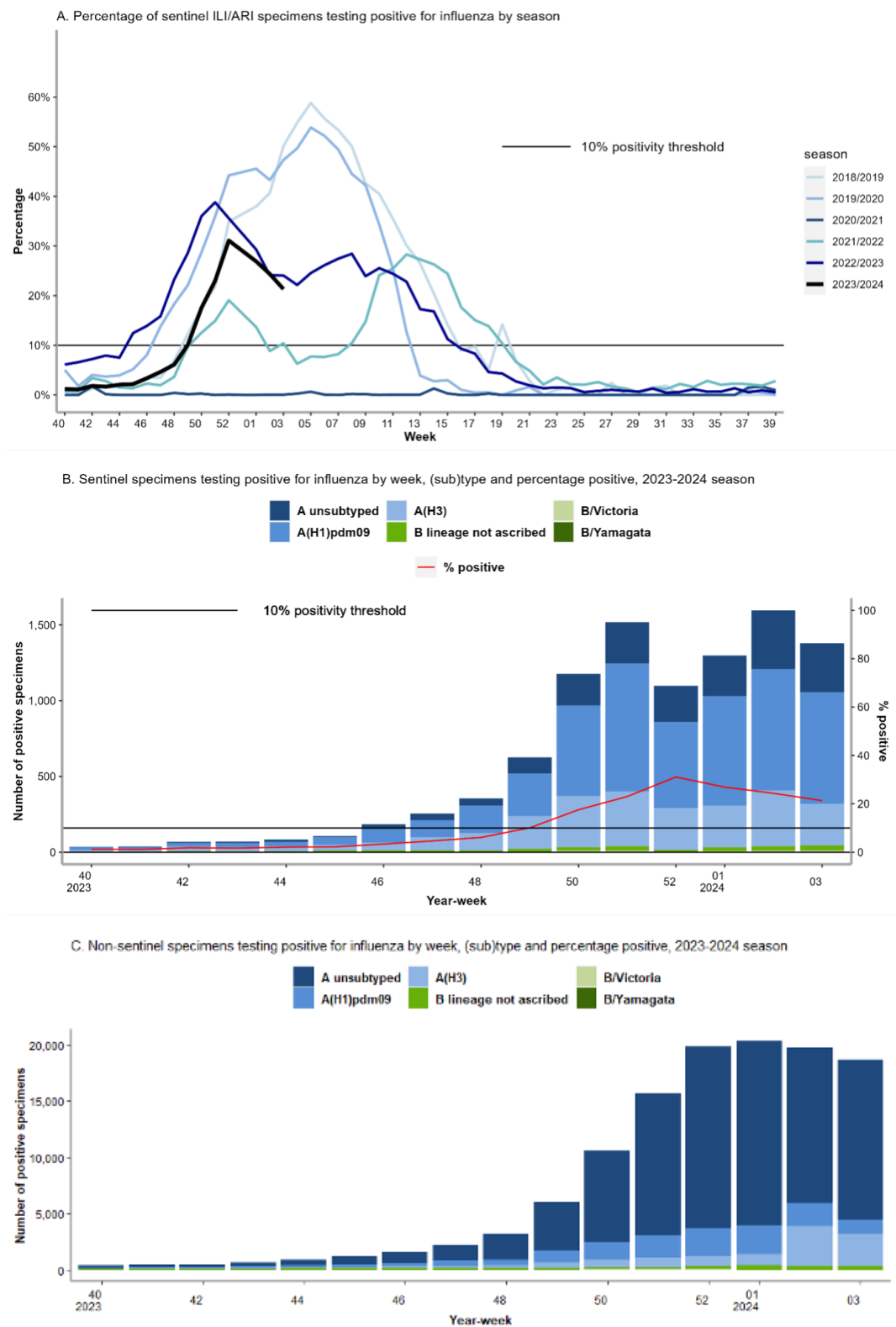


Figure E3. Sentinel specimens testing positive (bars) for influenza, RSV and SARS-CoV-2 (bars) by week and respective percentage positivity (lines), weeks 40/2023-03/2024

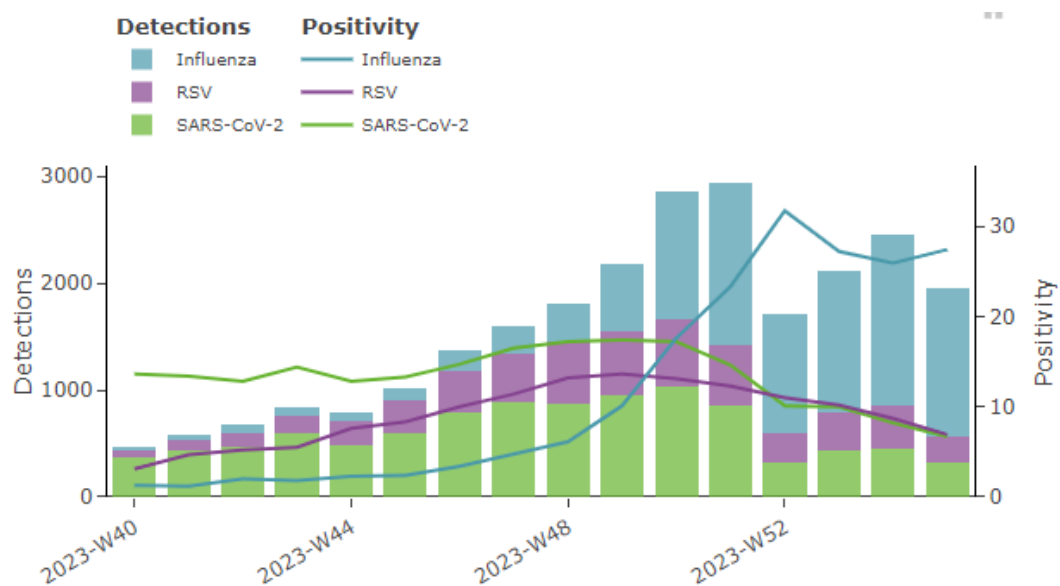
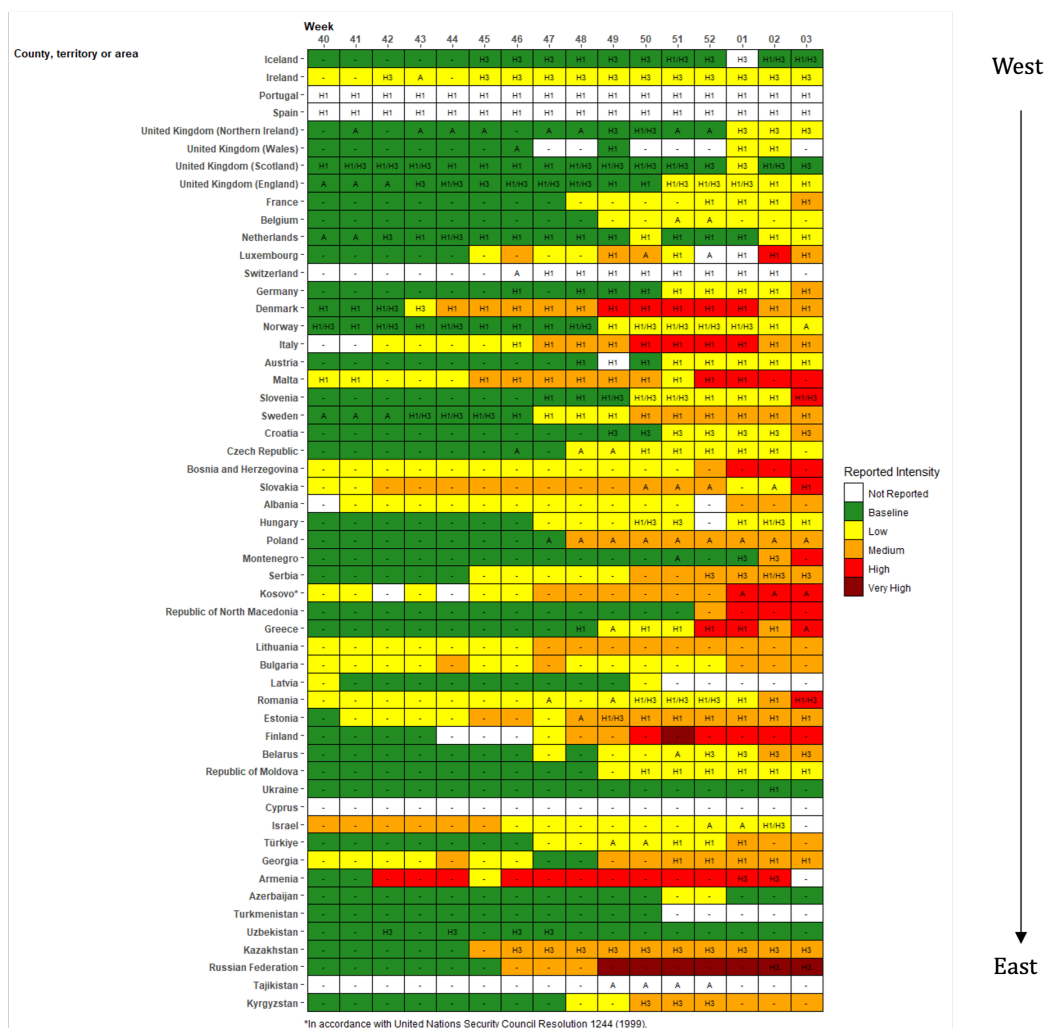


Figure E4. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and country, territory or area 2023-2024 season.

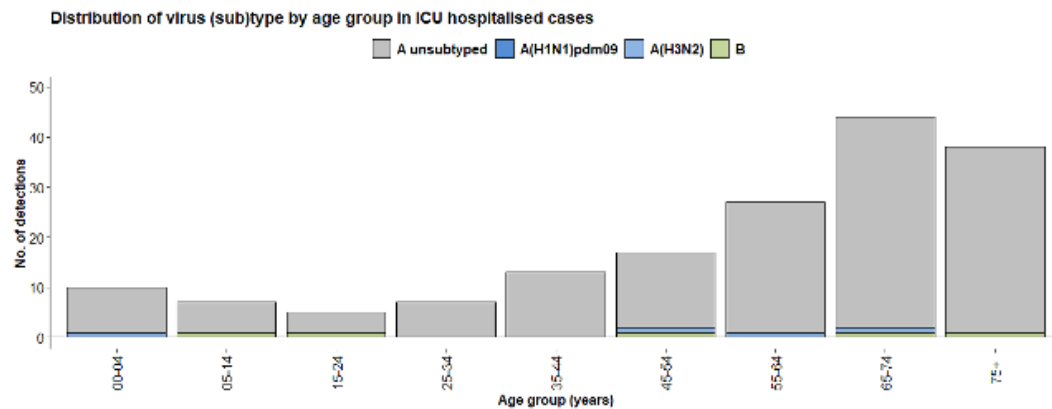


Notes: Baseline intensity influenza activity is the level at which clinical influenza activity remains throughout the summer. Countries, territory or area are ordered West to East. * in accordance with United Nations Security Council Resolution 1244 (1999).

Hospitalised cases

Between weeks 40/2023 and 03/2024, three countries and areas (Czech Republic, Sweden and United Kingdom (England)) reported a total of 510 laboratory-confirmed influenza cases from hospital settings to TESSy. Only the Czech Republic reported cases from both intensive care units (ICUs) and non-ICU hospital wards. Of these 510, 489 cases were admitted to intensive care units (ICUs): in the United Kingdom (England) (n = 321, 66%), Sweden (n = 144, 29%) and Czech Republic (n = 24, 5%). The remaining cases were reported from non-ICU hospital wards in Czech Republic (n = 21, 100%). Among the 489 cases admitted to ICUs, 465 (95%) were infected with type A viruses (of those subtyped, (n = 22), all were A(H3N2)) and 24 (5%) cases were infected with type B viruses (none ascribed to a lineage). A total of 168 cases had a recorded age, falling within different age groups: 82 were 65 years or older (49%), 60 were 30-64 years (36%), 10 were 0-4 years (6%), 9 were 15-29 years (5%) and 7 were 05-14 years (4%) (Figure E5).

Figure E5. Distribution of virus (sub)type in laboratory-confirmed hospitalized cases in intensive care units by age group in the WHO European Region, 2023-2024 season



A total of 27 countries, territories and areas (Albania, Armenia, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Croatia, Georgia, Germany, Ireland, Kazakhstan, Kyrgyzstan, Lithuania, Malta, North Macedonia, Republic of Moldova, Romania, Russian Federation, Serbia, Slovakia, Spain, Türkiye, Turkmenistan, Ukraine, United Kingdom (England), Uzbekistan and Kosovo (in accordance with United Nations Security Council Resolution 1244 (1999))) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 67,779 cases between weeks 40/2023 and 03/2024. Data on age was available for 57,818 of these patients. Overall, children aged 0-4 years accounted for the largest proportion of cases (39%), followed by those aged 65 years or older (28%), those aged 30-64 years (13%), those aged 5-14 years (9%), and those aged 15-29 years (6%). SARI cases were predominantly among children aged less than 5 years in seven countries (Armenia, Georgia, Kazakhstan, Kyrgyzstan, North Macedonia, Russian Federation, and Uzbekistan) And predominantly in those aged 65 years or older in countries (Croatia, Germany, Ireland, Slovakia, Spain and Kosovo (in accordance with United Nations Security Council Resolution 1244 (1999))) (Table E3). Of 30,128 SARI cases tested for influenza virus over the period, 12% (n = 3,500) were positive. The majority (n = 2,679, 76%) were infected with type A viruses and, of these 1,793 subtyped viruses, 1,151 (64%) were A(H3N2) and 642 (36%) were A(H1N1)pdm09). A total of 21 infections by influenza type B viruses were reported from Ukraine (n = 6), Russian Federation (n = 4), Spain (n = 4), Armenia (n = 2), Albania (n = 1), Bosnia and Herzegovina (n = 1), North Macedonia (n = 1), Serbia (n = 1) and Türkiye (n = 1). Of the 3 viruses ascribed to a lineage, all were B/Victoria.

Table E3. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and country, territory or area, and influenza positives, 2023-2024 season

Country, territory or area	SARI cases		Percentage (%) of total cases					SARI specimen s	Positive detection s (%)
	Total	With age data	0-4 years	5-14 years	15-29 years	30-64 years	>=65 years		
Albania	1,084	1,056	20	10	10	29	31	577	23 (4%)
Armenia	1,084	1,084	56	10	7	13	14	727	69 (9%)
Azerbaijan	3	3	0	0	0	0	0	-	-
Belarus	528	528	18	3	9	36	34	213	3 (1%)
Belgium	1,111	937	31	6	4	21	37	1,109	121 (11%)
Bosnia and Herzegovina	792	749	14	5	5	41	35	146	7 (5%)
Croatia	339	339	17	5	4	17	59	339	12 (4%)
Georgia	1,658	1,658	65	17	3	5	10	-	115 (-%) ¹
Germany	11,554	11,554	17	4	2	16	60	11,554	731 (6%)
Ireland	242	242	0	0	3	19	78	231	19 (8%)
Kazakhstan	23,691	17,559	61	13	13	9	3	1,785	372 (21%)
Kyrgyzstan	1,939	1,939	85	6	2	4	4	902	113 (13%)
Lithuania	0	0	-	-	-	-	-	0	0 (-%)
Malta	789	789	31	5	2	14	48	715	106 (15%)
North Macedonia	361	361	57	12	3	14	48	173	3 (2%)
Republic of Moldova	2,961	2,961	44	6	1	21	28	81	5 (6%)
Romania	231	231	43	6	4	19	28	226	76 (34%)
Russian Federation	1,109	1,109	60	41	43	40	24	1,193	209 (18%)
Serbia	811	782	41	9	4	11	36	363	42 (12%)
Slovakia	11	11	0	0	0	27	73	-	-
Spain	6,198	6,198	15	3	1	18	63	5,645	1,188 (21%)
Türkiye	3,282	3,282	31	11	0	0	24	1,664	66 (4%)
Turkmenistan	1,728	1,728	47	20	0	0	5	1,026	0 (0%)
Ukraine	2,044	2,044	27	17	11	24	20	790	65 (8%)
United Kingdom (England)	3,549	0	-	-	-	-	-	-	-
Uzbekistan	632	632	58	12	10	13	8	632	152 (24%)
Kosovo*	48	42	7	5	7	19	62	37	3 (8%)
Total	67,779	57,818	22,655 (39%)	5,228 (9%)	3,438 (6%)	7,480 (13%)	16,331 (28%)	30,128	3,500 (12%)

*In accordance with United Nations Security Council Resolution 1244 (1999).

¹ Likely reporting error: number of tests performed is missing

Mortality data

Pooled analyses of all-cause excess and influenza-attributable mortality from 27 European countries or sub-national regions participating in the EuroMOMO network (Austria, Belgium, Cyprus, Berlin (Germany), Denmark, England (United Kingdom (UK)), Estonia, Finland, France, Germany, Greece, Hesse (Germany), Hungary, Ireland, Israel, Italy, Luxemburg, Malta, the Netherlands, Northern Ireland (UK), Portugal, Scotland (UK), Slovenia, Spain, Sweden, Switzerland, and Wales (UK)) showed an overall substantially increased excess all-cause mortality for the participating European countries during the 2023-2024 season which is now

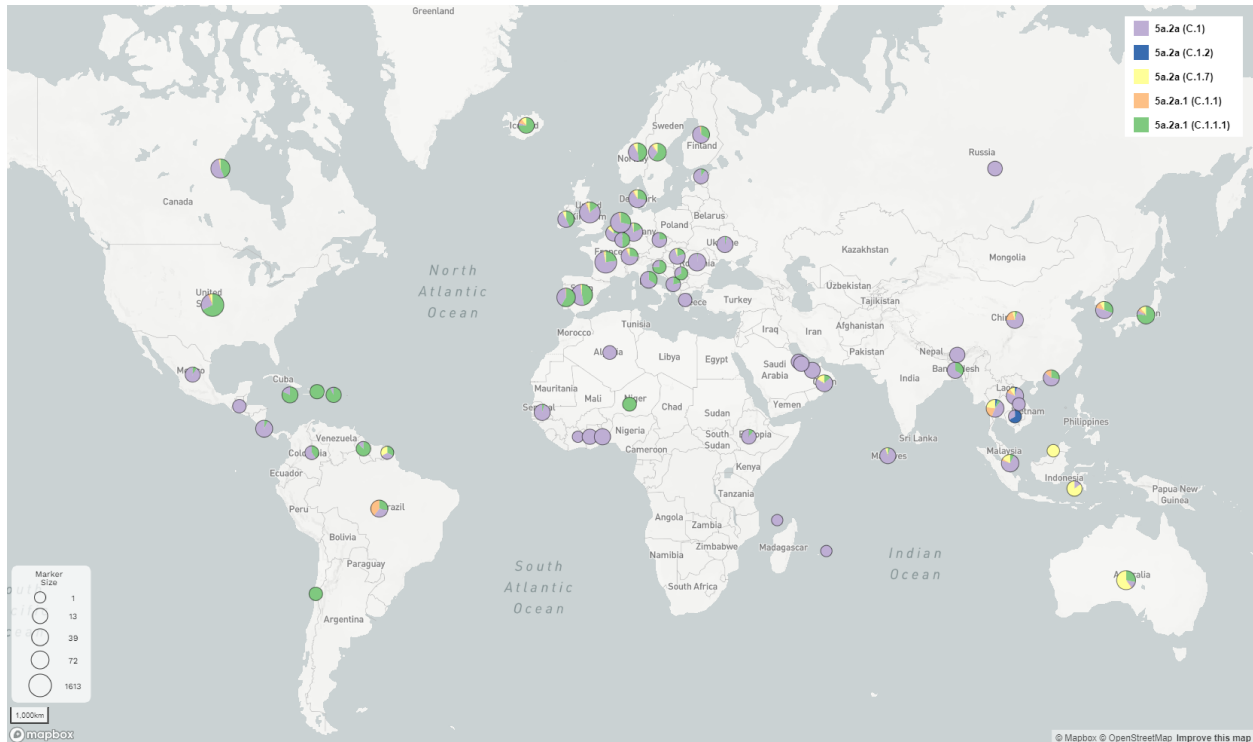
declining. This excess mortality was driven by a substantial elevated level of mortality, overall and in the age groups above 45 years of age. Please refer to the EuroMOMO website for a cautionary note relating to interpretation of these data.

All references *In accordance with United Nations Security Council Resolution 1244 (1999).

Influenza A H1N1

Genetic analyses: H1N1

Microreact: Subclade

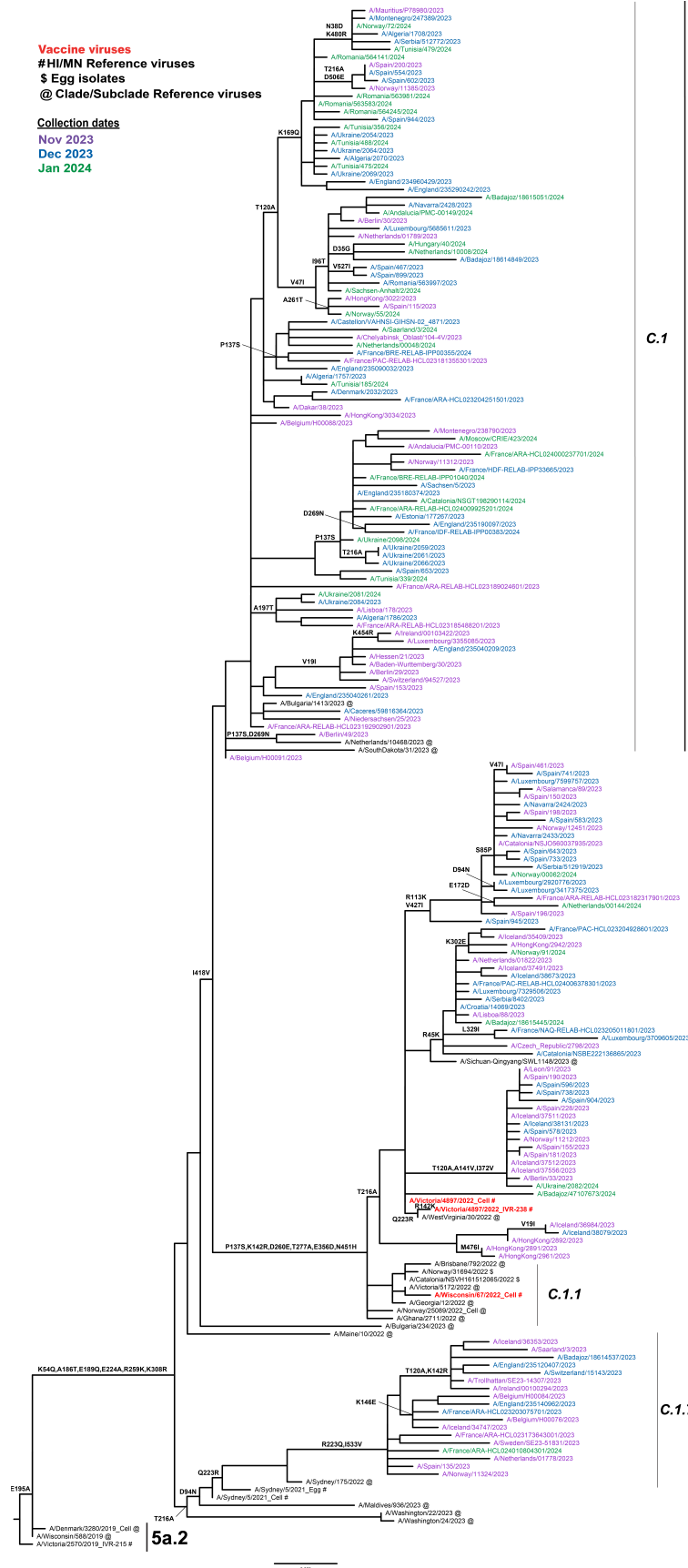


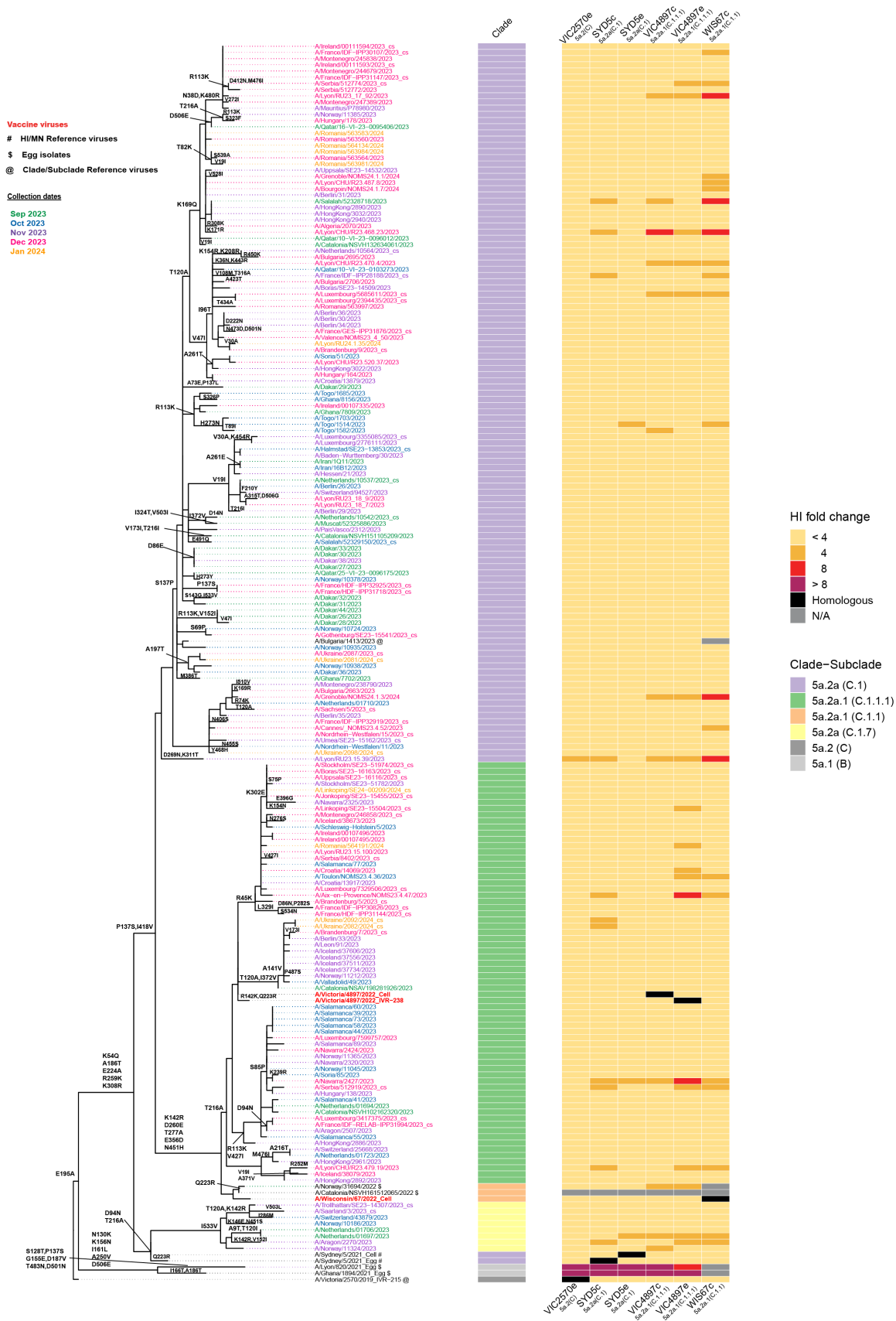
Maximum likelihood phylogenetic trees: H1N1

Maximum likelihood time-resolved phylogenetic tree inferred using IQTree2 from HA sequence data obtained from GISAID from 1st September onwards, manually curated and then downsampled using Treemer to retain a representative tree topology of 200 sequences with emphasis on European viruses and viruses sequenced at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

Vaccine viruses
HI/MN Reference viruses
\$ Egg isolates
@ Clade/Subclade Reference viruses

Collection dates
Nov 2023
Dec 2023
Jan 2024





Antigenic analyses: H1N1

Haemagglutination inhibition tables: H1N1

Table H1-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-10-24)

Viruses	Other information	Genetic group	Collection Date	Genetic group	Passage	Ferret number	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
							F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
							5a.2	5a.2a	5a.2a	5a.2a.1	5a.2a.1	5a.2a.1
REFERENCE VIRUSES												
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E3			1280	640	640	640	640	320
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	JCK3/MDCK4			1280	1280	2560	2560	1280	1280
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2022-10-31	E3/E3			1280	640	1280	1280	1280	640
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1	2022-10-02	IAT2/MDCK2			2560	1280	2560	2560	2560	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1	2022-10-02	E3/D6/E2			640	640	640	640	1280	320
A/Wisconsin/67/2022		5a.2a.1	2022-10-25	MDCK2			1280	640	1280	1280	1280	1280
TEST VIRUSES												
240010	A/Catalonia/NSVH151105209/2023	5a.2a	2023-09-05	IAT1/MDCK1			2560	2560	2560	2560	2560	1280
240006	A/Catalonia/NSVH132634061/2023	5a.2a	2023-09-06	IAT1/MDCK1			2560	2560	2560	1280	1280	1280
240008	A/Catalonia/NSAV198281926/2023	5a.2a.1	2023-09-05	IAT1/MDCK1			2560	1280	1280	2560	2560	2560
240002	A/Catalonia/NSVH102162320/2023	5a.2a.1	2023-09-13	IAT1/MDCK1			1280	640	1280	2560	1280	1280
< relates to the lowest dilution of antiserum used							Vaccine SH 2021	Vaccine SH 2023		Vaccine NH 2023-24 SH 2024		
ND = Not Done							NH 2021-22 SH 2022 NH 2022-23					

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 ≥ 160 (no homologous titre)

Table H1-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-11-21)

Viruses	Other information	Genetic group	Collection Date	Genetic group	Passage	Haemagglutination inhibition titre					
						Post-infection ferret antiserum					
						IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
						F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
					Ferret number	5a.2 (C)	5a.2a (C.1)	5a.2a (C.1)	5a.2a.1 (C.1.1.1)	5a.2a.1 (C.1.1.1)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES											
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2 (C)	2018-11-22	E4/D7/E3		1280	640	640	640	1280	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a (C.1)	2021-10-16	ICK3/MDCK4		2560	1280	2560	1280	2560	1280
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a (C.1)	2022-10-31	E3/E3		1280	640	640	640	1280	640
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1 (C.1.1.1)	2022-10-02	IAT2/MDCK2		1280	640	640	1280	640	1280
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1 (C.1.1.1)	2022-10-02	E3/D6/E1		1280	1280	1280	1280	2560	640
A/Wisconsin/67/2022		5a.2a.1 (C.1.1)	2022-10-25	MDCK2		640	640	640	1280	640	1280
TEST VIRUSES											
240055 A/iran/25H10/2023		5a.2a (C.1)	2023-08-07	IAT1/MDCK2		1280	1280	1280	1280	1280	640
240054 A/iran/18H10/2023		5a.2a (C.1)	2023-08-08	IAT1/MDCK1		1280	1280	1280	1280	1280	640
240053 A/iran/1Q11/2023		5a.2a (C.1)	2023-09-29	IAT1/MDCK1		1280	1280	1280	1280	1280	640
240073 A/Salalah/52329150/2023		5a.2a (C.1)	2023-10-03	MDCK2		2560	1280	1280	1280	1280	640
240052 A/iran/16B12/2023		5a.2a (C.1)	2023-10-09	MDCK		5120	2560	2560	2560	2560	1280
< relates to the lowest dilution of antiserum used ND = Not Done						Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23	Vaccine SH 2023		Vaccine NH 2023-24 SH 2024		

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 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-12-08)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
							F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
							5a.2	5a.2a	5a.2a	5a.2a.1	5a.2a.1	5a.2a.1
REFERENCE VIRUSES												
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E3	2560	1280	1280	1280	1280	640		
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	ICK3/MDCK4	1280	5120	5120	5120	2560	2560		
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2022-10-31	E3/E2	1280	1280	1280	1280	1280	640		
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1	2022-10-02	iAT2/MDCK2	2560	1280	2560	2560	2560	2560		
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1	2022-10-02	E3/D6/E1	1280	640	640	640	1280	640		
A/Wisconsin/67/2022		5a.2a.1	2022-10-25	MDCK2	1280	640	640	1280	640	1280		
TEST VIRUSES												
240177	A/Qatar/10-VI-23-0103273/2023	5a.2a	2023-10-15	MDCK1	2560	2560	2560	1280	2560	1280		
240181	A/Qatar/25-VI-23-0096175/2023	5a.2a	2023-09-19	MDCK1	2560	2560	2560	2560	2560	1280		
240182	A/Qatar/10-VI-23-0096012/2023	5a.2a	2023-09-19	MDCK1	2560	2560	2560	2560	2560	1280		
240184	A/Qatar/16-VI-23-0095406/2023	5a.2a	2023-09-17	MDCK2	2560	2560	2560	2560	2560	1280		
< relates to the lowest dilution of antiserum used ND = Not Done							Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23	Vaccine SH 2023		Vaccine NH 2023-24 SH 2024		

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Table H1-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2023-12-19)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
							F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
							5a.2	5a.2a	5a.2a	5a.2a.1	5a.2a.1	5a.2a.1
REFERENCE VIRUSES												
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)	5a.2	2018-11-22	E4/D7/E3	2560		640	1280	1280	1280	1280	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A	5a.2a	2021-10-16	ICK3/MDCK4	2560	1280	1280	2560	1280	1280	1280	1280
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)	5a.2a	2022-10-31	E3/E2	640	1280	640	1280	640	640	640	640
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A	5a.2a.1	2022-10-02	IA/T2/MDCK2	2560	1280	1280	2560	1280	2560	1280	2560
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)	5a.2a.1	2022-10-02	E3/D6/E1	1280	640	640	640	640	1280	1280	640
A/Wisconsin/67/2022		5a.2a.1	2022-10-25	MDCK2	1280	640	1280	1280	1280	1280	1280	1280
TEST VIRUSES												
240438	A/Berlin/26/2023	5a.2a	2023-10-01	P1/MDCK1	2560	2560	2560	2560	2560	1280	1280	1280
240303	A/Norway/10186/2023	5a.2a	2023-10-08	MDCK1	1280	1280	1280	1280	1280	1280	1280	1280
240302	A/Norway/10378/2023	5a.2a	2023-10-10	MDCK1	1280	1280	1280	1280	1280	1280	1280	1280
240301	A/Norway/10428/2023	5a.2a	2023-10-13	MDCK1	1280	1280	1280	1280	1280	1280	1280	640
240300	A/Norway/10724/2023	5a.2a	2023-10-19	MDCK1	2560	1280	1280	1280	1280	1280	1280	1280
240298	A/Norway/10935/2023	5a.2a	2023-10-24	MDCK1	1280	1280	1280	1280	1280	1280	1280	1280
240296	A/Norway/10938/2023	5a.2a	2023-10-28	MDCK1	1280	1280	1280	1280	1280	1280	1280	1280
240436	A/Nordrhein-Westfalen/11/2023	5a.2a	2023-10-30	P1/MDCK1	1280	1280	1280	1280	1280	1280	1280	1280
240291	A/Norway/11385/2023	5a.2a	2023-11-06	MDCK1	2560	1280	1280	1280	1280	1280	1280	1280
240435	A/Hessen/21/2023	5a.2a	2023-11-06	P1/MDCK1	2560	2560	2560	2560	2560	2560	1280	1280
240290	A/Norway/11324/2023	5a.2a	2023-11-07	MDCK1	1280	640	640	640	640	1280	640	640
240434	A/Berlin/30/2023	5a.2a	2023-11-07	P1/MDCK1	2560	2560	2560	2560	2560	2560	1280	1280
240433	A/Berlin/29/2023	5a.2a	2023-11-08	P1/MDCK1	2560	1280	2560	2560	2560	2560	2560	2560
240432	A/Berlin/31/2023	5a.2a	2023-11-10	P1/MDCK1	2560	1280	1280	1280	1280	1280	1280	1280
240430	A/Baden-Wuerttemberg/30/2023	5a.2a	2023-11-14	P1/MDCK1	2560	2560	2560	2560	2560	2560	2560	2560
240427	A/Berlin/34/2023	5a.2a		P1/MDCK1	1280	1280	1280	1280	1280	1280	1280	640
240428	A/Berlin/35/2023	5a.2a		P1/MDCK1	1280	1280	2560	2560	2560	2560	2560	2560
240429	A/Berlin/36/2023	5a.2a		P1/MDCK1	2560	1280	2560	2560	2560	2560	2560	1280
240439	A/Sachsen-Anhalt/8/2023	5a.2a.1	2023-08-15	P1/MDCK1	1280	640	1280	1280	1280	1280	1280	1280
240437	A/Schleswig-Holstein/5/2023	5a.2a.1	2023-10-19	P1/MDCK1	1280	1280	1280	1280	2560	1280	2560	2560
240297	A/Norway/11045/2023	5a.2a.1	2023-10-25	MDCK1	2560	1280	2560	2560	2560	2560	2560	2560
240295	A/Norway/10941/2023	5a.2a.1	2023-10-28	MDCK1	2560	1280	1280	2560	1280	1280	2560	2560
240293	A/Norway/11212/2023	5a.2a.1	2023-11-01	MDCK1	1280	640	1280	1280	2560	1280	2560	2560
240289	A/Norway/11365/2023	5a.2a.1	2023-11-07	MDCK1	1280	640	1280	1280	1280	1280	1280	1280
240431	A/Berlin/33/2023	5a.2a.1	2023-11-14	P1/MDCK1	1280	640	1280	2560	2560	2560	2560	2560
< relates to the lowest dilution of antiserum used ND = Not Done							Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23	Vaccine SH 2023		Vaccine NH 2023-24 SH 2024		

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 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table H1-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-01-16)

Viruses	Other information	Passage	Ferret number	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				18K-216 A/Vic/2079/19 Egg	A/Sydney/5/2021 8/21 MDCK	A/Sydney/5/2021 8/21 Egg	A/Vic/2079/19 4887/2022 MDCK	18K-238 A/Vic/4887/22 Egg	18K-238 A/Vic/4887/22 MDCK	67/2022 MDCK	
				F37/21	F46/22	F54/22	F59/23	F70/23	F17/23		
Genetic group		Collection Date	Genetic group		Sa.2 (C.1)	Sa.2a (C.1)	Sa.2a (C.1)	Sa.2a.1 (C.1.1.1)	Sa.2a.1 (C.1.1.1)	Sa.2a.1 (C.1.1.1)	
REFERENCE VIRUSES											
VR-216 (A/Vic/2079/2019)	K130N, N158K, L161L, V260A, E189Q (A195E_Q223R)	2019-11-22	E4D1E3	1280	640	640	640	1280	640	640	
A/Sydney/5/2021	K34Q, A198T, E224A, R299K, K308R, D34N, T216A	2020-10-31	ES1E3	1280	640	1280	1280	1280	1280	1280	
A/Sydney/5/2021	K34Q, A198T, E224A, R299K, K308R, D34N, T216A (Q223R)	2022-10-31	SIAT2MDCK2	2560	1280	1280	2560	1280	2560	2560	
A/Vic/4887/2022	P137S, K142R, T216A, D266E, T277A	2022-10-02	ES1E3E1	1280	640	1280	1280	1280	640	640	
VR-238 (A/Vic/4887/2022)	P137S, K142R, T216A, D266E, T277A (R142K_Q223R)	2022-10-02	MDCK2	1280	640	640	2560	1280	1280		
A/Wisconsin/97/2022		2022-10-25									
TEST VIRUSES											
240417	A/Netherlands/81623/2023	2023-06-26	hCK1/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240418	A/Netherlands/81628/2023	2023-06-01	hCK1/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240448	A/Dakar/17/2023	2023-07-17	MDCK/MDCK1	1280	1280	1280	1280	1280	1280	640	
240647	A/Dakar/19/2023	2023-07-21	MDCK1/MDCK1	1280	640	640	1280	1280	1280	640	
240648	A/Dakar/19/2023	2023-07-24	MDCK2/MDCK1	2560	1280	2560	2560	2560	2560	2560	
240644	A/Dakar/20/2023	2023-08-28	MDCK/MDCK1	1280	640	1280	1280	1280	1280	640	
240643	A/Dakar/23/2023	2023-08-28	MDCK/MDCK1	2560	2560	2560	2560	2560	2560	1280	
240414	A/Netherlands/81685/2023	2023-08-30	hCK1/MDCK1	2560	1280	2560	2560	2560	2560	2560	
240642	A/Dakar/26/2023	2023-09-01	MDCK1/MDCK1	1280	640	640	1280	640	640	640	
240641	A/Dakar/16/2023	2023-09-06	MDCK1/MDCK1	1280	1280	1280	2560	1280	1280	1280	
240640	A/Dakar/25/2023	2023-09-06	MDCK1/MDCK1	1280	1280	1280	1280	1280	1280	640	
240639	A/Dakar/22/2023	2023-09-07	MDCK1/MDCK1	1280	1280	1280	2560	1280	1280	640	
240638	A/Dakar/21/2023	2023-09-08	MDCK1/MDCK1	1280	640	1280	1280	1280	1280	640	
240637	A/Dakar/29/2023	2023-09-08	MDCK/MDCK1	1280	640	1280	1280	1280	1280	1280	
240636	A/Dakar/20/2023	2023-09-09	MDCK/MDCK1	1280	1280	1280	1280	1280	1280	1280	
240107	A/Muscato5232886/2023	2023-09-11	MDCK1	2560	2560	2560	2560	2560	2560	1280	
240634	A/Dakar/27/2023	2023-09-13	MDCK1	2560	2560	2560	2560	2560	2560	2560	
240632	A/Dakar/44/2023	2023-09-19	MDCK1/MDCK1	2560	1280	2560	1280	1280	1280	1280	
240629	A/Dakar/7/2023	2023-09-19	MDCK1	2560	1280	1280	2560	1280	1280	1280	
240631	A/Dakar/33/2023	2023-09-21	MDCK1/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240630	A/Dakar/30/2023	2023-09-22	MDCK1/MDCK1	2560	1280	2560	2560	1280	1280	1280	
240629	A/Dakar/34/2023	2023-09-22	MDCK1/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240638	A/Ghana/7809/2023	2023-09-22	MDCK1	2560	1280	2560	2560	2560	2560	1280	
240627	A/Dakar/13/2023	2023-09-28	MDCK/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240626	A/Dakar/32/2023	2023-09-27	MDCK1/MDCK1	2560	640	2560	2560	2560	2560	1280	
240410	A/Netherlands/81760/2023	2023-10-02	hCK1/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240386	A/Ghana/8156/2023	2023-10-11	MDCK1	2560	1280	2560	2560	2560	2560	1280	
240624	A/Dakar/8/2023	2023-10-16	MDCK/MDCK1	2560	2560	2560	2560	2560	2560	1280	
240600	A/Togo/1682/2023	2023-10-21	MDCK1	1280	640	640	640	640	640	640	
240787	A/Son/H1/2023	2023-10-23	MDCK1/MDCK1	1280	1280	2560	2560	1280	1280	640	
240486	A/Togo/1685/2023	2023-10-26	MDCK1	1280	1280	1280	1280	1280	1280	640	
240486	A/Togo/1793/2023	2023-10-30	MDCK1	1280	1280	1280	1280	1280	1280	640	
240609	A/HongKong/2360/2023	2023-11-09	MDCK1/MDCK1	2560	2560	6120	2560	2560	2560	1280	
240623	A/Dakar/38/2023	2023-11-10	MDCK2/MDCK1	2560	2560	2560	2560	2560	2560	1280	
240603	A/HongKong/2340/2023	2023-11-15	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
240317	A/Switzerland/84627/2023	2023-11-22	MDCK1	2560	1280	2560	2560	2560	2560	1280	
240677	A/HongKong/2322/2023	2023-11-23	MDCK1/MDCK1	2560	1280	2560	2560	2560	2560	2560	
240676	A/HongKong/2323/2023	2023-11-26	MDCK1/MDCK1	2560	1280	2560	2560	2560	2560	1280	
240673	A/HongKong/2324/2023	2023-11-28	MDCK1/MDCK1	2560	2560	2560	2560	2560	2560	1280	
240402	A/Croatia/2847/2023	2023-11-28	MDCK1	2560	2560	2560	2560	2560	2560	1280	
240412	A/Netherlands/81897/2023	2023-09-22	hCK1/MDCK1	640	640	320	640	320	320	320	
240411	A/Netherlands/81786/2023	2023-09-29	hCK1/MDCK1	2560	2560	2560	1280	2560	1280	1280	
240318	A/Switzerland/83978/2023	2023-10-27	MDCK1	1280	640	1280	1280	640	640	640	
240415	A/Netherlands/81685/2023	2023-07-06	hCK1/MDCK1	1280	640	1280	2560	1280	1280	1280	
240409	A/Netherlands/81722/2023	2023-10-09	hCK1/MDCK1	1280	640	1280	2560	1280	1280	1280	
240687	A/HongKong/2891/2023	2023-11-11	MDCK1/MDCK1	2560	640	2560	6120	2560	2560	2560	
240608	A/HongKong/2383/2023	2023-11-13	MDCK/MDCK1	2560	1280	2560	6120	2560	2560	2560	
240679	A/HongKong/2384/2023	2023-11-20	MDCK1/MDCK1	2560	1280	2560	6120	2560	2560	2560	
240316	A/Switzerland/25668/2023	2023-11-24	MDCK1	2560	1280	2560	2560	1280	2560	2560	
240664	A/Croatia/8879/2023	2023-12-02	MDCK1	1280	640	1280	2560	1280	2560	2560	
240413	A/Netherlands/81694/2023	2023-09-07	hCK1/MDCK1	1280	640	1280	2560	1280	1280	1280	
240785	A/Salamancas/80/2023	2023-10-23	BIAT1/MDCK1	2560	2560	2560	6120	2560	2560	2560	
240783	A/Salamancas/86/2023	2023-10-23	BIAT1/MDCK1	2560	1280	1280	6120	2560	2560	2560	
240781	A/Salamancas/80/2023	2023-10-24	BIAT1/MDCK1	2560	1280	1280	2560	2560	2560	2560	
240777	A/Salamancas/81/2023	2023-10-25	BIAT1/MDCK1	1280	1280	1280	6120	1280	1280	1280	
240771	A/Salamancas/90/2023	2023-10-26	BIAT1/MDCK1	640	640	2560	640	640	640	640	
240769	A/Salamancas/82/2023	2023-10-26	BIAT1/MDCK1	1280	640	1280	2560	1280	1280	1280	
240767	A/Salamancas/77/2023	2023-10-29	BIAT1/MDCK1	2560	1280	2560	6120	2560	2560	2560	
240768	A/Salamancas/70/2023	2023-10-31	BIAT1/MDCK1	2560	1280	2560	6120	1280	2560	2560	
240747	A/Salamancas/89/2023	2023-11-05	BIAT1/MDCK1	2560	1280	2560	6120	2560	2560	2560	
240691	A/HongKong/2360/2023	2023-11-08	MDCK1	1280	640	2560	2560	1280	1280	1280	
240401	A/HongKong/2340/2023	2023-11-15	MDCK1/MDCK1	640	640	1280	2560	1280	1280	1280	
240400	A/Croatia/8791/2023	2023-11-20	MDCK1	640	640	1280	2560	1280	1280	1280	
240408	A/Croatia/87912/2023	2023-11-20	MDCK1	1280	640	1280	1280	1280	1280	1280	
240407	A/Croatia/87968/2023	2023-11-20	MDCK1	640	640	2560	2560	1280	1280	1280	
240406	A/Croatia/87968/2023	2023-11-20	MDCK1	1280	1280	1280	1280	2560	2560	2560	
240405	A/Croatia/87734/2023	2023-11-20	MDCK1	1280	640	1280	2560	1280	1280	1280	
240401	A/Croatia/88873/2023	2023-12-06	MDCK1	1280	640	1280	2560	1280	1280	1280	
< relates to the lowest dilution of antiserum used				Vaccine							
ND = Not Done				NH 2021							
				NH 2021-22							
				NH 2022							
				NH 2022-23							
				NH 2023-24							
				NH 2024							

Table H1-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-01-24)

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre					
							Post-infection ferret antiserum					
							IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 MDCK	A/Sydney 5/21 Egg	A/Victoria/ 4897/2022 MDCK	IVR-238 A/Vic/4897/22 Egg	A/Wisconsin/ 67/2022 MDCK
							F37/21	F46/22	F04/22	F05/23	F07/23	F17/23
							5a.2 (C)	5a.2a (C.1)	5a.2a (C.1)	5a.2a.1 (C.1.1.1)	5a.2a.1 (C.1.1.1)	5a.2a.1 (C.1.1)
REFERENCE VIRUSES												
IVR-215 (A/Victoria/2570/2019)	K130N, N156K, L161I, V250A, E189Q (A195E, Q223R)			5a.2 (C)	2018-11-22	E4/D7/E3	1280	640	640	1280	1280	640
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A			5a.2a (C.1)	2021-10-16	MDCK3/MDCK4	2560	1280	2560	2560	1280	1280
A/Sydney/5/2021	K54Q, A186T, E224A, R259K, K308R, D94N, T216A (Q223R)			5a.2a (C.1)	2022-10-31	E3/E3	1280	640	1280	1280	1280	640
A/Victoria/4897/2022	P137S, K142R, T216A, D260E, T277A			5a.2a.1 (C.1.1.1)	2022-10-02	SIAT2/MDCK2	2560	1280	2560	2560	1280	1280
IVR-238 (A/Victoria/4897/2022)	P137S, K142R, T216A, D260E, T277A (R142K, Q223R)			5a.2a.1 (C.1.1.1)	2022-10-02	E3/D6/E1	1280	640	1280	1280	1280	640
A/Wisconsin/67/2022				5a.2a.1 (C.1.1)	2022-10-25	MDCK2	2560	1280	1280	2560	2560	2560
TEST VIRUSES												
240081 A/Salalah/523287/18/2023				5a.2a (C.1)	2023-09-26	MDCK2	640	320	640	640	640	320
240504 A/Togo/1514/2023				5a.2a (C.1)	2023-10-14	MDCK2	640	1280	320	1280	640	640
240873 A/Montenegro/238790/2023				5a.2a (C.1)	2023-11-27	MDCK1	1280	1280	2560	2560	2560	1280
240948 A/Hungary/164/2023				5a.2a (C.1)	2023-12-11	MDCK1/MDCK1	2560	2560	2560	2560	2560	1280
240946 A/Hungary/178/2023				5a.2a (C.1)	2023-12-11	MDCK1/MDCK1	1280	1280	1280	1280	1280	1280
240871 A/Montenegro/244679/2023				5a.2a (C.1)	2023-12-13	MDCK1	2560	2560	2560	2560	2560	1280
240870 A/Montenegro/245269/2023				5a.2a (C.1)	2023-12-13	MDCK1	2560	2560	2560	2560	2560	2560
240869 A/Montenegro/245728/2023				5a.2a (C.1)	2023-12-15	MDCK1	2560	2560	2560	2560	2560	1280
240868 A/Montenegro/245838/2023				5a.2a (C.1)	2023-12-16	MDCK1	2560	1280	2560	1280	1280	1280
240880 A/Algeria/2070/2023				5a.2a (C.1)	2023-12-19	MDCK1	2560	2560	2560	2560	2560	1280
240865 A/Montenegro/247389/2023				5a.2a (C.1)	2023-12-20	MDCK1	2560	1280	2560	1280	1280	1280
240805 A/Salamanca/40/2023				5a.2a (C.1.7)	2023-10-13	MDCK1/MDCK1	2560	1280	2560	1280	1280	1280
240807 A/Salamanca/39/2023				5a.2a.1 (C.1.1.1)	2023-10-11	MDCK1/MDCK1	2560	1280	2560	2560	2560	2560
240803 A/Salamanca/41/2023				5a.2a.1 (C.1.1.1)	2023-10-14	MDCK1/MDCK1	1280	1280	1280	2560	1280	1280
240801 A/Salamanca/42/2023				5a.2a.1 (C.1.1.1)	2023-10-14	MDCK1/MDCK1	1280	640	1280	2560	1280	640
240799 A/Salamanca/44/2023				5a.2a.1 (C.1.1.1)	2023-10-16	MDCK1/MDCK1	1280	640	1280	2560	1280	1280
240789 A/Valladolid/49/2023				5a.2a.1 (C.1.1.1)	2023-10-23	MDCK1/MDCK2	1280	640	1280	2560	1280	1280
240950 A/Hungary/138/2023				5a.2a.1 (C.1.1.1)	2023-11-16	MDCK1/MDCK1	2560	1280	1280	2560	2560	1280
240607 A/Ireland/00107495/2023				5a.2a.1 (C.1.1.1)	2023-12-05	MDCK1	2560	1280	1280	2560	1280	1280
240606 A/Ireland/00107496/2023				5a.2a.1 (C.1.1.1)	2023-12-05	MDCK2	1280	1280	1280	2560	2560	2560
240605 A/Ireland/00107335/2023				5a.2a.1 (C.1.1.1)	2023-12-08	MDCK2	1280	640	1280	1280	1280	1280
240867 A/Montenegro/246858/2023				5a.2a.1 (C.1.1.1)	2023-12-19	MDCK1	1280	640	1280	1280	640	1280
240757 A/Salamanca/79/2023				no sequence	2023-11-03	MDCK1/MDCK2	1280	1280	1280	2560	1280	1280
240947 A/Hungary/167/2023				no sequence	2023-12-11	MDCK1/MDCK1	1280	1280	2560	2560	2560	2560

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

Vaccine
SH 2023

Vaccine
NH 2023-24
SH 2024

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Visus	Other Information	Neuroanatomical attributes										
		Post-embolic lesion volume (mm ³)										
		DB-15	AT-15	AT-15	AT-15	DB-15	AT-15	AT-15	AT-15	DB-15	AT-15	
Patients		A00197919	S1	S1	S1	A00197919	S1	S1	S1	A00197919	S1	
Form number		F00217	F0022	F0023	F0023	F0023	F0023	F0023	F0023	F0023	F0023	
Genetic group		Collection Date	Genetic group	Se-1 (C)	Se-2 (C)	Se-3 (C)	Se-4 (C)	Se-5 (C)	Se-6 (C)	Se-7 (C)	Se-8 (C)	
REFERENCE VISUSES												
V00131 (A0019792019)		K15N, K15K, L161, V25A, E25B, E25C, E25D	Se-1 (C)	2018-01-12	ADAT023	1200	640	1200	640	1200	640	
V00132 (A0019792020)		K15N, K15T, E25A, E25B, K25B, E25C, E25D, T25A	Se-1 (C)	2018-01-12	ADAT023	1200	640	1200	640	1200	640	
V00133 (A0019792021)		K25A, K25L, K25M, K25N, K25P, E25C, E25D, T25A	Se-1 (C)	2018-01-21	E25E2	1200	640	1200	640	1200	640	
V00134 (A0019792022)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00135 (A0019792023)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00136 (A0019792024)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00137 (A0019792025)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00138 (A0019792026)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00139 (A0019792027)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00140 (A0019792028)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00141 (A0019792029)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00142 (A0019792030)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00143 (A0019792031)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00144 (A0019792032)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00145 (A0019792033)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00146 (A0019792034)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00147 (A0019792035)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00148 (A0019792036)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT023	1200	640	1200	640	1200	640	
V00149 (A0019792037)		P137E, K14Z, T25A, E25C, E25D, E25E, E25F	Se-1 (C)	2018-01-21	ADAT02							

< relates to the lowest dilution of antiserum used
 NO = Not Done

NH 2021-22
 SH 2022
 NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Vaccine	Vaccine	Vaccine
SH 2021	SH 2023	NH 2023-24
NH 2021-22		SH 2024
SH 2022		

Vaccine
2023-24

SPH 2024

Table H1-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2024-02-06)

Haemagglutination inhibition titre										
Viruses	Other information	Post-infection ferret antiserum								
		Passage	Ferret number	F37/21 F40/22 F04/22 F05/23 F07/23 F17/23						
				NR-218 AV/02/07/19	Atydney Egg	Atydney MOCK	Atydney MOCK	Atydney MOCK	NR-228 KVA/4887/22	Atydney MOCK
Genetic group		Collection Date	Genetic group	F37/21 (C)	F40/22 (C.1)	F04/22 (C.1)	F05/23 (C.1)	F07/23 (C.1.1.1)	F17/23 (C.1.1)	C.1
REFERENCE VIRUSES										
NR-218 (A/Victoria/237/2019)	K150N, N156K, L181I, V250A, E180Q (A/159C, Q22)	2019-11-22	E407(E)	1280	640	1280	1280	1280	1280	640
Atydney/2021	K54Q, A168T, E224A, R239K, K268L, D24N, T216I	2021-10-16	MOCK1/MOCK1	2560	2560	2560	2560	2560	2560	1280
Atydney/2021	K54Q, A168T, E224A, R239K, K268L, D24N, T216I	2022-10-31	E1E3	1280	1280	1280	1280	1280	1280	1280
AVictoria/4887/2022	P117S, K142R, T216A, D246E, T277A	2022-10-02	BIAT2/MOCK2	2560	1280	2560	2560	1280	1280	1280
NR-228 (A/Victoria/4887/2022)	P117S, K142R, T216A, D246E, T277A (H42K, Q22)	2022-10-02	E30NE1	1280	1280	1280	1280	1280	1280	640
AWicconin/97/2022	Sa.2a.1 (C.1.1)	2022-10-25	MOCK2	1280	1280	2560	2560	2560	2560	2560
TEST VIRUSES										
241997	A/France/DF-PP318/2023	Sa.2a (C)	2023-11-20	MOCK1/MOCK1	1280	640	1280	1280	1280	640
241993	A/France/DF-PP3107/2023	Sa.2a (C)	2023-12-04	MOCK1/MOCK1	2560	1280	1280	1280	1280	640
241993	A/France/DF-PP3171/2023	Sa.2a (C)	2023-12-11	MOCK1/MOCK1	2560	1280	2560	2560	2560	1280
241779	A/Northern-Westfalen/1/2023	Sa.2a (C)	2023-12-12	P1/MOCK1	2560	2560	9120	2560	2560	2560
241861	A/France/ES-APP3187/2023	Sa.2a (C)	2023-12-13	MOCK1/MOCK1	9120	2560	9120	2560	2560	2560
241779	A/France/DF-PP2325/2023	Sa.2a (C)	2023-12-14	P2/MOCK1	9120	2560	2560	2560	2560	2560
241977	A/France/DF-PP32929/2023	Sa.2a (C)	2023-12-20	MOCK1/MOCK1	2560	2560	2560	2560	2560	2560
241975	A/France/DF-PP32929/2023	Sa.2a (C)	2023-12-20	MOCK1/MOCK1	2560	2560	2560	2560	2560	2560
241777	A/Sachsen/5/2023	Sa.2a (C)	2023-12-20	P2/MOCK1	2560	2560	2560	2560	2560	2560
241438	A/Israel/0111/2023	Sa.2a (C)	2023-12-27	MOCK1	2560	1280	2560	1280	1280	1280
241469	A/Israel/001118/2023	Sa.2a (C)	2023-12-27	MOCK1	2560	1280	2560	2560	1280	1280
241339	A/Ukraine/2087/2023	Sa.2a (C)	2023-12-28	BIAT1	2560	2560	2560	2560	2560	1280
241338	A/Ukraine/2087/2024	Sa.2a (C)	2024-01-01	BIAT1	2560	2560	2560	2560	2560	1280
241335	A/Ukraine/2088/2024	Sa.2a (C)	2024-01-03	BIAT1	2560	2560	2560	2560	2560	1280
241784	A/Brazil/DF/2/2023	Sa.2a.1 (C.1.1)	2023-11-28	P1/MOCK1	2560	2560	9120	2560	2560	2560
241895	A/France/DF-APP30113/2023	Sa.2a.1 (C.1.1)	2023-12-02	MOCK1/MOCK1	1280	640	1280	640	1280	640
241783	A/Brazil/DF/2/2023	Sa.2a.1 (C.1.1)	2023-12-04	P1/MOCK1	2560	2560	9120	2560	2560	2560
241782	A/Brazil/DF/2/2023	Sa.2a.1 (C.1.1)	2023-12-06	P1/MOCK1	2560	1280	2560	9120	2560	2560
241891	A/France/DF-APP31144/2023	Sa.2a.1 (C.1.1)	2023-12-06	MOCK1/MOCK1	2560	1280	2560	9120	2560	2560
241889	A/France/DF-ASL-AB-PP31194/2023	Sa.2a.1 (C.1.1)	2023-12-06	MOCK1/MOCK2	2560	1280	2560	9120	2560	2560
241887	A/France/DF-PP30826/2023	Sa.2a.1 (C.1.1)	2023-12-11	MOCK1/MOCK1	2560	1280	2560	9120	2560	2560
241860	A/Brazil/Burg/24173/2023	Sa.2a.1 (C.1.1)	2023-12-18	BIAT1	2560	1280	2560	2560	2560	1280
241837	A/Ukraine/2082/2024	Sa.2a.1 (C.1.1)	2024-01-01	BIAT1	640	640	1280	2560	1280	1280
241836	A/Ukraine/2082/2024	Sa.2a.1 (C.1.1)	2024-01-02	BIAT1	1280	640	1280	2560	1280	1280
241472	A/Israel/0110907/2023	no sequence	2023-09-02	MOCK1	2560	1280	2560	2560	1280	2560
241807	A/England/3/2023	2023-09-02	MOCK1/MOCK1	2560	1280	2560	1280	2560	1280	2560
241743	A/Brazil/1630/2023	2023-09-10	BIAT1/MOCK1	2560	2560	2560	2560	2560	1280	1280
241908	A/England/3/2023	2023-09-11	BIAT1/MOCK1	2560	1280	1280	2560	2560	1280	1280
241427	A/Romania/96133/2023	2023-11-06	BIAT1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241881	A/Bulgum/4/2023	2023-11-18	MOCK1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241875	A/Bulgum/8/2023	2023-11-22	MOCK1/MOCK1	2560	1280	2560	2560	2560	2560	1280
241873	A/Bulgum/9/2023	2023-11-22	MOCK1/MOCK1	2560	1280	2560	2560	2560	2560	1280
241830	A/Panama/27/2023	2023-11-24	MOCK1/MOCK1	2560	1280	2560	2560	2560	2560	1280
241785	A/Brazil/Burg/1/2023	2023-11-27	P1/MOCK1	2560	1280	1280	1280	1280	1280	640
241629	A/Panama/20/2023	2023-11-27	MOCK3/MOCK1	2560	2560	2560	2560	2560	2560	1280
241626	A/Panama/1/2023	2023-11-27	MOCK3/MOCK1	2560	2560	2560	2560	2560	2560	1280
241785	A/Hessen/22/2023	2023-11-28	P1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241627	A/Torino/16/2023	2023-11-28	MOCK3/MOCK1	2560	2560	9120	2560	2560	2560	2560
241626	A/Torino/18/2023	2023-11-29	MOCK3/MOCK1	1280	640	2560	1280	1280	1280	1280
241825	A/Panama/25/2023	2023-12-01	MOCK4/MOCK1	1280	1280	1280	2560	2560	2560	1280
241824	A/Panama/1/2023	2023-12-03	MOCK3/MOCK1	9120	2560	9120	2560	2560	2560	2560
241869	A/Bulgum/13/2023	2023-12-04	MOCK1/MOCK1	2560	1280	2560	1280	2560	2560	1280
241823	A/Panama/42/2023	2023-12-06	MOCK3/MOCK1	2560	1280	2560	2560	2560	2560	1280
241822	A/Panama/6/2023	2023-12-11	MOCK3/MOCK1	9120	2560	2560	2560	2560	2560	1280
241821	A/Brazil/2/2023	2023-12-11	MOCK3/MOCK1	2560	1280	2560	1280	1280	1280	1280
241781	A/Hurtingen/10/2023	2023-12-11	P1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241780	A/Brazil/4/2023	2023-12-12	P1/MOCK1	9120	9120	640	2560	2560	2560	2560
241820	A/Panama/7/2023	2023-12-12	MOCK3/MOCK1	2560	2560	2560	2560	2560	2560	2560
241426	A/Romania/96306/2023	2023-12-13	BIAT1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241425	A/Romania/96306/2023	2023-12-16	BIAT1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241819	A/Panama/9/2023	2023-12-16	MOCK3/MOCK1	2560	2560	2560	2560	2560	2560	1280
241979	A/France/24487/2023	2023-12-18	MOCK1/MOCK1	2560	1280	1280	2560	2560	2560	2560
241741	A/Slovenia/2291/2023	2023-12-19	MOCK4/MOCK1	2560	2560	2560	2560	2560	2560	1280
241424	A/Romania/96340/2023	2023-12-20	BIAT1/MOCK1	9120	2560	2560	2560	2560	2560	2560
241423	A/Romania/96340/2023	2023-12-20	BIAT1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241422	A/Romania/96340/2023	2023-12-21	BIAT1/MOCK1	9120	2560	2560	2560	2560	2560	2560
241421	A/Romania/96340/2023	2023-12-21	BIAT1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241971	A/France/3381/2023	2023-12-27	MOCK1/MOCK1	2560	2560	2560	2560	2560	2560	1280
241782	A/Brazil/2377/2023	2023-12-27	MOCK1/MOCK1	2560	2560	9120	2560	2560	2560	2560
241776	A/Hurtingen/12/2023	2023-12-27	P2/MOCK1	1280	640	2560	2560	2560	2560	2560
241424	A/Ukraine/2088/2023	2023-12-27	BIAT1	2560	1280	1280	1280	1280	1280	640
241869	A/France/3185/2023	2023-12-29	MOCK1/MOCK1	2560	2560	2560	2560	2560	2560	1280
< relates to the lowest dilution of antiserum used										
ND = Not Done										
				Vaccine	Vaccine		Vaccine			
				SH 2021-22	SH 2023		NH 2023-24			
				SH 2022			SH 2024			
				NH 2022-23						

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Fold-reduction table: H1N1

Reference Virus	clade	<4-fold difference		4-fold difference		>4-fold difference		total
		number	percentage	number	percentage	number	percentage	
IVR-215 (A/Victoria/2570/2019) Egg	5a.2	278	99.6	1	0.4	0	0.0	279
A/Sydney/5/2021 Cell	5a.2a	265	95.0	14	5.0	0	0.0	279
A/Sydney/5/2021 Egg	5a.2a	276	98.9	3	1.1	0	0.0	279
A/Victoria/4897/2022 Cell	5a.2a.1	268	96.1	10	3.6	1	0.4	279
IVR-238 (A/Victoria/4897/2022) Egg	5a.2a.1	261	93.5	16	5.7	2	0.7	279
A/Wisconsin/67/2022 Cell	5a.2a.1	253	90.7	21	7.5	5	1.8	279

Susceptibility to antivirals: H1

Table AV1. Antiviral (NAI) testing of isolates with collection dates after 2023-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2023									
September	23		10		0				33
October	17	1	23		8				49
November	31		22		24				77
December	49		27		8				84
2024									
January	9		3		0				12
Totals	129	1	85	0	40	0	0	0	255

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-02-12

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Qatar/10-VI-23-0103273/2023	H1pdm	15-Oct-23	421		H275Y
Outliers					

¹ Clinical sample not available

Table AV2. Antiviral (BXM) testing of isolates with collection dates after 2023-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	
2023									
September	13		6		0				19
October	8		18		8				34
November	12		13		17				42
December	1		4		0				5
2024									
January	0		0		0				0
Totals	34	0	41	0	25	0	0	0	100

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-02-09

Virus name	Subtype/ Lineage	Collection date	Fold difference	NA substitution
			Baloxavir	
Outliers				

¹ Clinical sample not available

In-silico assessment for the presence of NA substitutions associated with reduced susceptibility to neuraminidase inhibitors (NAI) was conducted for 365 H1N1, 323 H3N2 and 40 B/Victoria NA gene sequences generated at the WIC and deposited in GISAID for viruses with collection dates from 1st September 2023. No substitutions associated with reduced susceptibility were detected, with the following exceptions: • A/H1N1 A/Qatar/10-VI-23-0103273/2023 showed presence of H275Y, which was confirmed by phenotypic testing with a Highly reduced inhibition profile. Based on whole genome sequencing at the WIC, full-length PA gene sequences were generated from 289 H1N1, 315 H3N2 viruses with collection dates from 1st September 2023. No sequences with mutations encoding amino acid markers associated with reduced susceptibility to Baloxavir marboxil (BXM) were identified.

A/H1N1: Egg isolates

Virus	Status	Genetic group/clade	Comment
A/Lyon/820/2021	CBER-49B, NIB-132, SAN-015	6B.1A.5a.1 (128T, P137S, G155E)	Egg adaptation: (X187V, T483N - CHO), 2-way pass
A/Norway/31694/2022	NIB-133 (2-way pass)	6B.1A.5a.2a.1 (P137S, K142R)	Egg adaptation: (Q223R)
A/Catalonia/NSVH161512065/2022	NIB-134 (2-way pass)	6B.1A.5a.2a.1 (P137S, K142R)	Egg adaptation: (Q223R)
A/Ghana/1894/2021	Reference only	6B.1A.5a.1 (I166T, A186T)	Egg adaptation: (no change)

A/H1N1: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
IVR-215 (A/Victoria/2570/2019)	5a.2 (C)	E4/D7/E3	F37/21
A/Sydney/5/2021	5a.2a (C.1)	MDCK3/MDCK4	F46/22
A/Sydney/5/2021	5a.2a (C.1)	E3/E3	F04/22
A/Victoria/4897/2022	5a.2a.1 (C.1.1.1)	SIAT2/MDCK2	F05/23
IVR-238 (A/Victoria/4897/2022)	5a.2a.1 (C.1.1.1)	E3/D6/E2	F07/23
A/Wisconsin/67/2022	5a.2a.1 (C.1.1)	MDCK2	F17/23

Summary: H1N1

Genetic analyses

6B.1A.5a.2a (C.1) and 6B.1A.5a.2a.1 (C.1.1. and C.1.1.1) clade viruses both continued to circulate with differing relative proportions depending on region, with a global predominance of 5a.2a viruses.

In Europe, both 5a.2a and 5a.2a.1 viruses were detected, with 5a.2a predominating in 2/3 of A/H1N1 sequenced viruses.

Within the 5a.2a viruses, characterised by substitutions K54Q, A186T, E224A, R259K and K308R, three subclades were observed: one minor subclade with D94N and T216A (C.1.7), with viruses predominating in Australia, New Zealand and Indonesia, and in minor proportions in Europe and Asia; a larger subclade defined by substitution I418V (C.1) which was predominating in the Middle East, Africa, South-East Asia, Central America and some countries in Europe, and circulated in significant proportions in other countries. This subclade included a cluster characterised by T120A with K169Q or V47I and another with P137S. Other subclades that were reported in previous weeks were not seen during this period, except for one with A48P (C.1.2) with viruses from Thailand and Vietnam.

Within the 5a.2a.1 viruses, characterised by substitutions P137S, K142R, D260E, T277A, E356D and N451H, there are two main groups of viruses: a major clade with T216A (C.1.1.1.) represented by A/Victoria/4897/2022, predominating in the US, the Caribbean, Japan and some countries in Europe, showing additional heterogeneity with a cluster with R45K; and a minor clade represented by A/Wisconsin/67/2022 (C.1.1), circulating in Brazil, Europe and South East Asia.

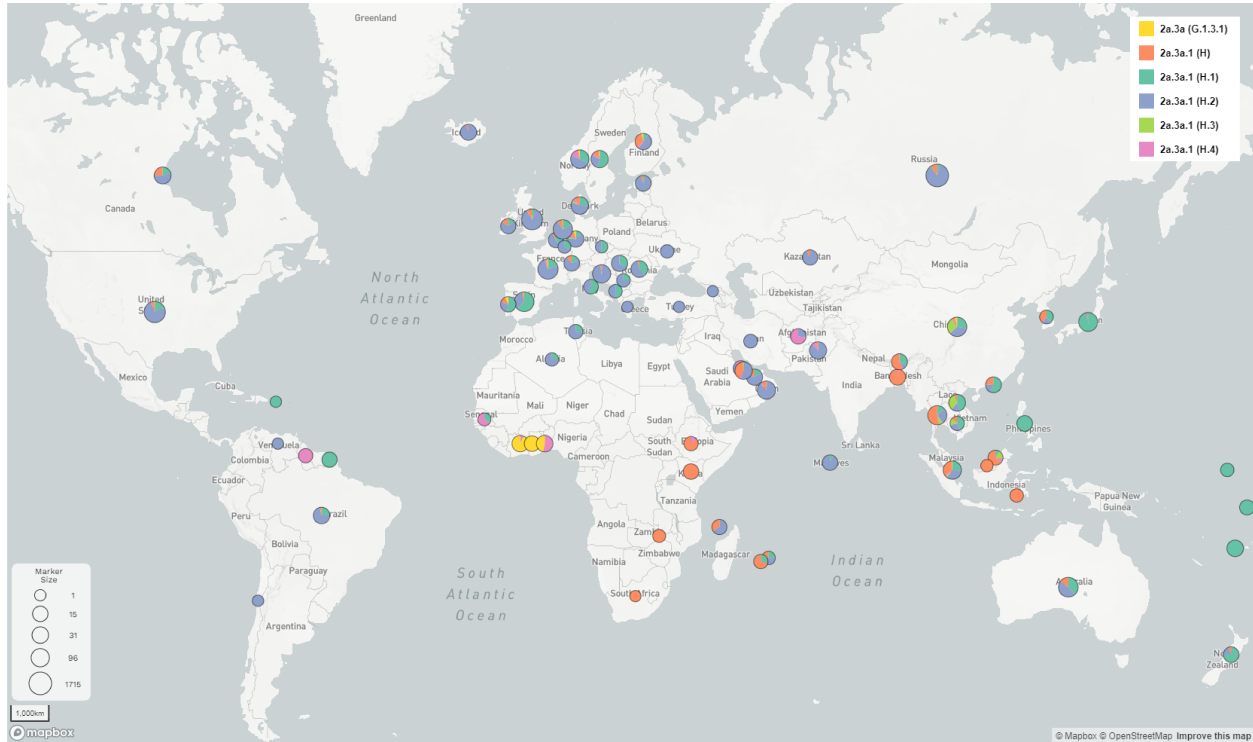
Antigenic analyses

HI titres shows that the SH2023 vaccine strain (A/Sydney/5/21 cell 5a.2), the cell- and egg-based NH 2023-24 strain (A/Victoria/4897/2022) recognises both 5a.2a and 5a.2a.1 well.

Influenza A H3N2

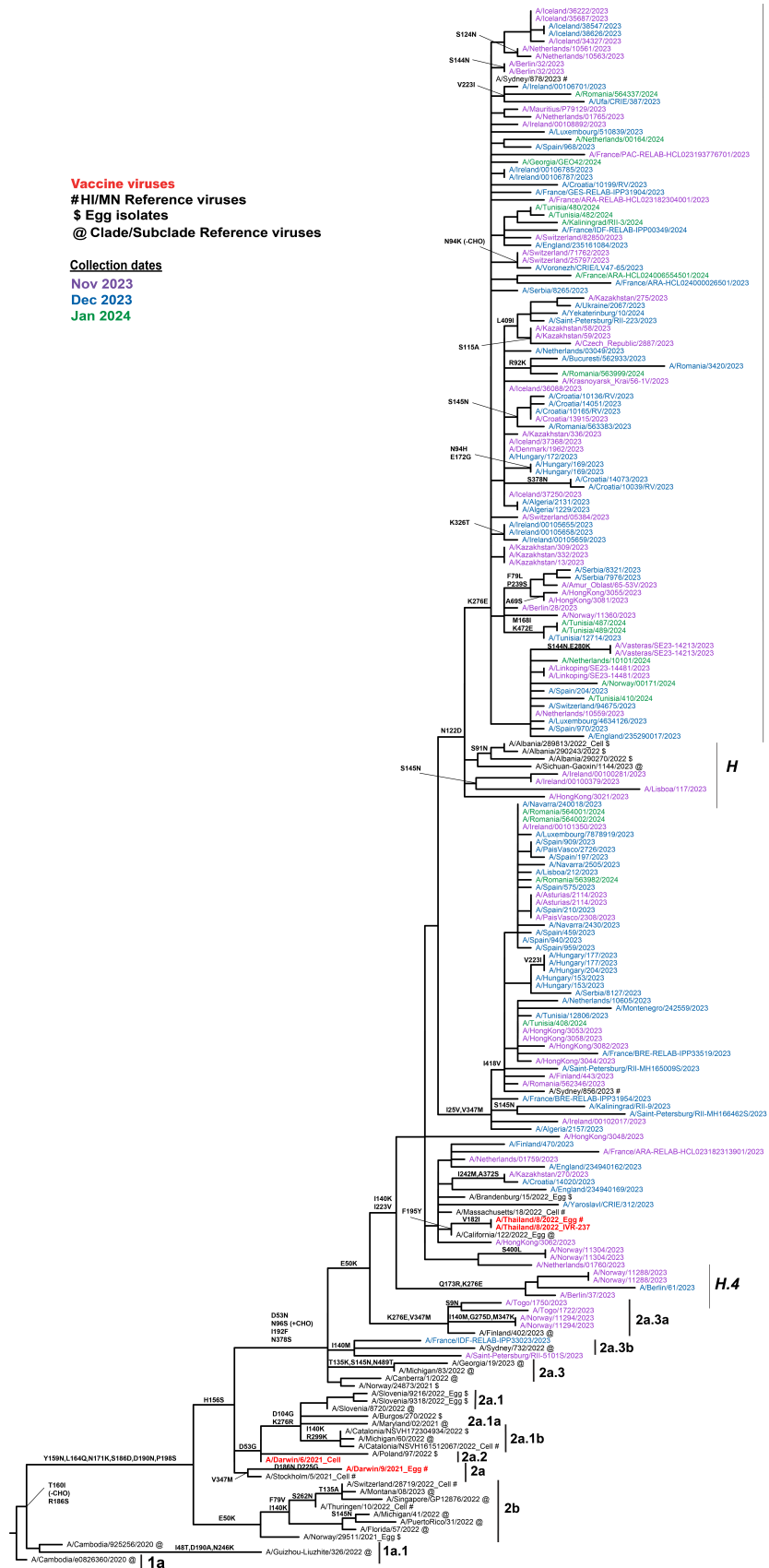
Genetic analyses: H3N2

Microreact: Subclade



Maximum likelihood phylogenetic tree: H3N2

Maximum likelihood time-resolved phylogenetic tree inferred using IQTree2 from HA sequence data obtained from GISAID from 1st September onwards, manually curated and then downsampled using Treemer to retain a representative tree topology of 200 sequences with emphasis on European viruses and viruses sequenced at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.



Vaccine viruses
HI/MN Reference viruses
\$ Egg isolates
@ Clade/Subclade Reference viruses

Collection dates
Nov 2023
Dec 2023
Jan 2024

Antigenic analyses: H3N2

Haemagglutination inhibition tables: H3N2

Table H3-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-10-20

Haemagglutination inhibition titre																
Viruses	Other information	Genetic group	Collection Date	Genetic group	Post-infection ferret antiserum											
					A/Thuringen 1/0/22 SIAT	A/Switzerland /28719/22 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Catal NSVH-2067/22 SIAT	A/Albania /289613/2022 SIAT	A/Brandenburg /15/2022 Egg	A/Massachusetts /18/2022 SIAT	A/California /12/22 Egg	A/Thailand /08/2022 Egg	IVR-237 /08/2022 Egg	
					Passage Ferret number	F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F18/23	F36/23	F33/23	F34/23	F35/23
					2b	2b	2a	2a	2a.1b	2a.3a.1	2a.3a.1	2a.3a.1	2a.3a.1	2a.3a.1	2a.3a.1	2a.3a.1
REFERENCE VIRUSES																
A/Thuringen/19/2022	S262N	2b	2022-04-01	P1/SIAT3	160	640	160	80	40	80	40	80	160	80	160	
A/Switzerland/28719/2022	T135A	2b	2022-12-19	SIAT1	160	1280	160	80	40	80	40	80	160	160	160	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	160	320	320	160	80	80	80	160	320	160	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	640	160	160	160	320	640	640	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	40	320	160	320	640	160	80	160	160	320	160	
A/Albania/289613/2022	N122D	2a.3a.1	2022-12-13	MDCK1	160	640	160	320	160	1280	320	640	640	1280	1280	
A/Brandenburg/15/2022		2a.3a.1		EviE5	320	1280	320	1280	320	1280	1280	1280	1280	2560	2560	
233467 A/Massachusetts/18/2022		2a.3a.1		SIAT3/SIAT1	80	640	160	320	160	1280	640	640	640	1280	1280	
235135 A/California/12/2022		2a.3a.1		E1/E1	160	640	160	320	160	640	320	320	640	1280	1280	
233361 A/Thailand/08/2022		2a.3a.1		E3/E1	160	640	320	320	160	640	320	320	640	1280	1280	
233363 IVR-237 (A/Thailand/08/2022)		2a.3a.1		E3/D7/E1	80	640	160	320	160	640	160	320	640	1280	1280	
TEST VIRUSES																
240028 A/Catalonia/NSVH10215361/2023		2a.3a.1	2023-09-02	SIAT1/SIAT1	80	320	80	320	160	160	88	160	640	320	320	
240026 A/Catalonia/NSVH10215629/2023		2a.3a.1	2023-09-03	SIAT1/SIAT1	80	320	160	320	80	320	160	320	640	640	640	
240024 A/Catalonia/NSVH102156615/2023		2a.3a.1	2023-09-04	SIAT1/SIAT1	80	640	160	320	80	320	160	320	640	640	640	
240022 A/Catalonia/NSVH160871625/2023		2a.3a.1	2023-09-05	SIAT1/SIAT1	80	320	80	160	80	320	88	160	320	320	320	
240017 A/Catalonia/NSVH151040742/2023		2a.3a.1	2023-09-06	SIAT1/SIAT1	80	320	80	160	80	320	320	320	640	640	640	
240019 A/Catalonia/NSVH102158148/2023		2a.3a.1	2023-09-06	SIAT1/SIAT1	160	640	160	320	80	640	320	640	640	1280	1280	
240014 A/Catalonia/NSVH102161296/2023		2a.3a.1	2023-09-11	SIAT1/SIAT1	80	320	160	320	80	320	160	320	320	640	640	
240012 A/Catalonia/NSVH102165041/2023		2a.3a.1	2023-09-17	SIAT1/SIAT1	160	640	160	320	160	1280	640	640	640	1280	1280	
< relates to the lowest dilution of antiserum used ND = Not Done																
Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24																
Vaccine SH 2024																
< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)																

Table H3-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-11-24

Viruses		Other information		Haemagglutination inhibition titre																							
				Post-infection ferret antiserum																NEW		NEW		NEW		NEW	
				A/Thuringen /10/22 SIAT	A/Switzerland /28719/22 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Catal NSVH-2087/22 /289813/2022 SIAT	A/Albania /18/2022 SIAT	A/Brandenburg /18/2022 Egg	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg	A/Thailand /18/2022 SIAT	A/Thailand /08/2022 Egg	A/Montana /08/2023 SIAT	A/Montana /08/2023 Egg	NEW	NEW	NEW	NEW							
				F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F18/23	F36/23	F34/23	F37/23	F38/23	F39/23	F40/23											
				Genetic group	Collection Date	Passage Ferret number	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2b	2b						
REFERENCE VIRUSES																											
A/Thuringen/10/2022		S262N	2b	2022-04-01	P1/SIAT3	320		640	320	320	40	80	80	80	160	160	320	160	320	160	320	160					
A/Switzerland/28719/2022		T135A	2b	2022-12-19	SIAT1	160		1280	320	160	40	80	80	<40	80	160	160	320	80	320	320	80					
A/Stockholm/5/2021			2a	2021-04-16	SIAT0/SIAT3	80		320	320	640	80	80	40	40	80	320	160	80	80	40	80	40					
A/Darwin/9/2021		D186N, D225G (egg)	2a	2021-04-17	E3/E4	160		640	640	640	160	160	160	160	160	320	640	320	160	80	160	80					
A/Catalonia/NSVH161512067/2022			2a.1b	2022-09-14	SIAT1/SIAT4	80		320	320	640	640	320	80	160	320	320	320	160	80	80	80	80					
A/Albania/289813/2022		N122D	2a.3a.1 (H)	2022-12-13	MDCK1	160		320	160	320	160	1280	320	320	640	1280	1280	640	160	80	160	80					
A/Brandenburg/18/2022			2a.3a.1 (H)		E3/E5	640		1280	1280	2560	640	1280	2560	1280	2560	2560	2560	1280	320	1280	320	1280					
A/Massachusetts/18/2022			2a.3a.1 (H)		SIAT3/SIAT1	160		640	320	640	160	1280	640	1280	640	1280	1280	1280	320	160	160	160					
233361 A/Thailand/08/2022			2a.3a.1 (H)		E3/E1	80		640	320	640	160	640	160	320	1280	640	640	640	160	160	160	160					
235139 A/Montana/06/2023			2a.3a.1 (H)		SIAT1/SIAT1	160		1280	320	160	<40	80	<40	160	160	160	160	160	640	320	640	320					
235598 A/Montana/06/2023			2a.3a.1 (H)		E1/E1	320		2560	640	1280	80	1280	160	640	1280	1280	1280	640	640	640	640	640					
TEST VIRUSES																											
240060 A/ram/23E11/2023			2a.3a.1 (H.2)	2023-09-13	SIAT1/SIAT1	40		80	80	80	160	<40	80	<40	40	160	160	80	40	<40	<40	<40					
240059 A/ram/11G11/2023			2a.3a.1 (H.2)	2023-09-17	SIAT1/SIAT1	160		320	160	320	80	320	80	320	80	320	640	320	320	80	80	80					
240058 A/ram/20L11/2023			2a.3a.1 (H.2)	2023-09-24	SIAT1/SIAT1	80		320	160	320	40	320	80	160	320	640	640	640	320	160	80	80					
240130 A/Salalah/S2328680/2023			2a.3a.1 (H.2)	2023-09-26	SIAT1	160		640	320	640	160	640	160	320	640	640	640	320	160	80	160	80					
240057 A/ram/24N11/2023			2a.3a.1 (H.2)	2023-09-27	SIAT1/SIAT1	40		160	80	320	<40	160	40	80	320	320	320	320	160	40	<40	<40					
240124 A/Sofhar/S2328337/2023			2a.3a.1 (H.2)	2023-09-29	SIAT2	80		320	320	640	80	320	80	160	640	640	640	320	80	40	80	80					
240121 A/Al Musanaah/7234645/2023			2a.3a.1 (H.2)	2023-10-01	SIAT1	160		320	320	320	80	640	80	320	640	640	640	320	80	80	80	80					
240122 A/As Suwaig/7234643/2023			2a.3a.1 (H.2)	2023-10-01	SIAT1	160		640	320	640	80	640	160	320	640	640	640	320	160	80	80	80					
240123 A/Nizwa/7234595/2023			2a.3a.1 (H.2)	2023-10-01	SIAT1	80		320	160	320	80	320	80	160	320	320	320	160	160	160	160	160					
240119 A/Muscat/7234671/2023			2a.3a.1 (H.2)	2023-10-03	SIAT1	160		320	160	640	<40	80	80	320	320	320	320	160	160	160	160	160					
240056 A/ram/25A12/2023			2a.3a.1 (H.2)	2023-10-09	SIAT1	80		320	160	320	80	320	40	160	320	320	320	160	80	40	80	80					
< relates to the lowest dilution of antiserum used ND = Not Done												Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24				Vaccine SH 2024											

Table H3-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-12-11

Viruses	Other information	Haemagglutination inhibition titre												
		Genetic group	Collection Date	Passage Ferret number	Genetic group	Post-infection ferret antiserum								
						A/Thuringen /10/22 SIAT	A/Switzerland /28719/22 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Catal NSVH-2067/22 SIAT	A/Albania /289813/2022 SIAT	A/Brandenburg /15/2022 Egg	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg
						F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F18/23	F36/23	F34/23
										2b	2b	2a	2a	2a.1b
REFERENCE VIRUSES														
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT3	320	640	320	640	40	160	80	320	160	
A/Switzerland/28719/2022	T135A	2b	2022-12-19	SIAT2	160	1280	320	160	<40	160	40	160	160	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	160	640	640	640	160	320	80	80	640	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	320	640	640	640	320	320	320	320	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	160	320	320	640	1280	640	160	160	640	
A/Albania/289813/2022	N122D	2a.3a.1	2022-12-13	MDCK1	320	640	320	640	320	2560	640	1280	2560	
A/Brandenburg/15/2022		2a.3a.1		Ex/E5	1280	2560	2560	2560	640	2560	2560	2560	2560	
A/Massachusetts/18/2022		2a.3a.1		SIAT3/SIAT1	320	1280	320	1280	320	2560	1280	2560	2560	
A/Thailand/08/2022		2a.3a.1		E3/E1	160	640	320	640	160	640	160	320	1280	
TEST VIRUSES														
240189	A/Qatar/25-VI-23-0107245/2023	2a.3a.1	2023-10-28	SIAT1	80	160	80	320	80	640	320	640	1280	
240190	A/Qatar/10-VI-23-0107179/2023	2a.3a.1	2023-10-28	SIAT1	80	320	160	320	160	1280	640	640	2560	
240191	A/Qatar/10-VI-23-0107015/2023	2a.3a.1	2023-10-26	SIAT1	160	320	160	640	<40	640	160	320	640	
240192	A/Qatar/23-VI-23-0106322/2023	2a.3a	2023-10-24	SIAT1	80	160	160	320	40	160	40	80	320	
240193	A/Qatar/10-VI-23-0106239/2023	2a.3a.1	2023-10-24	SIAT1	40	320	160	320	80	640	320	640	1280	
240194	A/Qatar/16-VI-23-0105945/2023	2a.3a.1	2023-10-23	SIAT1	160	320	160	640	80	640	80	320	640	
240195	A/Qatar/10-VI-23-0105504/2023	no sequence	2023-10-22	SIAT1	80	320	160	640	160	1280	640	640	2560	
240199	A/Qatar/10-VI-23-0103751/2023	2a.3a.1	2023-10-16	SIAT1	80	320	80	160	40	80	40	80	320	

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24

Vaccine
SH 2024

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2023-12-20

Viruses	Other information	Haemagglutination inhibition titre												
		Genetic group	Collection Date	Passage Ferret number	Genetic group	Post-infection ferret antiserum								
						A/Thuringen /10/22 SIAT F36/22	A/Switzerland /28719/22 SIAT F29/23	A/Stockholm 5/21 SIAT F15/22	A/Darwin 9/21 Egg F39/21	A/Catal /289813/2022 SIAT F41/22	A/Albania /289813/2022 SIAT F21/23	A/Brandenburg /15/2022 Egg F18/23	A/Massachusetts /18/2022 SIAT F36/23	A/Thailand /08/2022 Egg F34/23
						2b	2b	2a	2a	2a.1b	2a.3a.1	2a.3a.1	2a.3a.1	2a.3a.1
						REFERENCE VIRUSES								
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT3	320	1280	320	320	80	80	80	320	320	
A/Switzerland/28719/2022	T135A	2b	2022-12-19	SIAT2	80	1280	320	160	<40	80	<40	80	80	
A/Stockholm/5/2021		2a	2021-04-16	SIAT0/SIAT3	80	320	640	640	160	160	40	80	320	
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	640	640	160	320	160	320	640	
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	80	320	160	320	640	320	160	160	320	
A/Albania/289813/2022	N122D	2a.3a.1	2022-12-13	MDCK1	160	320	320	320	160	1280	640	640	1280	
A/Brandenburg/15/2022		2a.3a.1		Ex/E5	320	1280	1280	1280	640	1280	1280	1280	2560	
A/Massachusetts/18/2022		2a.3a.1		SIAT3/SIAT1	160	640	320	640	160	2560	640	1280	1280	
A/Thailand/08/2022		2a.3a.1		E3/E1	160	320	320	640	160	640	160	320	640	
TEST VIRUSES														
240447	A/Bayern/10/2023	2a.3a.1	2023-08-18	P1/SIAT1	80	320	160	320	80	640	320	640	1280	
240445	A/Brandenburg/3/2023	2a.3a.1 (H)	2023-10-09	P1/SIAT1	40	160	80	80	40	320	160	160	320	
240443	A/Hessen/20/2023	2a.3a.1 (H)	2023-10-11	P1/SIAT1	80	160	80	160	40	640	160	320	640	
240305	A/Norway/11304/2023	2a.3a.1 (H)	2023-11-03	SIAT1	40	160	80	320	40	640	160	320	640	
240446	A/Bayern/11/2023	2a.3a.1 (H.2)	2023-09-18	P1/SIAT1	40	160	80	160	40	160	40	80	320	
240444	A/Berlin/27/2023	2a.3a.1 (H.2)	2023-10-09	P1/SIAT1	80	160	80	160	40	320	80	160	320	
240304	A/Norway/11360/2023	2a.3a.1 (H.2)	2023-11-03	SIAT1	80	160	80	320	40	160	80	160	320	
240442	A/Berlin/28/2023	2a.3a.1 (H.2)	2023-11-06	P1/SIAT1	40	160	160	160	40	160	40	80	320	
240441	A/Berlin/32/2023	2a.3a.1 (H.2)	2023-11-13	P1/SIAT1	40	160	160	160	40	320	80	160	320	
240312	A/Norway/10423/2023	2a.3a.1 (H.4)	2023-10-16	SIAT1	40	160	80	160	40	160	40	160	160	
240306	A/Norway/11288/2023	2a.3a.1 (H.4)	2023-11-03	SIAT1	40	160	80	160	40	160	40	160	320	
240440	A/Berlin/37/2023	2a.3a.1 (H.4)		P1/SIAT1	40	160	80	160	<40	160	40	80	160	
< relates to the lowest dilution of antiserum used ND = Not Done										Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24				Vaccine SH 2024

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2024-01-12

Viruses		Other information	Haemagglutination inhibition titre										
			Post-infection ferret antiserum										
			A/Thuringen /10/22 SIAT	A/Switzerland /28719/22 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Catal NSVH-2067/22 SIAT	A/Albania /289813/22 SIAT	A/Brandenburg /15/2022 Egg	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg		
			Passage Ferret number	F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F18/23	F36/23	F34/23	
		Genetic group	Collection Date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)
REFERENCE VIRUSES													
A/Thuringen/10/2022	S262N		2b	2022-04-01	P1/SIAT3	320	640	320	160	40	80	160	160
A/Switzerland/28719/2022	T135A		2b	2022-12-19	SIAT2	160	2560	160	160	<40	80	<40	80
A/Stockholm/5/2021			2b	2021-04-16	SIAT0/SIAT5	80	320	640	640	160	160	80	320
A/Switzerland/2021	D186N, D225G (egg)		2a	2021-04-17	E3/E4	160	320	640	320	80	160	160	320
A/Catalonia/NSVH161512067/2022			2a.1b	2022-09-14	SIAT1/SIAT4	80	320	160	320	320	160	80	160
A/Albania/289813/2022	N122D		2a.3a.1 (H)	2022-12-13	MDCK1/SIAT1	160	320	160	320	80	640	320	640
A/Brandenburg/15/2022			2a.3a.1 (H)		Ex/E5	320	1280	640	1280	320	2560	1280	2560
A/Massachusetts/18/2022			2a.3a.1 (H)		SIAT3/SIAT1	160	640	320	640	320	1280	640	1280
A/Thailand/08/2022			2a.3a.1 (H)		E3/E1	160	640	320	640	160	640	160	320
TEST VIRUSES													
240425	A/Netherlands/01617/2023		2a.1b	2023-05-23	hCK1/SIAT1	80	160	80	320	160	160	40	80
240423	A/Netherlands/01685/2023		2a.3a.1 (H)	2023-09-08	hCK1/SIAT1	80	160	80	160	80	640	160	320
240422	A/Netherlands/01732/2023		2a.3a.1 (H)	2023-10-09	hCK1/SIAT1	80	640	160	320	80	640	320	640
240845	A/Burgos/64/2023		2a.3a.1 (H)	2023-10-31	SIAT1/SIAT1	80	160	80	160	80	640	160	320
240711	A/HongKong/3021/2023		2a.3a.1 (H)	2023-11-23	MDCK1/SIAT1	160	320	160	320	80	640	320	640
240705	A/HongKong/3048/2023		2a.3a.1 (H)	2023-11-27	MDCK2/SIAT1	160	320	160	320	160	640	320	1280
240693	A/HongKong/3062/2023		2a.3a.1 (H)	2023-11-29	MDCK1/SIAT1	160	640	160	320	160	1280	640	640
240853	A/Burgos/63/2023		2a.3a.1 (H.1)	2023-10-13	SIAT1/SIAT1	80	320	160	320	80	640	320	320
240310	A/Norway/11072/2023		2a.3a.1 (H.1)	2023-10-29	SIAT2	80	320	160	320	160	640	320	640
240709	A/HongKong/3044/2023		2a.3a.1 (H.1)	2023-11-26	MDCK1/SIAT1	160	320	160	320	80	640	320	640
240703	A/HongKong/3053/2023		2a.3a.1 (H.1)	2023-11-27	MDCK1/SIAT1	80	320	160	320	80	640	320	320
240701	A/HongKong/3058/2023		2a.3a.1 (H.1)	2023-11-27	MDCK2/SIAT1	80	320	160	320	80	640	320	640
240695	A/HongKong/3082/2023		2a.3a.1 (H.1)	2023-11-28	MDCK1/SIAT1	160	640	160	320	80	640	320	320
240421	A/Netherlands/01729/2023		2a.3a.1 (H.2)	2023-10-13	hCK1/SIAT1	80	160	80	320	80	160	40	160
240420	A/Netherlands/01733/2023		2a.3a.1 (H.2)	2023-10-18	hCK1/SIAT1	80	320	160	320	80	320	80	160
240419	A/Netherlands/01735/2023		2a.3a.1 (H.2)	2023-10-20	hCK1/SIAT1	160	320	160	320	80	320	80	320
240707	A/HongKong/3072/2023		2a.3a.1 (H.2)	2023-11-26	MDCK1/SIAT1	160	320	160	320	80	320	80	160
240699	A/HongKong/3081/2023		2a.3a.1 (H.2)	2023-11-27	MDCK1/SIAT1	80	320	160	320	80	320	80	320
240697	A/HongKong/3055/2023		2a.3a.1 (H.2)	2023-11-28	MDCK1/SIAT1	160	320	160	320	80	320	80	160
240424	A/Netherlands/01643/2023		2a.3a.1 (H.4)	2023-07-25	hCK1/SIAT1	80	320	160	320	80	320	80	160
240418	A/Netherlands/01744/2023		2a.3a.1 (H.4)	2023-10-29	hCK1/SIAT1	160	320	160	320	80	320	80	160
< relates to the lowest dilution of antiserum used ND = Not Done								Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24	Vaccine SH 2024				

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24

Vaccine
SH 2024

 < 4-fold  4-fold  8-fold  > 8-fold  < not recognised by the antiserum  ≥ 160 (no homologous titre)

Table H3-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2024-01-17

Viruses	Other Information		Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre									
						Passage Ferret number	Post-infection ferret antiserum								
							A/Thuringen /19/22 SIAT	A/Switzerland /28719/22 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Catal NSVH-2067/22 SIAT	A/Albania 1/8/2022 SIAT	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg	
							F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F36/23	F34/23	
						2b	2b	2a	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)		
REFERENCE VIRUSES															
A/Thuringen/10/2022	S262N	2b	2022-04-01	P1/SIAT3	320		640	320	320	40	80	160	160		
A/Switzerland/28719/2022	T135A	2b	2022-12-19	SIAT2	320	1280	160	160	<40	160	80	160	160		
A/Stockholm/5/2021		2a	2021-04-16	SIAT3/SIAT5	160	320	640	640	160	320	160	320	320		
A/Darwin/9/2021	D186N, D225G (egg)	2a	2021-04-17	E3/E4	160	640	320	640	160	160	160	320	320		
A/Catalonia/NSVH161512067/2022		2a.1b	2022-09-14	SIAT1/SIAT4	80	320	160	320	320	160	160	320	320		
A/Albania/28813/2022	N122D	2a.3a.1 (H)	2022-12-13	MOCK1/SIAT1	160	320	160	320	40	640	320	640	640		
A/Massachusetts/15/2022		2a.3a.1 (H)		SIAT3/SIAT1	160	640	320	640	320	1280	640	1280	1280		
A/Thailand/08/2022		2a.3a.1 (H)		E3/E1	160	640	320	640	160	640	320	640	640		
TEST VIRUSES															
240402 A/Ghana/7529/2023		2a.3a	2023-09-12	SIAT1	80	160	160	320	80	320	80	320	320		
240403 A/Ghana/7409/2023		2a.3a	2023-09-12	SIAT1	80	160	80	320	80	160	80	320	320		
240401 A/Ghana/7625/2023		2a.3a	2023-09-17	SIAT1	160	320	320	320	80	320	160	640	640		
240396 A/Ghana/2186/2023		2a.3a	2023-10-23	SIAT1	80	320	160	320	80	320	160	320	320		
240328 A/Switzerland/62567/2023		2a.3a.1 (H)	2023-10-28	SIAT1	320	640	320	640	160	1280	640	1280	1280		
240327 A/Switzerland/62589/2023		2a.3a.1 (H)	2023-10-28	SIAT1	80	320	160	320	80	640	320	640	640		
240494 A/Iceland/34327/2023		2a.3a.1 (H)	2023-11-01	SIAT1	80	160	80	160	80	640	320	640	640		
240621 A/Ireland/00100379/2023		2a.3a.1 (H)	2023-11-05	SIAT1	80	160	80	320	80	640	320	640	640		
240620 A/Ireland/00100281/2023		2a.3a.1 (H)	2023-11-06	SIAT1	80	160	80	160	80	640	160	640	640		
240863 A/Burkina Faso/27/2023		2a.3a.1 (H.1)	2023-09-24	SIAT1/SIAT3	160	640	320	320	160	1280	640	1280	1280		
240861 A/Burkina Faso/35/2023		2a.3a.1 (H.1)	2023-09-24	SIAT1/SIAT1	160	320	80	320	80	640	320	640	640		
240859 A/Burkina Faso/28/2023		2a.3a.1 (H.1)	2023-09-27	SIAT1/SIAT1	320	320	160	320	80	640	320	640	640		
240849 A/Serbia/54/2023		2a.3a.1 (H.1)	2023-10-22	SIAT1/SIAT1	160	320	160	160	80	640	320	640	640		
240325 A/Switzerland/28412/2023		2a.3a.1 (H.1)	2023-11-14	SIAT1	160	320	160	320	160	1280	640	1280	1280		
240619 A/Ireland/00101350/2023		2a.3a.1 (H.1)	2023-11-15	SIAT1	160	640	320	640	160	1280	640	1280	1280		
240618 A/Ireland/00102017/2023		2a.3a.1 (H.1)	2023-11-16	SIAT1	160	640	320	640	160	640	320	1280	1280		
240493 A/Iceland/34967/2023		2a.3a.1 (H.2)	2023-11-07	SIAT1	160	640	320	640	160	640	320	640	640		
240492 A/Iceland/35570/2023		2a.3a.1 (H.2)	2023-11-12	SIAT1	160	320	160	320	80	640	160	320	320		
240491 A/Iceland/35687/2023		2a.3a.1 (H.2)	2023-11-13	SIAT1	160	640	160	640	160	640	320	640	640		
240490 A/Iceland/35741/2023		2a.3a.1 (H.2)	2023-11-13	SIAT1	80	160	80	320	80	320	160	320	320		
240326 A/Switzerland/05364/2023		2a.3a.1 (H.2)	2023-11-13	SIAT1	80	320	160	320	80	320	160	320	320		
240489 A/Iceland/35921/2023		2a.3a.1 (H.2)	2023-11-14	SIAT1	160	640	160	640	160	640	320	640	640		
240488 A/Iceland/36089/2023		2a.3a.1 (H.2)	2023-11-15	SIAT1	80	160	80	160	40	320	80	320	320		
240487 A/Iceland/36119/2023		2a.3a.1 (H.2)	2023-11-16	SIAT1	80	160	80	160	40	160	160	320	320		
240486 A/Iceland/36222/2023		2a.3a.1 (H.2)	2023-11-17	SIAT1	160	640	160	640	160	640	160	640	640		
240485 A/Iceland/36373/2023		2a.3a.1 (H.2)	2023-11-18	SIAT1	80	160	80	320	80	320	160	320	320		
240484 A/Iceland/36433/2023		2a.3a.1 (H.2)	2023-11-19	SIAT1	80	320	80	320	80	320	160	320	320		
240324 A/Switzerland/03069/2023		2a.3a.1 (H.2)	2023-11-20	SIAT1	160	640	320	640	160	640	320	640	640		
240323 A/Switzerland/02050/2023		2a.3a.1 (H.2)	2023-11-21	SIAT1	160	640	160	640	80	640	320	640	640		
240322 A/Switzerland/25797/2023		2a.3a.1 (H.2)	2023-11-23	SIAT1	80	160	80	160	80	160	160	320	320		
240483 A/Iceland/37250/2023		2a.3a.1 (H.2)	2023-11-24	SIAT1	80	160	80	160	40	320	160	320	320		
240321 A/Switzerland/0649/2023		2a.3a.1 (H.2)	2023-11-24	SIAT1	160	320	160	320	80	640	160	640	640		
240482 A/Iceland/37255/2023		2a.3a.1 (H.2)	2023-11-25	SIAT1	160	320	160	640	160	640	320	640	640		
240481 A/Iceland/37274/2023		2a.3a.1 (H.2)	2023-11-25	SIAT1	160	160	80	320	80	320	160	320	320		
240480 A/Iceland/37349/2023		2a.3a.1 (H.2)	2023-11-27	SIAT1	320	1280	320	1280	320	1280	640	1280	1280		
240479 A/Iceland/37410/2023		2a.3a.1 (H.2)	2023-11-27	SIAT1	160	640	160	640	80	640	320	640	640		
240478 A/Iceland/37528/2023		2a.3a.1 (H.2)	2023-11-28	SIAT1	80	320	160	320	80	640	160	640	640		
240477 A/Iceland/37811/2023		2a.3a.1 (H.2)	2023-11-29	SIAT1	80	160	80	320	80	320	160	320	320		
240320 A/Switzerland/71162/2023		2a.3a.1 (H.2)	2023-11-29	SIAT1	160	640	160	640	80	640	320	640	640		
240476 A/Iceland/37943/2023		2a.3a.1 (H.2)	2023-11-30	SIAT1	80	320	80	320	40	160	160	320	320		
240475 A/Iceland/38351/2023		2a.3a.1 (H.2)	2023-12-04	SIAT1	80	160	160	320	80	320	160	320	320		
240474 A/Iceland/38354/2023		2a.3a.1 (H.2)	2023-12-04	SIAT1	160	160	80	320	80	320	160	320	320		
240473 A/Iceland/38547/2023		2a.3a.1 (H.2)	2023-12-05	SIAT1	80	160	80	160	80	320	160	320	320		
240471 A/Iceland/38721/2023		2a.3a.1 (H.2)	2023-12-06	SIAT1	80	160	80	320	80	320	160	320	320		
240615 A/Ireland/00106707/2023		2a.3a.1 (H.2)	2023-12-06	SIAT1	160	640	320	640	160	640	320	640	640		
240614 A/Ireland/00107299/2023		2a.3a.1 (H.2)	2023-12-09	SIAT1	160	640	160	320	160	640	320	640	640		
240613 A/Ireland/00107900/2023		2a.3a.1 (H.2)	2023-12-11	SIAT1	160	640	320	640	160	640	320	640	640		

< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24

Vaccine
SH 2024

< 4-fold 4-fold 8-fold > 8-fold - - not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2024-01-25

Viruses	Other information	Haemagglutination inhibition titre																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
		Genetic group	Collection Date	Passage Ferret number	Post-infection ferret antiserum																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
					Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H.1)	2a.3a.1 (H.2)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
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< relates to the lowest dilution of antiserum used
ND = Not Done

Vaccine
SH 2022
SH 2023
NH 2023-24

Vaccine
SH 2024

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (no homologous titre)

Table H3-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20mM Oseiltamivir) 2024-02-02

Viruses		Other information	Haemagglutination inhibition titre															
			Post-infection level estimate															
			Antigenic difference from A/Thuringen/10/2022															
			Formal number		F36/22		F29/23		F10/22		F39/21		F41/22		F21/23		F36/23	
Passes		SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT		
Genetic group		Collection date	Genetic group	2b	2b	2a	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)		
REFERENCE VIRUSES																		
A/Thuringen/10/2022		SD2N	2b	2022-04-01	P18/23	320	640	320	320	40	80	80	320	640				
A/Berlin/20/17/2022		T1/24	2a	2022-10-19	SIAT2	320	160	160	160	<40	80	80	160	160	320			
A/Berlin/20/17/2022			2a	2022-04-16	SIAT/USAT5	160	320	640	640	160	160	160	320	320				
A/Berlin/20/17/2022		D18N, D2250 (egg)	2a	2022-04-17	E3/24	160	640	320	640	160	320	160	320	160	320			
A/Catalonia/NS/16151/2007/2022			2a.1b	2022-09-14	SIAT/USAT4	80	320	160	320	320	320	160	320	320	320			
A/Berlin/20/17/2022		NT220	2a.1 (H)	2022-10-13	MOCK/USAT1	80	320	160	320	40	80	320	640	640	320			
A/Hongkong/4801/2023			2a.1 (H)	2022-09-14	SIAT/USAT1	160	640	320	640	160	1280	640	1280	1280	640			
A/Sydney/05/2023			2a.3a.1 (H.1)	2023-09-12	SIAT/USAT1	320	640	320	640	320	1280	1280	2560	2560	1280			
A/Sydney/05/2023			2a.3a.1 (H.2)	2023-10-02	SIAT/USAT1	320	640	320	640	160	440	320	1280	1280	1280			
TEST VIRUSES																		
241324		A/Netherlands/1056/2023	2a.3a	2023-10-22	MOCK-MOCK/USAT1	80	320	160	160	160	160	320	320	320	ND	ND		
241889		A/lyon/CHU/823.440.8/2023	2a.3a	2023-11-14	MOCK/USAT1	80	320	320	320	80	320	160	320	320	ND	ND		
241332		A/Netherlands/1056/2023	2a.3a.1 (H)	2023-08-07	MOCK-MOCK/USAT2	160	640	160	320	160	1280	640	1280	640	1280	ND		
241331		A/Netherlands/1056/2023	2a.3a.1 (H)	2023-08-12	MOCK-MOCK/USAT1	80	320	160	320	160	440	320	1280	640	1280	ND		
240989		A/Netherlands/1056/2023	2a.3a.1 (H)	2023-10-17	SIAT2	80	320	160	320	80	320	160	320	160	320	ND		
240933		A/Croatia/14/2023	2a.3a.1 (H)	2023-10-01	SIAT1	160	320	320	320	160	1280	640	1280	640	1280	ND		
241326		A/Netherlands/1056/2023	2a.3a.1 (H.1)	2023-10-19	MOCK-MOCK/USAT1	160	320	160	320	160	1280	640	1280	640	1280	ND		
241163		A/Netherlands/1056/2023	2a.3a.1 (H.1)	2023-10-24	SIAT/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241162		A/Netherlands/1056/2023	2a.3a.1 (H.1)	2023-10-26	SIAT/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241311		A/Bulgaria/46/2023	2a.3a.1 (H.1)	2023-10-26	SIAT/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241386		A/Linking/BE23.4408/2023	2a.3a.1 (H.1)	2023-11-08	SIAT/USAT1	80	320	160	320	160	1280	640	1280	640	1280	ND		
241382		A/Ontario/BE23.4408/2023	2a.3a.1 (H.1)	2023-11-02	SIAT/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241380		A/Israel/BE23.5164/2023	2a.3a.1 (H.1)	2023-11-04	SIAT/USAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
241388		A/Turkey/14/2023	2a.3a.1 (H.1)	2023-11-20	SIAT/USAT1	160	640	320	320	160	1280	640	1280	640	1280	ND		
240861		A/Bulgaria/14/2023	2a.3a.1 (H.1)	2023-11-23	MOCK/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241434		A/Hongkong/4801/2023	2a.3a.1 (H.1)	2023-11-24	SIAT/USAT1	160	640	160	640	160	1280	640	1280	640	1280	ND		
241134		A/Hongkong/4801/2023	2a.3a.1 (H.1)	2023-11-25	SIAT1	160	320	160	320	80	440	320	640	640	320	ND		
241133		A/Hongkong/4801/2023	2a.3a.1 (H.1)	2023-12-08	SIAT1	160	320	160	320	80	440	320	640	640	320	ND		
241131		A/Hongkong/4801/2023	2a.3a.1 (H.1)	2023-12-11	SIAT/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241888		A/Berlin/20/17/2023	2a.3a.1 (H.1)	2023-12-15	SIAT1	160	320	160	320	80	440	320	640	640	320	ND		
241886		A/Catalonia/BE23.1407/2023	2a.3a.1 (H.1)	2023-12-15	SIAT/USAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
241888		A/Luxembourg/787819/2023	2a.3a.1 (H.1)	2023-12-18	SIAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
240984		A/lyon/CHU/823.440.8/2023	2a.3a.1 (H.1)	2023-12-19	MOCK/USAT1	160	640	160	320	160	1280	640	1280	640	1280	ND		
240993		A/lyon/CHU/823.440.8/2023	2a.3a.1 (H.1)	2023-12-27	MOCK/USAT1	160	320	160	320	80	320	160	320	320	640	ND		
241132		A/Bulgaria/14/2023	2a.3a.1 (H.1)	2023-12-27	SIAT1	160	640	160	320	160	1280	320	1280	640	1280	ND		
241428		A/Hongkong/4801/2024	2a.3a.1 (H.1)	2024-01-02	SIAT/USAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
241429		A/Hongkong/4801/2024	2a.3a.1 (H.1)	2024-01-08	SIAT/USAT1	160	640	160	320	160	1280	640	1280	640	1280	ND		
240372		A/Netherlands/1056/2023	2a.3a.1 (H.2)	2023-08-23	SIAT2	80	160	160	160	80	640	320	1280	640	1280	ND		
241916		A/Bulgaria/46/2023	2a.3a.1 (H.2)	2023-09-29	SIAT/USAT1	160	320	160	320	80	320	160	320	160	320	ND		
241914		A/Bulgaria/46/2023	2a.3a.1 (H.2)	2023-09-29	SIAT/USAT1	160	320	160	320	80	320	160	320	160	320	ND		
241329		A/Netherlands/1056/2023	2a.3a.1 (H.2)	2023-09-29	MOCK-MOCK/USAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
241913		A/Bulgaria/46/2023	2a.3a.1 (H.2)	2023-10-06	SIAT/USAT1	80	320	160	320	80	320	160	320	160	320	ND		
241328		A/Netherlands/1056/2023	2a.3a.1 (H.2)	2023-10-12	MOCK-MOCK/USAT1	320	640	320	640	160	640	320	1280	640	1280	ND		
241902		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-10-23	SIAT/USAT1	160	320	160	320	80	440	320	640	640	320	ND		
241912		A/Bulgaria/46/2023	2a.3a.1 (H.2)	2023-10-23	SIAT/USAT1	80	320	160	320	80	320	160	320	160	320	ND		
241910		A/Bulgaria/46/2023	2a.3a.1 (H.2)	2023-11-01	SIAT/USAT1	80	160	160	160	80	320	160	320	160	320	ND		
241322		A/Netherlands/1056/2023	2a.3a.1 (H.2)	2023-11-02	MOCK-MOCK/USAT1	320	640	320	640	160	640	320	1280	640	1280	ND		
241320		A/Netherlands/1056/2023	2a.3a.1 (H.2)	2023-11-03	MOCK-MOCK/USAT1	160	320	160	320	80	320	160	320	160	320	ND		
241388		A/Vietnam/BE23.4413/2023	2a.3a.1 (H.2)	2023-11-07	SIAT/USAT1	80	320	160	320	80	440	160	640	320	640	ND		
241386		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-11-08	MOCK/USAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
241384		A/Linking/BE23.4408/2023	2a.3a.1 (H.2)	2023-11-09	SIAT/USAT1	160	640	320	640	160	640	320	640	320	640	ND		
241316		A/Netherlands/1056/2023	2a.3a.1 (H.2)	2023-11-09	MOCK-MOCK/USAT1	160	640	320	640	160	1280	640	1280	640	1280	ND		
241001		A/Luxembourg/787819/2023	2a.3a.1 (H.2)	2023-11-13	MOCK/USAT1	320	320	320	640	320	640	320	640	320	640	ND		
240986		A/Bulgaria/46/2023	2a.3a.1 (H.2)	2023-11-18	MOCK/USAT1	80	320	320	640	80	320	160	320	160	320	ND		
240988		A/Ontario/BE23.4408/2023	2a.3a.1 (H.2)	2023-11-17	MOCK/USAT1	80	320	320	640	80	320	160	320	160	320	ND		
240986		A/lyon/CHU/823.440.8/2023	2a.3a.1 (H.2)	2023-11-25	MOCK/USAT1	160	160	160	160	80	320	160	320	160	320	ND		
241881		A/Bulgaria/14/2023	2a.3a.1 (H.2)	2023-11-27	SIAT/USAT1	160	320	160	320	160	640	160	640	160	640	ND		
240932		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-12-01	SIAT1	160	640	320	320	160	640	320	640	320	640	ND		
240931		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-12-02	SIAT1	80	320	160	320	80	320	160	320	160	320	ND		
240929		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-12-04	SIAT1	80	160	160	160	80	320	160	320	160	320	ND		
240930		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-12-04	MOCK/USAT1	80	160	160	320	80	320	160	320	160	320	ND		
240996		A/lyon/CHU/823.440.8/2023	2a.3a.1 (H.2)	2023-12-04	MOCK/USAT1	160	320	160	320	80	320	160	640	320	640	ND		
241423		A/Hongkong/4801/2023	2a.3a.1 (H.2)	2023-12-04	SIAT/USAT1	80	320	160	320	160	1280	320	1280	640	1280	ND		
240926		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-12-05	SIAT1	160	320	160	320	80	440	320	640	320	640	ND		
240927		A/Croatia/14/2023	2a.3a.1 (H.2)	2023-12-05	SIAT1	160	320	160	320	160	640	320	640	320	640	ND		
241889		A/Berlin/20/17/2023	2a.3a.1 (H.2)	2023-12-07	SIAT1	160	320	160	320	80	320	160	640	160	640	ND		
240995		A/lyon/CHU/823.440.8/2023	2a.3a.1 (H.2)	2023-12-13	MOCK/USAT1	160	320	160	320	80	320	160	640	320	640	ND		
240996		A/B																

Table H3-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nm Oseltamivir) 2024-02-09

Viruses	Other information	Passage	Ferret number	Genetic group	Collection Date	Haemagglutination inhibition titre							
						Post-infection ferret antiserum							
						A/Thuringen /10/22 SIAT	A/Switzerland /207/19/22 SIAT	A/Stockholm 5/21 SIAT	A/Darwin 9/21 Egg	A/Catal /SVH-2067/22 SIAT	A/Albania 289813/2022 SIAT	A/Massachusetts /18/2022 SIAT	A/Thailand /08/2022 Egg
						F36/22	F29/23	F15/22	F39/21	F41/22	F21/23	F36/23	F34/23
						2b	2b	2a	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)
REFERENCE VIRUSES													
A/Thuringen/10/2022	S282N		2b	2022-04-01	P1/SIAT3	320	640	160	160	40	80	160	80
A/Switzerland/28719/2022	T135A		2b	2022-12-19	SIAT2	160	1280	160	160	<40	80	80	160
A/Stockholm/5/2021			2a	2021-04-16	SIAT0/SIAT5	160	320	320	640	160	160	160	320
A/Darwin/9/2021	D186N, D225G (egg)		2a	2021-04-17	E3/E4	160	640	320	640	160	160	160	320
A/Catalonia/NSVH161512047/2022			2a.1b	2022-09-14	SIAT1/SIAT4	80	320	160	640	320	320	160	320
A/Albania/289813/2022	N122D		2a.3a.1 (H)	2022-12-13	MDCK1/SIAT1	160	320	160	320	80	640	320	1280
A/Massachusetts/18/2022			2a.3a.1 (H)		SIAT3/SIAT1	160	640	320	640	160	1280	640	1280
A/Thailand/08/2022			2a.3a.1 (H)		E3/E1	160	640	320	640	160	640	320	1280
TEST VIRUSES													
240577 A/Togo/1356/2023			2a.3a	2023-09-30	SIAT1	80	160	160	320	80	160	80	160
240570 A/Togo/1459/2023			2a.3a	2023-10-11	SIAT1	80	320	160	320	80	160	80	320
241623 A/France/GES-IPP26703/2023			2a.3a.1 (H.1)	2023-11-06	MDCK1/SIAT1	80	320	160	320	80	640	320	640
241611 A/France/IDF-RELAB-IPP32965/2023			2a.3a.1 (H.1)	2023-12-13	MDCK1/SIAT1	160	320	160	320	160	640	320	640
241607 A/France/IDF-RELAB-IPP32969/2023			2a.3a.1 (H.1)	2023-12-13	MDCK1/SIAT1	160	320	160	320	160	640	160	1280
241627 A/France/IDF-IPP21897/2023			2a.3a.1 (H.2)	2023-09-27	MDCK1/SIAT1	160	320	160	320	80	320	160	320
241615 A/France/HDF-IPP30115/2023			2a.3a.1 (H.2)	2023-12-02	MDCK1/SIAT1	160	320	160	320	80	320	160	320
241791 A/Beyern/13/2023			2a.3a.1 (H.2)	2023-12-07	P1/SIAT1	80	160	160	320	80	320	160	320
241789 A/Nordrhein-Westfalen/16/2023			2a.3a.1 (H.2)	2023-12-12	P1/SIAT1	160	320	160	320	80	640	160	640
241790 A/Rheinland-Pfalz/11/2023			2a.3a.1 (H.2)	2023-12-12	P1/SIAT1	160	320	160	640	80	640	160	640
241609 A/France/IDF-RELAB-IPP32966/2023			2a.3a.1 (H.2)	2023-12-13	MDCK1/SIAT1	80	320	160	320	160	320	160	320
241605 A/France/IDF-RELAB-IPP32874/2023			2a.3a.1 (H.2)	2023-12-15	MDCK1/SIAT1	160	320	160	320	80	320	160	320
241787 A/Nordrhein-Westfalen/1/2024			2a.3a.1 (H.2)	2024-01-02	P1/SIAT1	160	320	160	320	80	320	160	320
241792 A/Berlin/40/2023			2a.3a.1 (H.3)	2023-12-02	P1/SIAT1	160	320	160	320	80	640	320	640
241768 A/Slovenia/1634/2023			pending	2023-09-28	SIATx/SIAT1	160	320	160	320	80	320	160	320
241700 A/Belgium/94/2023			pending	2023-11-10	SIAT1/SIAT1	160	640	160	640	160	640	320	640
241697 A/Belgium/83/2023			pending	2023-11-16	SIAT1/SIAT1	160	640	320	640	80	640	320	640
241619 A/France/28187/2023			pending	2023-11-20	MDCK1/SIAT1	160	320	160	320	80	640	320	1280
241766 A/Slovenia/1968/2023			pending	2023-11-20	SIATx/SIAT1	160	640	320	640	160	1280	640	1280
241695 A/Belgium/93/2023			pending	2023-11-21	SIAT1/SIAT1	160	640	320	320	160	1280	640	1280
241693 A/Belgium/96/2023			pending	2023-11-22	SIAT1/SIAT1	160	640	160	320	160	1280	640	1280
241689 A/Belgium/110/2023			pending	2023-11-29	SIAT1/SIAT1	160	320	160	320	80	640	320	640
241687 A/Belgium/106/2023			pending	2023-11-30	SIAT1/SIAT1	160	320	160	320	80	320	160	640
241531 A/Roma/1/2023			pending	2023-12-09	SIAT3/SIAT1	160	320	160	320	80	640	160	640
241667 A/Lisboa/216/2023			pending	2023-12-15	SIAT1/SIAT1	160	320	160	320	80	640	160	320
241665 A/Lisboa/217/2023			pending	2023-12-20	SIAT1/SIAT1	160	320	160	320	80	320	160	640
241601 A/France/33176/2023			pending	2023-12-21	MDCK1/SIAT1	160	640	320	640	160	640	320	640
241788 A/Baden-Wuerttemberg/37/2023			pending	2023-12-27	P2/SIAT1	160	320	160	320	80	640	320	640
241599 A/France/33503/2023			pending	2023-12-27	MDCK1/SIAT1	80	320	160	320	80	320	160	320
241761 A/Slovenia/8/2024			pending	2023-12-28	MDCKx/SIAT1	160	320	160	320	80	640	320	640
241759 A/Slovenia/21/2024			pending	2024-01-01	MDCKx/SIAT1	160	320	160	320	80	320	160	320
241754 A/Slovenia/49/2024			pending	2024-01-06	MDCKx/SIAT1	80	320	80	160	80	160	160	320
241752 A/Slovenia/62/2024			pending	2024-01-08	SIATx/SIAT1	160	320	160	320	80	320	160	320
241749 A/Slovenia/119/2024			pending	2024-01-10	MDCKF/SIAT1	160	320	160	320	80	320	160	640
241747 A/Slovenia/145/2024			pending	2024-01-12	MDCKF/SIAT1	80	320	160	320	80	320	160	320
241745 A/Slovenia/180/2024			pending	2024-01-15	MDCKF/SIAT1	80	160	80	320	80	160	80	160

< relates to the lowest dilution of antiserum used

1 hyperimmune sheep serum; ND = Not Done

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Microneutralisation tables: H3N2

Viruses				Neutralisation titre								
				Post-infection ferret antisera								
				A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 28/19/2022 SIAT F41/22	A/Stockholm 9/2021 SIAT F10/22	A/Catalonia NSVH-2087/2022 SIAT F21/22	A/Albania 28/08/13/2022 SIAT F21/22	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 SIAT F34/23		
				Passage history Ferret number Passage history	2b	2b + T135A	2a	2a.1b	2a.3a.1+H122D	2a.3a.1		
REFERENCE VIRUSES												
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT 10 ⁶	119	199	84	<	<	<	<	<	
A/Switzerland/28/19/2022	2b + T135A	2022-12-19	SIAT1 10 ⁶	56	375	82	<	<	<	<	55	
A/Stockholm/9/2021	2a	2021-04-16	SIAT9/SIAT5 10 ⁶	<	112	95	58	<	<	<	88	
A/Catalonia/NSVH1512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁶	<	86	51	64	<	<	<	51	
A/Albania/28/08/13/2022	2a.3a.1+H122D	2022-12-13	MDCK1/SIAT 10 ⁶	76	274	117	55	450	269		600	
A/Massachusetts/18/2022	2a.3a.1	2022-06-04	SIAT3/SIAT 10 ⁶	97	485	201	68	570	448		685	
A/Thailand/8/2022	2a.3a.1		E3/E1 10 ⁶	346	1887	988	466	1728	735		2082	
TEST VIRUSES												
A/Catalonia/NSVH102165941/2023	2a.3a.1	2023-09-17	SIAT1/SIAT1	80	366	158	60	441	320		483	
A/Catalonia/NSVH151040742/2023	2a.3a.1	2023-09-06	SIAT1/SIAT1	118	576	283	70	731	419		612	
A/Catalonia/NSVH102158148/2023	2a.3a.1	2023-09-06	SIAT1/SIAT1	287	1696	720	195	1991	1280		2485	
A/Catalonia/NSVH102151296/2023	2a.3a.1 + H122D	2023-09-11	SIAT1/SIAT1	110	454	240	60	454	199		378	
A/Catalonia/NSVH10871625/2023	2a.3a.1 + H122D	2023-09-05	SIAT1/SIAT1	94	393	191	57	304	273		413	
A/Catalonia/NSVH102156515/2023	2a.3a.1 + H122D	2023-09-04	SIAT1/SIAT1	99	541	233	63	440	210		418	
A/Catalonia/NSVH102156229/2023	2a.3a.1 + H122D	2023-09-03	SIAT1/SIAT1	109	557	243	60	379	264		403	
A/Catalonia/NSVH102155361/2023	2a.3a.1 + H122D	2023-09-02	SIAT1/SIAT1	<	209	79	56		197		486	

<=2-fold

>2<=4-fold

>4<=8-fold

>8-fold

not recognised by the antiserum

Table MN-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2023-12-11

Viruses			Neutralisation titre							
			Post-infection ferret antisera							
			A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 287/19/2022 SIAT F09/23	A/Stockholm 9/2021 SIAT F19/22	A/Catania NSVH/2087/2022 SIAT F41/22	A/Albania 2898/13/2022 SIAT F21/23	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	
Genetic group	Collection Date	Passage history Ferret number Passage History	2b	2b + T135A	2a	2a.1b	2a.3a.1-NY122D	2a.3a.1	2a.3a.1	
REFERENCE VIRUSES										
A/Thuringen/10/2022	2b	A/27/2022-04-01	P1/SIAT3 10 ⁶	119	199	84	<	<	<	<
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10 ⁶	56	375	82	<	<	<	55
A/Stockholm/9/2021	2a	2021-04-16	SIAT0/SIAT5 10 ⁶	<	112	95	58	<	<	88
A/Catania/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁶	<	86	51	64	<	<	51
A/Albania/2898/13/2022	2a.3a.1-NY122D	2022-12-13	MDCK1/SIAT1 10 ⁶	76	274	117	55	450	269	600
A/Massachusetts/18/2022	2a.3a.1	2022-06-04	SIAT3/SIAT1 10 ⁶	97	485	201	68	570	448	685
A/Thailand/8/2022	2a.3a.1		E3E1 10 ⁶	346	1887	968	466	1728	735	2682
TEST VIRUSES										
A/Iran/20A12/2023	2a.3a.1-NY122D	2023-10-09	SIAT1	142	579	272	74	459	309	555
A/Iran/24N11/2023	2a.3a.1-NY122D	2023-09-27	SIAT1/SIAT1	106	520	166	62	320	234	486
A/Iran/26L11/2023	2a.3a.1-NY122D	2023-09-24	SIAT1/SIAT1	105	410	178	56	293	320	409
A/Iran/11G11/2023	2a.3a.1-NY122D	2023-09-17	SIAT1/SIAT1	103	298	143	52	269	287	452
A/Iran/22E11/2023	2a.3a.1-NY122D	2023-09-13	SIAT1/SIAT1	48	172	97	<	210	145	228
A/Muscot/7234671/2023	2a.3a.1-NY122D	2023-10-03	SIAT1	43	150	<	<	<	<	179
A/Al Musanah/7234645/2023	2a.3a.1-NY122D	2023-10-01	SIAT1	80	311	164	51	320	263	430
A/As Sowiq/7234643/2023	2a.3a.1-NY122D	2023-10-01	SIAT1	107	491	260	59	387	152	367

<=2-fold >2<=4-fold >4<=8-fold >8-fold not recognised by the antiserum

Table MN-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-01-11

Viruses			Neutralisation titre							
			Post-infection ferret antisera							
			A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 287/19/2022 SIAT F09/23 2b + T135A	A/Stockholm 9/2021 SIAT F19/22	A/Catalonia NSVH-2087/2022 SIAT F41/22	A/Albania 2898/13/2022 SIAT F21/23	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	
Genetic group	Collection Date	Passage history Ferret number	Passage history	2b	2b + T135A	2a	2a.1b	2a.3a.1+H122D	2a.3a.1	2a.3a.1
REFERENCE VIRUSES										
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT3 10 ⁴	119	199	84	<	<	<	<
A/Switzerland/287/19/2022	2b + T135A	2022-12-19	SIAT1 10 ⁴	56	375	82	<	<	<	55
A/Stockholm/9/2021	2a	2021-04-16	SIAT0/SIAT5 10 ⁴	<	112	95	58	<	<	88
A/Catalonia/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁴	<	86	51	64	<	<	51
A/Albania/2898/13/2022	2a.3a.1+H122D	2022-12-13	MDCK1/SIAT1 10 ⁵	76	274	117	55	450	269	600
A/Massachusetts/18/2022	2a.3a.1	2022-06-04	SIAT3/SIAT1 10 ⁵	97	485	201	68	570	448	685
A/Thailand/8/2022	2a.3a.1		E3/E1 10 ⁴	346	1887	968	466	1728	735	2082
TEST VIRUSES										
A/Qatar/23-VI-23-0106322/2023	2a.3a	2023-10-24	SIAT1	<	72	56	<	61	42	111
A/Qatar/25-VI-23-0107245/2023	2a.3a.1+H122D	2023-10-28	SIAT1	<	183	51	49	149	186	360
A/Qatar/10-VI-23-0107179/2023	2a.3a.1+H122D	2023-10-28	SIAT1	<	152	41	43	144	153	263
A/Qatar/10-VI-23-0107015/2023	2a.3a.1+H122D	2023-10-26	SIAT1	64	419	114	43	282	157	359
A/Qatar/10-VI-23-0106239/2023	2a.3a.1+H122D	2023-10-24	SIAT1	<	121	41	45	124	115	238
A/Qatar/16-VI-23-0109945/2023	2a.3a.1+H122D	2023-10-23	SIAT1	<	277	83	<	244	152	297
A/Qatar/10-VI-23-0109504/2023	2a.3a.1+H122D	2023-10-22	SIAT1	<	138	51	45	130	130	242
A/Qatar/10-VI-23-0103751/2023	2a.3a.1+H122D	2023-10-16	SIAT1	54	236	<	<	<	52	213

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Table MN-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-01-15

Viruses				Neutralisation titre							
				Post-infection ferret antisera							
				A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 28719/2022 SIAT F09/23 2b + T135A	A/Stockholm 9/2021 SIAT F19/22	A/Catalonia NSVH-2087/2022 SIAT F41/22	A/Albania 2898/13/2022 SIAT F21/23	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	
Genetic group	Collection Date	Passage history	Passage number	2b	2b + T135A	2a	2a.1b	2a.3a.1+H122D	2a.3a.1	2a.3a.1	
REFERENCE VIRUSES											
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT3 10 ⁶	119	199	84	<	<	<	<	
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10 ⁶	56	375	82	<	<	<	55	
A/Stockholm/9/2021	2a	2021-04-16	SIAT5/SIAT15 10 ⁶	<	112	95	58	<	<	88	
A/Catalonia/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁶	<	86	51	64	<	<	51	
A/Albania/2898/13/2022	2a.3a.1+H122D	2022-12-13	MDCK1/SIAT1 10 ⁶	76	274	117	55	450	269	600	
A/Massachusetts/18/2022	2a.3a.1	2022-06-04	SIAT3/SIAT1 10 ⁶	97	485	201	68	570	448	685	
A/Thailand/8/2022	2a.3a.1		E3/E1 10 ⁶	346	1887	968	466	1728	735	2682	
TEST VIRUSES											
A/Norway/11304/2023	2a.3a.1	2023-11-03	SIAT1	58	300	108	49	312	211	341	
A/Norway/11289/2023	2a.3a.1	2023-11-03	SIAT1	40	207	69	<	151	132	181	
A/Norway/10429/2023	2a.3a.1	2023-10-16	SIAT1	45	299	86	<	202	111	246	
A/Berlin/37/2023	2a.3a.1	ND	P1/SIAT1	54	330	108	41	195	152	240	
A/Brandenburg/32023	2a.3a.1	2023-10-09	P1/SIAT1	<	128	55	<	155	94	135	
A/Norway/11360/2023	2a.3a.1+H122D	2023-11-03	SIAT1	68	347	103	57	216	167	243	
A/Berlin/32/2023	2a.3a.1+H122D	2023-11-13	P1/SIAT1	<	285	117	<	252	163	250	
A/Berlin/20/2023	2a.3a.1+H122D	2023-11-06	P1/SIAT1	64	283	127	8	242	195	296	
A/Hessen/20/2023	2a.3a.1+H122D	2023-10-11	P1/SIAT1	133	541	181	71	789	341	614	
A/Berlin/27/2023	2a.3a.1+H122D	2023-10-09	P1/SIAT1	78	291	121	<	259	143	265	
A/Bayern/11/2023	2a.3a.1+H122D	2023-09-18	P1/SIAT1	63	295	114	43	209	158	272	

<=2-fold >2-4-fold >4-8-fold >8-fold not recognised by the antiserum

Table MN-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-01-22

Viruses			Neutralisation titre									
			Post-infection ferret antisera									
			A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 28/19/2022 SIAT F09/23	A/Stockholm 5/2021 SIAT F19/22	A/Catalonia NSVH-2087/2022 SIAT F41/22	A/Albania 28/08/2022 SIAT F21/23	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23			
Genetic group	Collection Date	Passage history Ferret number Passage History	2b	2b + T135A	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)			
REFERENCE VIRUSES												
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT2 10 ⁵	119	199	84	<	<	<	<		
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10 ⁵	56	375	82	<	<	<	55		
A/Stockholm/5/2021	2a	2021-04-16	SIAT0/SIAT5 10 ⁵	<	112	95	58	<	<	88		
A/Catalonia/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁵	<	86	51	64	<	<	51		
A/Albania/28/08/2022	2a.3a.1 (H)	2022-12-13	MDCK1/SIAT1 10 ⁵	76	274	117	55	450	269	600		
A/Massachusetts/18/2022	2a.3a.1 (H)	2022-06-04	SIAT3/SIAT1 10 ⁵	97	485	201	68	570	448	685		
A/Thailand/8/2022	2a.3a.1 (H)	-	E3E1 10 ⁵	346	1887	968	466	1728	735	2082		
TEST VIRUSES												
A/Netherlands/01732/2023	2a.3a.1 (H)	2023-10-09	hCK1/SIAT1	<	254	76	<	311	158	263		
A/Netherlands/01685/2023	2a.3a.1 (H)	2023-09-08	hCK1/SIAT1	<	200	68	<	240	126	233		
A/HongKong/3062/2023	2a.3a.1 (H)	2023-11-29	MDCK1/SIAT2	46	233	79	<	268	176	266		
A/HongKong/3040/2023	2a.3a.1 (H)	2023-11-27	MDCK1/SIAT2	41	183	88	<	225	160	251		
A/Norway/11672/2023	2a.3a.1 (H.1)	2023-10-29	SIAT2	55	247	103	<	295	230	407		
A/HongKong/3082/2023	2a.3a.1 (H.1)	2023-11-28	MDCK1/SIAT2	44	291	105	<	306	154	293		
A/HongKong/3058/2023	2a.3a.1 (H.1)	2023-11-27	MDCK1/SIAT2	44	250	86	<	354	235	392		
A/HongKong/3050/2023	2a.3a.1 (H.1)	2023-11-27	MDCK1/SIAT2	<	203	80	<	251	127	215		
A/Netherlands/01735/2023	2a.3a.1 (H.2)	2023-10-20	hCK1/SIAT1	52	251	105	<	222	162	158		
A/Netherlands/01733/2023	2a.3a.1 (H.2)	2023-10-18	hCK1/SIAT1	50	229	73	<	158	114	156		
A/Netherlands/01729/2023	2a.3a.1 (H.2)	2023-10-13	hCK1/SIAT1	<	106	46	<	61	43	68		
A/HongKong/3059/2023	2a.3a.1 (H.2)	2023-11-28	MDCK1/SIAT2	<	174	74	<	198	132	155		
A/HongKong/3081/2023	2a.3a.1 (H.3)	2023-11-27	MDCK1/SIAT2	<	140	62	<	127	75	153		
A/Netherlands/01744/2023	2a.3a.1 (H.4)	2023-10-29	hCK1/SIAT1	44	245	89	<	158	136	223		

<=2-fold

>=2-4-fold

>=4-8-fold

>=8-fold

not recognised by the antiserum

<x2-fold >2x4-fold >4x8-fold >8-fold not recognised by the antiserum

Table MN-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-01-29

Viruses			Neutralisation titre							
			Post-infection ferret antisera							
			A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 287/19/2022 SIAT F10/23 2b + T135A	A/Stockholm 9/2021 SIAT F19/22	A/Catalonia NSVH-2087/2022 SIAT F41/22	A/Albania 2898/13/2022 SIAT F21/23	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	
Genetic group	Collection Date	Passage history Ferret number Passage history		2b	2b + T135A	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	
REFERENCE VIRUSES										
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT3 10 ⁶	119	199	84	<	<	<	<
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10 ⁶	56	375	82	<	<	<	55
A/Stockholm/9/2021	2a	2021-04-16	SIAT0/SIAT5 10 ⁶	<	112	95	58	<	<	88
A/Catalonia/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁶	<	86	51	64	<	<	51
A/Albania/2898/13/2022	2a.3a.1 (H)	2022-12-13	MDCK1/SIAT1 10 ⁶	76	274	117	55	450	269	600
A/Massachusetts/18/2022	2a.3a.1 (H)	2022-06-04	SIAT1/SIAT1 10 ⁶	97	485	201	68	570	448	685
A/Thailand/8/2022	2a.3a.1 (H)	-	E3E1 10 ⁶	346	1887	968	466	1728	735	2652
TEST VIRUSES										
A/Ghana/2186/2023	2a.3a (S.1.3.1)	2023-10-23	SIAT1	<	95	67	<	56	<	106
A/HongKong/3021/2023	2a.3a.1 (H)	2023-11-23	MDCK1/SIAT2	64	259	130	<	280	198	366
A/BurkinaFaso/64/2023	2a.3a.1 (H)	2023-10-31	SIAT1/SIAT1	<	149	70	<	173	98	141
A/Switzerland/62289/2023	2a.3a.1 (H)	2023-10-30	SIAT1	73	201	135	<	309	243	340
A/HongKong/3044/2023	2a.3a.1 (H.1)	2023-11-26	MDCK1/SIAT2	59	278	138	40	400	378	409
A/BurkinaFaso/63/2023	2a.3a.1 (H.1)	2023-10-13	SIAT1/SIAT1	66	311	160	52	320	184	405
A/Switzerland/28412/2023	2a.3a.1 (H.1)	2023-11-14	SIAT1	90	233	112	<	283	272	354
A/HongKong/397/2023	2a.3a.1 (H.2)	2023-11-26	MDCK1/SIAT2	68	236	118	40	162	130	345
A/Switzerland/6849/2023	2a.3a.1 (H.2)	2023-11-24	SIAT1	54	258	135	<	206	108	246
A/Switzerland/25797/2023	2a.3a.1 (H.2)	2023-11-23	SIAT1	75	373	157	<	283	155	379
A/Switzerland/95384/2023	2a.3a.1 (H.2)	2023-11-13	SIAT1	74	349	186	<	238	144	234
A/Iceland/39771/2023	2a.3a.1 (H.2)	2023-12-06	SIAT1	74	420	202	<	293	256	318
A/Iceland/38547/2023	2a.3a.1 (H.2)	2023-12-05	SIAT1	72	403	180	55	276	141	475
A/Iceland/38354/2023	2a.3a.1 (H.2)	2023-12-04	SIAT1	62	320	180	<	283	198	280
A/Iceland/37831/2023	2a.3a.1 (H.2)	2023-12-04	SIAT1	95	362	179	<	283	157	289
A/Iceland/37843/2023	2a.3a.1 (H.2)	2023-11-30	SIAT1	58	308	141	<	204	151	275

<=2-fold >2-4-fold >4-8-fold >8-fold not recognised by the antiserum

Table MN-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-02-05

Viruses				Neutralisation titre							
				Post-infection ferret antisera							
				A/Thuringen 10/2022 SIAT F36/22	A/Switzerland 28/19/2022 SIAT F09/23	A/Stockholm 9/2021 SIAT F19/22	A/Catalonia NSVH-2087/2022 SIAT F41/22	A/Albania 28/08/2022 SIAT F21/23	A/Massachusetts 18/2022 SIAT F36/23	A/Thailand 8/2022 Egg F34/23	
Genetic group	Collection Date	Passage history Ferret number Passage history		2b	2b + T135A	2a	2a.1b	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	
REFERENCE VIRUSES											
A/Thuringen/10/2022	2b	2022-04-01	P1(SIAT) 10 ⁵	119	199	84	<	<	<	<	
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10 ⁴	56	375	82	<	<	<	55	
A/Stockholm/9/2021	2a	2021-04-16	SIAT0/SIAT5 10 ⁴	<	112	95	58	<	<	88	
A/Catalonia/NSVH161512067/2022	2a.1b	2022-09-14	SIAT1/SIAT4 10 ⁴	<	86	51	64	<	<	51	
A/Albania/28813/2022	2a.3a.1 (H)	2022-12-13	MDCK1/SIAT1 10 ⁵	76	274	117	55	450	269	600	
A/Massachusetts/18/2022	2a.3a.1 (H)	2022-06-04	SIAT3/SIAT1 10 ⁵	97	485	201	68	570	162	685	
A/Thailand/8/2022	2a.3a.1 (H)	-	E3E1 10 ⁴	346	1887	968	466	1728	735	1709	
TEST VIRUSES											
A/Iceland/94327/2023	2a.3a.1 (H)	2023-11-01	SIAT1	80	309	138	45	461	212	450	
A/Sagevia/54/2023	2a.3a.1 (H.1)	2023-10-22	SIAT1/SIAT2	40	176	71	<	259	199	310	
A/Burgos/28/2023	2a.3a.1 (H.1)	2023-09-27	SIAT1/SIAT1	50	242	113	<	291	156	294	
A/Burgos/15/2023	2a.3a.1 (H.1)	2023-09-24	SIAT1/SIAT1	<	160	<	<	158	116	207	
A/Burgos/27/2023	2a.3a.1 (H.1)	2023-09-24	SIAT1/SIAT1	55	257	112	<	266	193	289	
A/Iceland/937811/2023	2a.3a.1 (H.2)	2023-11-29	SIAT1	57	242	99	<	228	139	309	
A/Iceland/937274/2023	2a.3a.1 (H.2)	2023-11-25	SIAT1	66	301	146	<	282	205	270	
A/Iceland/937759/2023	2a.3a.1 (H.2)	2023-11-24	SIAT1	61	300	134	<	222	152	227	
A/Iceland/934533/2023	2a.3a.1 (H.2)	2023-11-19	SIAT1	71	299	150	<	232	217	288	
A/Iceland/936373/2023	2a.3a.1 (H.2)	2023-11-18	SIAT1	99	400	202	45	314	173	320	
A/Iceland/936888/2023	2a.3a.1 (H.2)	2023-11-15	SIAT1	68	267	122	<	273	229	331	
A/Iceland/935741/2023	2a.3a.1 (H.2)	2023-11-13	SIAT1	40	209	101	<	219	120	261	
A/Iceland/939710/2023	2a.3a.1 (H.2)	2023-11-12	SIAT1	80	292	120	<	244	227	320	

<=2-fold

>2-4-fold

>4-8-fold

>8-fold

not recognised by the antiserum

<=2-fold >2-4-fold >4-8-fold >8-fold not recognised by the antiserum

Table MN-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-02-08

Viruses	Genetic group	Collection Date	Passage history Parent number Passage history	Neutralisation titre									
				Post-infection ferret antisera									
				A/Thuringen 20/2021 SIAT F1022	A/Belarus/2019/2022 SIAT F0933	A/Belarus/2021 SIAT F1522	A/Catalonia N009-2017/2022 SIAT F4122	A/Alaska 2001/2022 SIAT F2123	A/Massachusetts 19/2022 SIAT F0923	A/Thailand 10/2022 Egg F4223	A/Sydney 05/2022 SIAT 1009429	A/Sydney 01/2023 SIAT 1009439	
				2b	2b + T135A	2a	2a 1b	2a 3a.1 (H)	2a 3a.1 (H)	2a 3a.1 (H)	2a 3a.1 (H.1)	2a 3a.1 (H.2)	
				REFERENCE VIRUSES									
A/Thuringen/10/2022	2b	2022-04-01	P1/SIAT3 10*	119	199	84	<	<	<	<	-	-	
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10*	96	375	82	<	<	<	55	-	-	
A/Buckhake/5/2021	2a	2021-04-16	SIAT9/SIAT5 10*	<	112	95	58	<	<	88	-	-	
A/Catalonia/N0191615120/2022	2a 1b	2022-09-14	SIAT1/SIAT4 10*	<	86	51	64	<	<	51	-	-	
A/Alaska/289813/2022	2a 3a.1 (H)	2022-12-13	MDCK1/SIAT1 10*	76	274	117	55	450	289	600	-	-	
A/Massachusetts/18/2022	2a 3a.1 (H)	2022-08-04	SIAT3/SIAT1 10*	97	485	201	68	970	162	685	871	824	
A/Thailand/8/2022	2a 3a.1 (H)	-	ESIE1 10*	346	1887	988	466	1728	735	1709	2015	2065	
A/Sydney/896/2022	2a 3a.1 (H.1)	2022-09-12	SIAT1/SIAT1 10*	71	343	116	40	411	234	352	901	804	
A/Sydney/879/2022	2a 3a.1 (H.2)	2022-10-02	SIAT2/SIAT1 10*	125	382	176	58	311	227	368	992	821	
TEST VIRUSES													
A/Montenegro/24258/2023	2a 3a.1 (H.1)	2023-12-07	SIAT1	43	120	87	<	154	99	191	489	372	
A/Montenegro/2229/2023	2a 3a.1 (H.1)	2023-10-16	SIAT1	89	239	183	<	274	189	293	775	623	
A/Hungary/153/2023	2a 3a.1 (H.1)	2023-12-04	MDCK1/SIAT1	64	316	120	<	287	225	288	819	701	
A/Montenegro/244877/2023	2a 3a.1 (H.2)	2023-12-13	SIAT1	134	310	183	66	591	236	434	960	1264	
A/Montenegro/245115/2023	2a 3a.1 (H.2)	2023-12-10	SIAT1	131	498	198	74	493	295	473	1166	1387	
A/Algeria/1767/2023	2a 3a.1 (H.2)	2023-12-13	SIAT1	109	416	154	60	272	210	364	987	912	
A/Algeria/1229/2023	2a 3a.1 (H.2)	2023-12-19	SIAT1	92	320	146	26	275	165	292	698	805	
A/Hungary/148/2023	2a 3a.1 (H.2)	2023-12-11	MDCK1/SIAT1	134	448	204	48	716	268	354	949	883	
A/Hungary/148/2023	no seq	2023-11-30	MDCK1/SIAT1	69	270	115	<	219	188	261	744	687	

<=2.6td >2-4.6td >4-8.6td >8.6td not recognised by the antiserum

Table MN-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2024-02-12

Viruses				Neutralisation titre										
				Post-infection ferret antisera										
				A/Thailand/10/2021	A/Belarus/2019/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022	A/Belarus/2021/2022
				SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023	SIAT F1023
Genetic group	Collection Date	Passage history	Parent number	2a	2b + T135A	2a	2a	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)	2a.3a.1 (H)
REFERENCE VIRUSES														
A/Thailand/10/2022	2a	2022-04-01	P1/SIAT3 10 ⁴	119	199	84	<	<	<	<	-	-	-	-
A/Switzerland/28719/2022	2b + T135A	2022-12-19	SIAT1 10 ⁴	96	375	82	<	<	<	55	-	-	-	-
A/Belarus/2021/2022	2a	2021-04-16	SIAT3/SIAT5 10 ⁴	<	112	95	58	<	<	88	-	-	-	-
A/Belarus/2021/2022	2a.3a.1 (H)	2022-05-14	SIAT1/SIAT4 10 ⁴	<	86	51	64	<	<	51	-	-	-	-
A/Belarus/2021/2022	2a.3a.1 (H)	2022-12-13	MDCK/SIAT1 10 ⁴	76	274	117	55	450	289	600	-	-	-	-
A/Belarus/2021/2022	2a.3a.1 (H)	2022-06-04	SIAT3/SIAT1 10 ⁴	97	485	201	68	570	162	685	671	624	624	624
A/Thailand/10/2022	2a.3a.1 (H)	-	ESIE1 10 ⁴	346	1887	988	466	1728	735	1709	2015	2065	2065	2065
A/Sydney/056/2023	2a.3a.1 (H.1)	2023-05-12	SIAT1/SIAT1 10 ⁴	71	343	116	40	411	234	352	901	804	804	804
A/Sydney/079/2023	2a.3a.1 (H.2)	2023-10-02	SIAT2/SIAT1 10 ⁴	125	382	176	58	311	227	366	992	821	821	821
TEST VIRUSES														
A/Guang/729/2023	2a.3a.1 (H)	2023-05-12	SIAT2	<	176	85	<	91	82	166	501	382	382	382
A/Belarus/2021/2022	2a.3a.1 (H)	2023-11-08	SIAT2	<	72	49	<	98	112	150	472	332	332	332
A/Russia/219/2023	2a.3a.1 (H.1)	2023-12-19	SIAT1	66	248	131	<	301	194	290	896	517	517	517
A/Hongkong/199/2023	2a.3a.1 (H.1)	2023-12-20	MDCK/SIAT1	<	100	56	<	99	61	85	302	243	243	243
A/Hongkong/204/2023	2a.3a.1 (H.1)	2023-12-19	MDCK/SIAT1	<	88	46	<	75	59	75	302	243	243	243
A/Hongkong/177/2023	2a.3a.1 (H.1)	2023-12-11	MDCK/SIAT1	<	85	47	<	77	49	75	230	209	209	209
A/Hongkong/206/2023	2a.3a.1 (H.2)	2023-12-18	MDCK/SIAT1	57	143	88	<	137	78	130	584	476	476	476
A/Hongkong/147/2023	2a.3a.1 (H.2)	2023-11-26	MDCK/SIAT1	<	89	59	<	68	47	87	286	219	219	219
A/Belarus/2021/2023	2a.3a.1 (H.2)	2023-11-16	SIAT1	78	294	138	44	302	148	320	811	640	640	640
A/Hongkong/202/2023	not seq	2023-12-20	MDCK/SIAT1	68	243	130	<	213	137	220	831	698	698	698

<2-fold >2-4-fold 4-8-fold 8-16-fold not recognised by the antisera

Fold-reduction table: H3N2

Reference Virus	clade	<4-fold difference		4-fold difference		>4-fold difference		total
		number	percentage	number	percentage	number	percentage	
A/Thuringen/10/2022 Cell	2b	146	61.6	80	33.8	11	4.6	237
A/Switzerland/28719/2022 Cell	2b	55	23.2	121	51.1	61	25.7	237
A/Stockholm/5/2021 Cell	2a	96	40.5	103	43.5	38	16.0	237
A/Darwin/9/2021 Egg	2a	206	86.9	30	12.7	1	0.4	237
A/Catalonia/NSVH161512067/2022 Cell	2a.1b	57	24.1	129	54.4	51	21.5	237
A/Albania/289813/2022 Cell	2a.3a.1	194	81.9	31	13.1	12	5.1	237
A/Brandenburg/15/2022 Egg	2a.3a.1	2	3.3	13	21.7	45	75.0	60
A/Massachusetts/18/2022 Cell	2a.3a.1	96	40.5	99	41.8	42	17.7	237
A/Thailand/08/2022 Egg	2a.3a.1	174	73.4	59	24.9	4	1.7	237

A/H3N2: Egg isolates

Virus	Status	Genetic group/clade	Comment
A/Norway/29511/2021	SAN-017/017A	3C.2a1b.2a.2 (E50K)	Egg adaptation: (D186N, D225G), 1-way pass
A/Norway/24873/2021	sent out	3C.2a1b.2a.2 (N96S, I192F, N378S)	Egg adaptation: (D225G)
A/Slovenia/9216/2022	NIB-137A/B (1-way pass, 2-way in progress)	3C.2a1b.2a.2 (H156S, D53G, D104G)	Egg adaptation: (S219F, D225G)
A/Slovenia/9318/2022	sent out	3C.2a1b.2a.2 (H156S, D53G, D104G)	Egg adaptation: (D186N, D225G)
A/Catalonia/NSVH161512067/2022	sent out	3C.2a1b.2a.2 (I140K, R299K)	Egg adaptation: (D225G)
A/Catalonia/NSVH172304934/2022	available to send out	3C.2a1b.2a.2 (I140K, R299K)	Egg adaptation: (D225G)
A/Poland/97/2022	Reference only	3C.2a1b.2a.2 (S145N)	Egg adaptation: (D225G)
A/Norway/24873/2021	sent out	2a.3 (E50K, D53N, N96S, I192F)	Egg adaptation: (D225G)
A/Brandenburg/15/2022	sent out recently	2a.3a.1 (E50K, I223V, I140R, R140K)	Egg adaptation: (D186G)
A/Albania/290243/2022	sent out recently	2a.3a.1 (E50K, I223V, I140R, R140K, S91N, N122D)	Egg adaptation: (F195Y)
A/Albania/290270/2022	NYMC X-411A (2-way pass)	2a.3a.1 (E50K, I223V, I140R, R140K, S91N, N122D)	Egg adaptation: (N190T, S228G)

Virus	Status	Genetic group/clade	Comment
A/Albania/289813/2022	NIB-138 (2-way pass)	2a.3a.1 (E50K, I223V, I140R, R140K, S91N, N122D)	Egg adaptation: (F195Y, S228T)
A/Switzerland/28719/2022	NIB-136A (2-way pass), SAN-024 (2-way pass)	2b (T135A, S262N - Thuringen group)	Egg adaptation: (H183F), 2-way pass
A/Burgos/270/2022	available to send out	2a.1 (T135A - CHO)	Egg adaptation: (D225G)

□

A/H3N2: HI reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/ <u>Thuringen</u> /10/2022	2b	P1/SIAT3	F36/22
A/Switzerland/28719/2022	2b	SIAT1	F29/23
A/Stockholm/5/2021	2a	SIAT0/SIAT3	F15/22
A/Darwin/9/2021	2a	E3/E4	F39/21
A/Catalonia/NSVH161512067/2022	2a.1b	SIAT1/SIAT4	F41/22
A/Albania/289813/2022	2a.3a.1 (H)	MDCK4	F21/23
A/ <u>Brandenberg</u> /15/2022	2a.3a.1 (H)	Ex/E5	F18/23
A/Massachusetts/18/2022	2a.3a.1 (H)	SIAT3/SIAT1	F36/23
A/California/122/2022	2a.3a.1 (H)	E1/E1	F33/23
A/Thailand/08/2022	2a.3a.1 (H)	E3/E1	F34/23
IVR-237 (A/Thailand/08/2022)	2a.3a.1 (H)	E3/D7/E1	F35/23

A/H3N2: MN reagents and references

Virus	Genetic group	Virus passage	Ferret ID
A/ <u>Thuringen</u> /10/2022	2b	P1/SIAT3	F36/22
A/Switzerland/28719/2022	2b	SIAT1	F09/23
A/Stockholm/5/2021	2a	SIAT0/SIAT5	F15/22
A/Catalonia/NSVH161512067/2022	2a.1b	SIAT1/SIAT4	F41/22
A/Albania/289813/2022	2a.3a.1 (H)	MDCK1/SIAT1	F21/23
A/Massachusetts/18/2022	2a.3a.1 (H)	SIAT3/SIAT1	F36/23
A/Thailand/8/2022	2a.3a.1 (H)	E3/E1	F34/23

Susceptibility to antivirals: H3

Table AV1. Antiviral (NAI) testing of isolates with collection dates after 2023-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	
2023									
September	23		10		0				33
October	17	1	23		8				49
November	31		22		24				77
December	49		27		8				84
2024									
January	9		3		0				12
Totals	129	1	85	0	40	0	0	0	255

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-02-12

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Qatar/10-VI-23-0103273/2023	H1pdm	15-Oct-23	421		H275Y
Outliers					

¹ Clinical sample not available

Table AV2. Antiviral (BXM) testing of isolates with collection dates after 2023-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	
2023									
September	13		6		0				19
October	8		18		8				34
November	12		13		17				42
December	1		4		0				5
2024									
January	0		0		0				0
Totals	34	0	41	0	25	0	0	0	100

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-02-09

Virus name	Subtype/ Lineage	Collection date	Fold difference	NA substitution
			Baloxavir	
Outliers				

¹ Clinical sample not available

In-silico assessment for the presence of NA substitutions associated with reduced susceptibility to neuraminidase inhibitors (NAI) was conducted for 365 H1N1, 323 H3N2 and 40 B/Victoria NA gene sequences generated at the WIC and deposited in GISAID for viruses with collection dates from 1st September 2023. No substitutions associated with reduced susceptibility were detected, with the following exceptions: • A/H1N1 A/Qatar/10-VI-23-0103273/2023 showed presence of H275Y, which was confirmed by phenotypic testing with a Highly reduced inhibition profile. Based on whole genome sequencing at the WIC, full-length PA gene sequences were generated from 289 H1N1, 315 H3N2 viruses with collection dates from 1st September 2023. No sequences with mutations encoding amino acid markers associated with reduced susceptibility to Baloxavir marboxil (BXM) were identified.

These H3N2 viruses from Netherlands have a deletion on the NA at position 30, two of them received and sequence-confirmed at the WIC. See NL report for further details.

A/Netherlands/01733/2023 A/Netherlands/01735/2023 A/Netherlands/01736/2023 A/Netherlands/01738/2023

Summary: H3N2

Genetic analyses

Clade 3C.2a1b.2a.2 (renamed as **2** since February 2023) predominated since 1st February in all geographic regions where H3N2 circulated.

During this reporting period, the great majority of H3 viruses detected belong to clade **2a.3a.1**, which share substitutions E50K with clade 2a.3a and present additional substitutions I140K and I223V. Within clade 2a.3a.1, viruses with I25V, V347M and some with I418V (subclade H.1) were seen in Europe, South-East Asia and Australia, whereas viruses with N122D (potential loss of N-glycosylation) and K276E (subclade H.2) were predominating in Europe, the US and the Middle East. Subclade H included a subclade with

N122D and V347M viruses from US, Australia and Qatar (the latter characterised by V112I and S145N) whereas the rest of subclade H was detected in Africa, South East Asia and a minority in Europe and the US, with no subclade-specific amino acids. Minor subclade H.4 was seen in Norway, West Africa and Central America.

A few viruses predominating in West Africa with substitutions K276E and V347M cluster within clade 2a.3a.

Antigenic analyses

We note variable recognition by current 2a.3a.1 reference and vaccine antisera for a number of 2a.3a.1 viruses that have been analysed since September.

Influenza B

Genetic analyses: B/Victoria

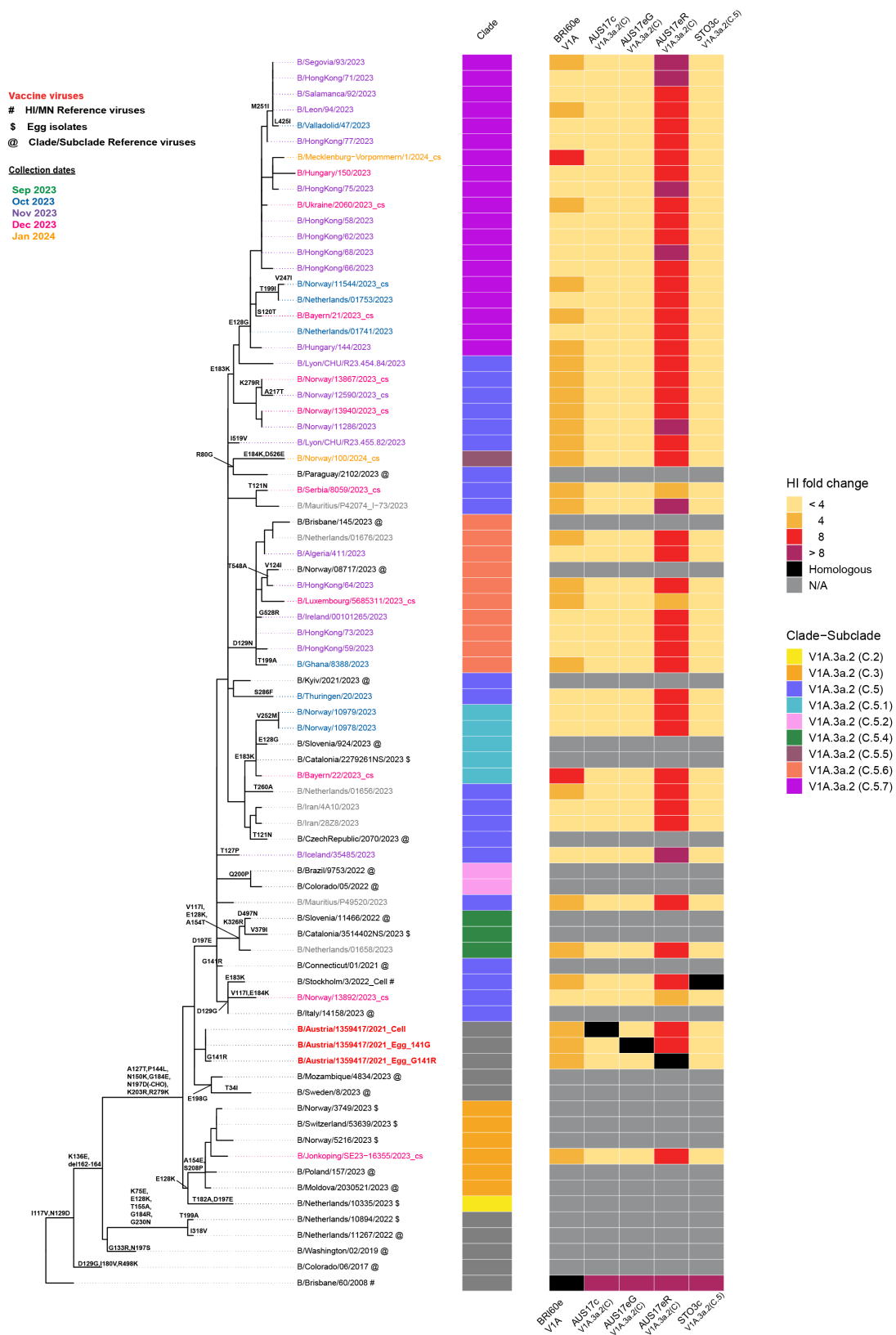
Microreact: Subclade



Maximum likelihood phylogenetic tree: B/Victoria

Maximum likelihood time-resolved phylogenetic tree inferred using IQTree2 from HA sequence data obtained from GISAID from 1st September onwards, manually curated and then downsampled using Treemer to retain a representative tree topology of 200 sequences with emphasis on European viruses and viruses sequenced at the WIC. Amino acid signatures annotated with Treetime. References and CVVs are marked as Cell or Egg.

Jan 2024



Antigenic analyses: B/Victoria

Haemagglutination inhibition tables: B/Victoria

Table BV-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2023-11-21

Viruses	Other information	Genetic group	Collection Date	Genetic group	Passage	Ferret number	Haemagglutination inhibition titre				
							Post-infection ferret antiserum				
							B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 5741	B/Austria 1359417/21 MDCK NIB F01/21	B/Austria 1359417/21 Egg G141 F40/21	B/Austria 1359417/21 Egg G141R F44/21	B/Stock 3/22 MDCK F28/22
							V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES											
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4		1280	40	<40	<40	<40	
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4		320	1280	640	320	320	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4		640	1280	640	320	320	
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5		160	1280	320	2560	320	
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3		640	640	640	320	640	
TEST VIRUSES											
240050	B/Iran/28Z8/2023	V1A.3a.2 (C.5)	2023-06-06	SIAT1/MDCK1		640	1280	640	320	640	
240049	B/Iran/4A10/2023	V1A.3a.2 (C.5)	2023-07-19	SIAT1/MDCK1		640	1280	320	320	640	
< relates to the lowest dilution of antiserum used								Vaccine			
1 hyperimmune sheep serum; ND = Not Done								SH 2022			
								NH 2022-23			
								SH 2023			
								NH 2023-24			

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

Vaccine
 SH 2022
 NH 2022-23
 SH 2023
 NH 2023-24

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table BV-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2023-12-20

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre						
					Passage	Ferret number	Post-infection ferret antiserum				
							B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK
							Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22
					V1A	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2		
REFERENCE VIRUSES											
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40		
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	SIAT1/MDCK4	320	1280	320	320	320		
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2	2021-01-09	E3/E4	320	1280	640	320	640		
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2	2021-01-09	E3/E5	160	1280	320	2560	320		
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2	2022-03-22	SIAT1/MDCK3	640	1280	640	320	1280		
TEST VIRUSES											
240426	B/Thuringen/20/2023	V1A.3a.2 (C.5)	2023-10-23	P1/MDCK1	640	1280	320	320	640		
240286	B/Norway/10979/2023	V1A.3a.2 (C.5.1)	2023-10-25	MDCK1	640	640	320	320	640		
240287	B/Norway/10978/2023	V1A.3a.2 (C.5.1)	2023-10-25	MDCK1	640	640	320	320	640		
< relates to the lowest dilution of antiserum used ¹ hyperimmune sheep serum; ND = Not Done							Vaccine SH 2022 NH 2022-23 SH 2023 NH 2023-24				

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table BV-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-01-23

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre										
					Post-infection ferret antiserum					B/Stock 3/22 MDCK					
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R							
					Passage Sh 539, 540, 543, 544, 570, 571, 5741	Ferret number NIB F01/21	F40/21	F44/21	F28/22						
											V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES															
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40						
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	320	1280	320	320	320						
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E4	640	1280	640	640	1280						
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	1280	640	2560	640						
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	1280	640	320	640						
TEST VIRUSES															
240363 B/Mauritius/P42074_1-73/2023		V1A.3a.2 (C.5)	2023-07-01	MDCKx/MDCK1	320	640	320	160	640						
240408 B/Netherlands/01656/2023		V1A.3a.2 (C.5)	2023-07-28	hCK1/MDCK1	320	640	320	320	640						
240282 B/Norway/11286/2023		V1A.3a.2 (C.5)	2023-11-03	MDCK2	320	640	320	160	320						
240450 B/Iceland/35485/2023		V1A.3a.2 (C.5)	2023-11-10	MDCK1	640	640	320	160	320						
240407 B/Netherlands/01658/2023		V1A.3a.2 (C.5.4)	2023-08-11	hCK1/MDCK1	320	640	320	320	640						
240406 B/Netherlands/01676/2023		V1A.3a.2 (C.5.6)	2023-08-25	hCK1/MDCK1	320	640	320	320	640						
240384 B/Ghana/8388/2023		V1A.3a.2 (C.5.6)	2023-10-25	MDCK1	320	640	320	320	640						
240669 B/HongKong/59/2023		V1A.3a.2 (C.5.6)	2023-11-10	MDCK1/MDCK1	640	640	640	320	640						
240603 B/Ireland/00101265/2023		V1A.3a.2 (C.5.6)	2023-11-15	MDCK1	640	640	320	320	640						
240665 B/HongKong/64/2023		V1A.3a.2 (C.5.6)	2023-11-19	MDCK1/MDCK1	320	640	320	320	640						
240879 B/Algeria/411/2023		V1A.3a.2 (C.5.6)	2023-11-21	MDCK1	640	640	320	320	640						
240655 B/HongKong/73/2023		V1A.3a.2 (C.5.6)	2023-11-23	MDCK1/MDCK1	640	640	320	320	640						
240405 B/Netherlands/01753/2023		V1A.3a.2 (C.5.7)	2023-10-06	hCK1/MDCK1	640	640	320	320	640						
240741 B/Valladolid/47/2023		V1A.3a.2 (C.5.7)	2023-10-17	MDCK1/MDCK2	640	640	320	320	640						
240404 B/Netherlands/01741/2023		V1A.3a.2 (C.5.7)	2023-10-22	hCK1/MDCK1	640	640	320	320	640						
240671 B/HongKong/58/2023		V1A.3a.2 (C.5.7)	2023-11-09	MDCK1/MDCK1	640	640	320	320	640						
240735 B/Leon/94/2023		V1A.3a.2 (C.5.7)	2023-11-15	MDCK1/MDCK2	320	640	320	320	640						
240737 B/Salamanca/92/2023		V1A.3a.2 (C.5.7)	2023-11-15	MDCK1/MDCK1	640	640	320	320	320						
240733 B/Segovia/93/2023		V1A.3a.2 (C.5.7)	2023-11-17	MDCK1/MDCK1	320	640	320	160	320						
240667 B/HongKong/62/2023		V1A.3a.2 (C.5.7)	2023-11-18	MDCK1/MDCK1	640	640	320	320	640						
240663 B/HongKong/66/2023		V1A.3a.2 (C.5.7)	2023-11-19	MDCK1/MDCK1	640	640	320	320	640						
240661 B/HongKong/68/2023		V1A.3a.2 (C.5.7)	2023-11-20	MDCK1/MDCK1	640	640	320	160	640						
240659 B/HongKong/71/2023		V1A.3a.2 (C.5.7)	2023-11-20	MDCK1/MDCK1	640	640	320	160	640						
240657 B/HongKong/77/2023		V1A.3a.2 (C.5.7)	2023-11-22	MDCK1/MDCK1	640	640	640	320	640						
240653 B/HongKong/75/2023		V1A.3a.2 (C.5.7)	2023-11-24	MDCK1/MDCK1	640	640	320	160	320						
240944 B/Hungary/150/2023		V1A.3a.2 (C.5.7)	2023-12-05	MDCK1/MDCK1	640	640	320	320	640						
< relates to the lowest dilution of antiserum used															
1 hyperimmune sheep serum; ND = Not Done															

< relates to the lowest dilution of antiserum used
¹ hyperimmune sheep serum; ND = Not Done

Vaccine
SH 2022
NH 2022-23
SH 2023
NH 2023-24

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (no homologous titre)

Table BV-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2024-02-07

Viruses	Other information	Genetic group	Collection Date	Genetic group	Haemagglutination inhibition titre				
					Post-infection ferret antiserum				
					B/Bris 60/08 Egg	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Stock 3/22 MDCK
					Passage Ferret number Sh 539, 540, 543, 544, 570, 571, 5741	NIB F01/21	F40/21	F44/21	F28/22
					V1A	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C)	V1A.3a.2 (C.5)
REFERENCE VIRUSES									
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	40	<40	<40	<40
B/Austria/1359417/2021	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	SIAT1/MDCK4	160	640	320	160	160
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141)	V1A.3a.2 (C)	2021-01-09	E3/E3	320	640	320	320	640
B/Austria/1359417/2021 Isolate 2	A127T, P144L, K302R (G141R)	V1A.3a.2 (C)	2021-01-09	E3/E5	320	640	320	1280	320
B/Stockholm/3/2022	D129G, E183K, D197E	V1A.3a.2 (C.5)	2022-03-22	SIAT1/MDCK3	320	640	320	320	320
TEST VIRUSES									
241355 B/Jonkoping/SE23-16355/2023		V1A.3a.2 (C.3)	2023-12-26	SIAT1/MDCK1	320	640	320	160	320
240354 B/Mauritius/P49520/2023		V1A.3a.2 (C.5)	2023-07-12	MDCK1	320	320	160	160	320
240971 B/Lyon/CHU/R23.454.84/2023		V1A.3a.2 (C.5)	2023-11-24	MDCK2/MDCK1	320	640	320	160	320
240970 B/Lyon/CHU/R23.455.82/2023		V1A.3a.2 (C.5)	2023-11-25	MDCK2/MDCK1	320	640	320	160	640
241536 B/Norway/12590/2023		V1A.3a.2 (C.5)	2023-11-30	MDCK1	320	320	160	160	320
241061 B/Serbia/8059/2023		V1A.3a.2 (C.5)	2023-12-11	MDCK1	320	640	320	320	640
241535 B/Norway/13867/2023		V1A.3a.2 (C.5)	2023-12-27	MDCK1	320	320	160	160	320
241534 B/Norway/13940/2023		V1A.3a.2 (C.5)	2023-12-29	MDCK1	320	320	320	160	320
241533 B/Norway/13892/2023		V1A.3a.2 (C.5)	2023-12-30	MDCK1	640	640	320	320	640
241772 B/Bayern/22/2023		V1A.3a.2 (C.5.1)	2023-12-21	P1/MDCK1	160	320	320	160	320
241532 B/Norway/100/2024		V1A.3a.2 (C.5.5)	2024-01-05	MDCK1	320	640	320	160	320
241044 B/Luxembourg/5685311/2023		V1A.3a.2 (C.5.6)	2023-12-27	MDCK1	320	640	320	320	640
241537 B/Norway/11544/2023		V1A.3a.2 (C.5.7)	2023-10-18	MDCK1	320	640	320	160	320
240945 B/Hungary/144/2023		V1A.3a.2 (C.5.7)	2023-11-24	MDCK1/MDCK2	320	640	320	160	640
241773 B/Bayern/21/2023		V1A.3a.2 (C.5.7)	2023-12-18	P1/MDCK1	320	320	160	160	320
241353 B/Ukraine/2060/2023		V1A.3a.2 (C.5.7)	2023-12-25	MDCK1	320	640	320	160	320
241769 B/Mecklenburg-Vorpommern/1/2024		V1A.3a.2 (C.5.7)	2024-01-08	P1/MDCK1	160	320	160	160	320
241518 B/Roma/1/2023		pending	2023-11-22	MDCK3/MDCK1	320	640	320	160	320
241631 B/Lisboa/59/2023		pending	2023-11-29	MDCK1/MDCK1	320	640	320	160	320
241629 B/Lisboa/60/2023		pending	2023-11-30	MDCK1/MDCK1	320	640	320	320	640
241517 B/Padova/7/2023		pending	2023-12-13	MDCK3/MDCK1	640	640	320	160	320
241725 B/Slovenia/2285/2023		pending	2023-12-15	MDCKF/MDCK1	320	640	320	160	320
241775 B/Thuringen/21/2023		pending	2023-12-18	P1/MDCK1	320	320	320	160	320
241774 B/Thuringen/22/2023		pending	2023-12-18	P2/MDCK1	320	320	160	160	320
241771 B/Thuringen/23/2023		pending	2023-12-27	P1/MDCK1	320	320	160	160	320
241770 B/Thuringen/1/2024		pending	2024-01-08	P1/MDCK1	320	320	320	160	320

< relates to the lowest dilution of antiserum used

¹ hyperimmune sheep serum; ND = Not Done

Vaccine

SH 2022

NH 2022-23

SH 2023

NH 2023-24

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (no homologous titre)

Fold-reduction table: B/Vic

Reference Virus	clade	<4-fold difference		4-fold difference		>4-fold difference		total
		number	percentage	number	percentage	number	percentage	
B/Brisbane/60/2008 Egg	V1A	24	40.7	32	54.2	3	5.1	59
B/Austria/1359417/2021 Cell	V1A.3a.2	59	100	0	0	0	0	59
B/Austria/1359417/2021 Egg G141	V1A.3a.2	59	100	0	0	0	0	59
B/Austria/1359417/2021 Egg G141R	V1A.3a.2	0	0	4	6.8	55	93.2	59
B/Stockholm/3/2022 Cell	V1A.3a.2	59	100	0	0	0	0	59

B/Victoria: Egg isolates

Virus	Status	Genetic group/clade	Comment
B/Netherlands/10894/2022	BX-119 (2-way pass)	V1A.3 (G184R)	Egg adaptation: (T199A), 2-way pass
B/Catalonia/3514402NS/2023	sent out	V1A.3a.2 (D197E, E128K, A154T, V117I, K326R)	Egg adaptation: (no change)
B/Catalonia/2279261NS/2023	BX-123 (1-way pass)	V1A.3a.2 (E183K)	Egg adaptation: (no change)
B/Netherlands/10335/2023	BX-127/127A (1-way pass)	V1A.3a.2 (E128K, A221T, T182A, D197E, T221A)	Egg adaptation: (no change)
B/Norway/5216/2023	available to send out	V1A.3a.2 (E128K, A154E, S208P)	Egg adaptation: (no change)
B/Norway/3749/2023	available to send out	V1A.3a.2 (E128K, A154E, S208P)	Egg adaptation: (no change)
B/Switzerland/53639/2023	available to send out	V1A.3a.2 (E128K, A154E, S208P)	Egg adaptation: (no change)

B/Victoria: Reagents and references

Virus	Genetic group	Virus passage	Ferret ID
B/Brisbane/60/2008	V1A	E4/E4	sheep pool
B/Austria/1359417/2021	V1A.3a.2 (C)	SIAT1/MDCK4	NIB F01/21
B/Austria/1359417/2021 G141	V1A.3a.2 (C)	E3/E4	F40/21
B/Austria/1359417/2021 G141R	V1A.3a.2 (C)	E3/E5	F44/21
B/Stockholm/3/2022	V1A.3a.2 (C.5)	SIAT1/MDCK3	F28/22

Susceptibility to antivirals: B/Victoria

Table AV1. Antiviral (NAI) testing of isolates with collection dates after 2023-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2023									
September	23		10		0				33
October	17	1	23		8				49
November	31		22		24				77
December	49		27		8				84
2024									
January	9		3		0				12
Totals	129	1	85	0	40	0	0	0	255

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-02-12

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Qatar/10-VI-23-0103273/2023	H1pdm	15-Oct-23	421		H275Y
Outliers					

¹ Clinical sample not available

Table AV2. Antiviral (BXM) testing of isolates with collection dates after 2023-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	NI	R/I/HRI	
2023									
September	13		6		0				19
October	8		18		8				34
November	12		13		17				42
December	1		4		0				5
2024									
January	0		0		0				0
Totals	34	0	41	0	25	0	0	0	100

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2024-02-09

Virus name	Subtype/ Lineage	Collection date	Fold difference	NA substitution
			Baloxavir	
Outliers				

¹ Clinical sample not available

In-silico assessment for the presence of NA substitutions associated with reduced susceptibility to neuraminidase inhibitors (NAI) was conducted for 365 H1N1, 323 H3N2 and 40 B/Victoria NA gene sequences generated at the WIC and deposited in GISAID for viruses with collection dates from 1st September 2023. No substitutions associated with reduced susceptibility were detected, with the following exceptions: • A/H1N1 A/Qatar/10-VI-23-0103273/2023 showed presence of H275Y, which was confirmed by phenotypic testing with a Highly reduced inhibition profile. Based on whole genome sequencing at the WIC, full-length PA gene sequences were generated from 289 H1N1, 315 H3N2 viruses with collection dates from 1st September 2023. No sequences with mutations encoding amino acid markers associated with reduced susceptibility to Baloxavir marboxil (BXM) were identified.

Summary: B/Victoria

Genetic analyses

Clade V1A.3a.2 viruses predominated since 1st February 2023 in geographic regions where B/Victoria-lineage viruses were detected.

No Clade V1A.3 viruses were detected since 1st February 2023.

No B/Yamagata lineage viruses have been detected since March 2020.

Antigenic analyses

All V1A.3a.2 viruses tested were well-recognised by antisera raised against B/Austria/1359417/2021 and -like viruses.